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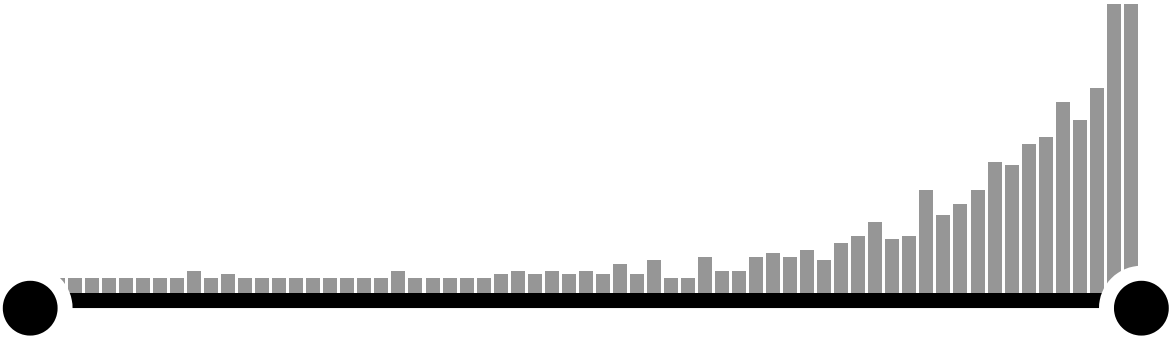
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	Document title	Authors	Source	Year	Citations
1	Article • *Open access* **Adaptive acoustic comfort: Evidence from an acoustic neutrality approach in naturally ventilated office buildings**	Yuksel Dicle S.,Aletta F., Kang J.	Applied Acoustics, 257, 111627	2027	0

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Acoustic comfort is commonly assessed using fixed *sound*-level criteria that assume stable exposure–response relationships. However, in naturally ventilated buildings, occupants' evaluations may vary with changing *acoustic* exposure and adaptive responses. Although such adaptive relationships are well established in *thermal* comfort research, they remain underexplored in *acoustics*. This study examines an adaptive perspective on *acoustic* comfort assessment through a field study conducted in six naturally ventilated or mixed-mode open-plan offices involving 72 participants and 480 observations. Repeated in situ perceptual evaluations, concurrent indoor and *outdoor sound*-level measurements, and self-reported adaptive behaviours were collected and analysed using mixed-effects modelling approaches. For level-based evaluations, an *Acoustic* Sensation Vote (ASV) of 0 ("neither loud nor quiet") emerged as the operational *acoustic* neutral reference, representing a threshold-like transition to dissatisfaction. Office-specific neutral levels ranged from 47.2 to 53.3 dB and tended to be higher in offices with higher indoor *sound*-level exposure. Adaptive actions were more likely at higher preceding-hour indoor L_{Aeq} (OR = 1.55 per 5 dB), and observations with reported adaptive behaviour were slightly more quiet-oriented after accounting for concurrent indoor *sound* level and office. *Outdoor* L_{Aeq} did not provide additional explanatory information for ASV; however, among neutral evaluations, each 1-dB increase in *outdoor* L_{Aeq} was associated with an approximately 0.16-dB increase in concurrent indoor *sound* level. Overall, the findings provide empirical support for a context-sensitive adaptive *acoustic* comfort perspective in which *acoustic* evaluations are considered in relation to indoor exposure, office context, behavioural responses and *outdoor acoustic* conditions. The findings should be interpreted within the scope of six London offices and a field-based design in which *acoustic* and contextual conditions could not be fully standardised across participants. For office design and assessment, the results suggest that fixed *sound*-level criteria can be complemented by consideration of office-specific *acoustic* exposure and occupants' opportunities for behavioural adjustment, particularly in naturally ventilated workplaces.

2	Article **Thermal resilience and well-being in Alpine social housing under heatwave conditions: a mixed-methods study for climate-adaptive retrofit**	Paoletti G.,Gupta A., Carnelli F.,Toledo L., Babich F.	Energy and Buildings, 374, 118442	2027	0

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Climate change is intensifying heatwaves in cold-climate regions, increasing indoor overheating risks for vulnerable groups, including elderly and low-income residents in social housing. Evidence on *thermal* resilience and climate-adaptive retrofit for Alpine social housing remains limited, particularly at dwelling scale and under real occupancy conditions. This study investigates indoor overheating as both measured exposure and lived experience in twelve social housing apartments in Casette Inglesi district of Bolzano, Italy, during summer 2024. A mixed-methods

Document title	Authors	Source	Year	Citations

approach combined indoor environmental monitoring, architectural survey, and resident interviews on perceived comfort and heat-related well-being, adaptive practices, and constraints. Over valid monitored periods, time above EN 16798–1 Category II adaptive comfort threshold ranged from 0% to 47.48%, reaching 99.74% during July heatwave in a west-facing upper-floor living room beneath an uninsulated pitched roof, despite window replacement and reported A/C use in an adjacent room. Interviews showed that residents relied on multiple adaptive strategies, including daytime window closure, shading, cross-ventilation, fans, and A/C. However, their practical applicability varied due to health conditions, energy-cost concerns, limited control over building-level interventions, comfort preferences, and non-*thermal* constraints such as *outdoor noise*, mosquitoes, and safety concerns. Prioritised interventions ranged from external shading, insect screens, and fans to envelope improvements and accessibility upgrades. Findings suggest that overheating risk in Alpine social housing is shaped not only by envelope performance, but also by the interaction between dwelling exposure, adaptive opportunities, resident vulnerability, and institutional feasibility. The study proposes a transferable mixed-methods approach for identifying evidence-informed retrofit priorities in vulnerable Alpine social housing.

Article • *Open access*

3 **Exploring interactions of thermal, acoustic, and aesthetic environments on thermal experience in parks** — Zhang Y.,Uchiyama Y., Sato M. — Scientific Reports, 16(1), 8564 — 2026 — 0

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The *thermal* comfort of urban parks is crucial for residents' well-being and urban livability. There is limited research on the integration of different environmental factors in urban parks, with most studies focusing on hot, sunny days in summer. This study explored the interactions of multi-sensory factors, including *thermal*, *acoustic*, and aesthetic, on *thermal* experience in parks across various weather types. The results reveal that *thermal* conditions directly dominate *thermal* experience. *Acoustic* comfort has a relatively weak positive effect on *thermal* comfort, and this effect disappears under strong heat-stress conditions. Natural *sounds* help increase *thermal* acceptability, whereas mechanically dominated soundscapes have the opposite effect. Aesthetic satisfaction also contributes to higher *thermal* acceptability. Compared with blue spaces, aesthetic satisfaction with green spaces exerts a more consistently positive influence on *thermal* experience. The negative effect of PET is significantly weaker than the positive effects of perceived *acoustic* comfort and aesthetic satisfaction on overall satisfaction. The revealed multi-sensory interactions yield practical recommendations for urban planners and designers in addressing climate change and enhancing urban livability.

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4 **Evaluation of heat stress mitigation potentials of climate change adaptation measures in cities: Scenario analyses for technical- and nature-based solutions using 3D microclimate modelling** — Eingruber N., Zimmermann D., Korres W.,Lohnert U., Schneider K. — Climate Risk Management , 54, 100894 — 2026 — 0

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Urban areas are expected to warm significantly more than the global average according to climate projections. This will lead to many adverse effects such as increasing health risks and mortality rates, making climate change adaptation crucial for future urban planning. Identifying effective and suitable adaptation measures requires assessment of cooling potentials, societal acceptance and cost effectiveness based on *sound*-scientific analyses. High-resolution microclimate modelling and scenario analyses are a promising approach to evaluate *thermal* effects of climate adaptation measures. A physically-based 3D-gridded ENVI-met model was parameterized based on field measurements and remote sensing data to simulate the microclimate of a 16-ha urban high-density area in

Document title	Authors	Source	Year	Citations

Cologne/Germany. The model was successfully validated using a densely-distributed NETATMO sensor network in the study area. We evaluated 92 adaptation scenarios for a 20-year extreme heat event to quantify heat mitigation potentials. Scenarios included technical- and nature-based solutions at buildings as well as in street corridors. Measures were simulated individually and in combinations, across varying idealized and more realistic implementation degrees. Results showed a significant mean air temperature reduction by -3.0 K up to -6.3 K as well as significant mean cooling effects on physiological equivalent temperature (PET) of -4.3 K up to -11.1 K in pedestrian height level, particularly for scenarios with high greenery coverage. Combinations of different nature-based and technical solutions performed better than single measures. Even partial implementations (e.g., 25% greened buildings) provided significant cooling effects in the entire quarter, while high-reflective surfaces and shading structures had limited impact on *thermal outdoor* comfort.

Article

5 **Environmental regulation of sunken courtyards in standalone underground spaces: field measurements and ventilation-oriented design strategies** — Wen Y.,Ouyang M., Wang Y.,... Guo T.,Wei H. — Tunnelling and Underground Space Technology , 178, 108030 — 2026 — 0

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While sunken courtyards are widely recognised for enhancing underground environmental quality, modern architectural practices frequently prioritise aesthetic and circulation. This deficiency is critical in standalone underground spaces, where the absence of adjacent aboveground massing restricts wind pressure differentials and downdraughts. Using a standalone underground museum as a case study, this research correlates early design strategies with post-construction field measurements to evaluate compliance with wind, *thermal*, *acoustic*, and air-quality regulations. Furthermore, Computational Fluid Dynamics simulations are utilised to analyse the aerodynamic impact of courtyard morphology on natural ventilation. The study identifies an optimal length-to-depth (L/D) ratio of 1.5, balancing ventilation performance with land-use efficiency. However, under low ambient wind conditions, supplementary strategies are necessary to enhance underground natural ventilation, thereby elevating average indoor and *outdoor* wind velocities to approximately 0.380 m/s and 0.650 m/s, respectively. Consequently, this research quantitatively verifies the efficacy of spatial loops, windcatchers, and wind deflectors. Results demonstrate that spatial loops significantly enhance natural ventilation by enriching the distribution of pressure differentials and facilitating global cross-ventilation. Conversely, windcatchers and wind deflectors only offer moderate enhancements for sunken wells (L/D = 0.5), while exerting attenuation effects or negligible impacts on the sunken courtyard (L/D = 1.5), such as a paradoxical 34.44 % reduction in indoor wind velocity caused by aerodynamic conflicts. These findings facilitate the design transition towards open morphology and performance integration, both of which are essential for modern underground spaces.

Article

6 **Assessing impacts of human activity on indoor air pollutants using noise and CO2-based variables with LSTM-LightGBM: a library case study** — Zhang H.,Srinivasan R., Chen C.,... Ganesan V., Zhou R. — Atmospheric Environment , 385, 122350 — 2026 — 0

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This study demonstrates the limitations of using carbon dioxide (CO_2) alone as a proxy for human activity intensity in multi-pollutant indoor air quality modeling. In a university library, an integrated indoor–*outdoor* monitoring system collected two weeks of continuous measurements of CO_2, *noise*, temperature, relative humidity, $PM_{2.5}$, PM_{10}, CO, O_3, NO_2, SO_2, and eTVOCs. A hybrid time-series framework combining long short-term memory (LSTM) networks and Light Gradient Boosting Machine (LightGBM) was developed to compare six feature-set scenarios based on different combinations of CO_2, *noise*, temperature, and relative humidity. Association analysis and feature extraction showed that the combined CO_2, *noise*, temperature, and relative humidity factors had markedly stronger relationships with most indoor pollutants than CO_2 or *noise* alone ($R = 0.59–0.91$, $p < 0.05$). CO_2 was more informative for particulate matter, whereas *noise* was better suited to several gaseous pollutants. Validation with the LSTM–LightGBM framework supported these results. Beyond the general benefit of adding predictors, leave-one-predictor-out ablation comparisons indicated pollutant-specific and partially non-redundant predictor roles. *Noise* provided

	Document title	Authors	Source	Year	Citations

independent information beyond CO_2 for NO_2, CO, and eTVOCs, whereas temperature and relative humidity primarily improved prediction through interactions with activity-related proxies rather than serving as universally informative variables. The full-variable model is therefore interpreted as a reference upper-bound scenario, while the main finding is that pollutant prediction depends on distinct combinations of CO_2, *noise*, and *thermo*-hygrometric variables, without implying direct causal effects of human activity. These results provide a practical basis for improving feature engineering and indoor air quality control in similar built environments.

Article

7 **Robust Thermal Stability, Flame Retardancy, and Weather Resistance Cross-Linked Cellulose Aerogel toward High-Performance Passive Radiative Cooling**

Zhao Z.,Zhu S.,Zhang C., ... Li Z.,Wu X.

ACS Applied Materials and Interfaces , 18(43)

2026

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The frequent occurrence of high temperatures has driven the development of green passive cooling technologies. Traditional passive radiative cooling materials have single performance, insufficient weather resistance and structural stability, and are unable to adapt to *outdoor* multifunctional *thermal* management. In this study, cellulose nanofibers were used as the substrate, and silica and methyltrimethoxysilane were incorporated to prepare a composite aerogel. Tests show that the optimal sample has excellent radiation cooling, heat insulation, flame retardancy, hydrophobicity, and *sound* absorption performance. With a solar reflectance of 93.5% and a mid-infrared emissivity of 94.3%, it provides a maximum *outdoor* temperature reduction of 15.7 °C. Its minimum *thermal* conductivity reaches 0.034 W m^{-1} K^{-1}, offering excellent *thermal* insulation performance. The total heat release and the rate of heat release were significantly reduced, and flame retardancy was greatly improved. In addition, the sample exhibits a water contact angle of 137.2°, endowing it with favorable self-cleaning properties. The average *sound* absorption coefficient reaches 0.78 in the broadband range of 1000–6300 Hz, and the *noise* reduction performance is superior to that of commercial materials. This multifunctional composite aerogel overcomes the performance limitations of traditional materials and offers new insights into the development and application of *outdoor* integrated *thermal* management functional materials.

Article

8 **Numerical simulation and experimental comparison of strain measurements within a vertical bifacial photovoltaic noise barrier demonstrator**

Casasola M.,Engelen T., Kyranaki N., ... Poortmans J.,Daenen M.

Solar Energy Materials and Solar Cells , 305, 114438

2026

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A photovoltaic *noise* barrier (PVNB) is an example of infrastructure-integrated photovoltaics (IIPV) that serves as a conventional roadside infrastructure element while also generating solar energy. These integrated applications of photovoltaic (PV) modules are particularly relevant in areas where there is limited land area available for the installation of PV plants. Like any engineering structure, a PVNB is influenced by environmental factors such as temperature variations, wind loads and other external stressors. This study focuses on temperature fluctuations and the resulting *thermomechanical* strains within a PVNB demonstrator in Genk, Belgium. To achieve this, optical fibre Bragg grating (FBG) sensors for temperature and strain monitoring are implemented within the PV modules on the inner surface of the glass cover at three different locations. Both the trend of increases and decreases in monitored temperatures and strains are in good agreement with the data recorded by a nearby meteorological station. Subsequently, numerical simulations are performed and compared with the experimentally measured values, demonstrating a strong correlation with only one stressor input, i.e. the monitored temperature within the PV modules. Furthermore, the influence of using alternative encapsulant materials, such as *thermoplastic* polyolefin (TPO) or polyvinyl butyral (PVB) instead of the more commonly used ethylene vinyl acetate (EVA), on the strains and stress distributions on the cells is studied. This work proves, therefore, the possibility of real-time monitoring of internal strains within PV-integrated *outdoor* applications and strengthens the use of numerical simulations in forecasting internal stresses for the use in degradation models.

Article

	Document title	Authors	Source	Year	Citations
9	**A framework for assessing smart thermostat fault detection and diagnosis under unmeasurable building load disturbances**	Ejenakevwe K.A., Tiamiyu N.,Hurt R., Wang G.,Song L.	Energy and Buildings , 368, 117856	2026	0

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Smart *thermostat*-based fault detection and diagnosis (FDD) for residential air-conditioning (AC) systems is gaining traction due to its low-cost prospect. However, smart *thermostats* measure only a limited set of variables, raising concerns about the reliability and applicability of *thermostat*-driven FDD. This study therefore proposes a representative co-simulation testbed and six quantitative performance metrics for assessing smart *thermostat* FDD methods. The testbed integrates models of a residential building, a vapor compression AC system, and unmeasurable building load disturbances (UBLD) arising from occupant behavior and weather variability. For representativeness, a co-simulation approach was used with EnergyPlus for building modeling and a Dymola for AC modeling. Additionally, components from three specialized Modelica-based libraries were used to develop the testbed, making it more representative. To demonstrate the testbed's capability to capture UBLDs due to irregular occupancy patterns and/or weather conditions (e.g., windspeed and *outdoor* temperature), 15 case studies of faults and UBLDs were designed and simulated. The UBLD scenarios were designed from two survey data. Simulation results were then used to analyze the impact of UBLDs and common AC faults on FDD performance. The analyses show that UBLDs could cause false alarm rates of up to 30% even with a 15% deviation threshold, and that this can be reduced with a cumulative sum (CUSUM) logic. Furthermore, a signal-to-*noise* ratio metric – incorporating both false positive and false negative rates – is applied to assess a commonly used FDD feature, duty factor, and revealed its limited effectiveness. Overall, this study demonstrates the value of the proposed co-simulation testbed and evaluation metrics for assessing the performance and limitations of smart *thermostat*-based data-driven FDD methods in residential AC systems.

	Article				
10	**Material Selection for Components of an On-Campus Study Pod**	Odah E.,Abdullah R., Alsswey A.,Mostafa A.	Jordan Journal of Civil Engineering , 20(4)	2026	0

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On-campus study pods offer private, flexible learning spaces designed to provide controlled *thermal* and *acoustic* environments within busy university settings. This study presents a systematic material selection process for the key components of a compact, modular, weather-exposed study pod designed for 2–4 students. Using the Ashby methodology, six application-specific performance indices were derived to optimize stiffness-to-weight, strength-to-weight, wear resistance, *thermal* insulation, *thermal* expansion compatibility, and cost efficiency. Realistic design constraints, including self-weight, occupancy loads, wind pressure, and deflection limits, were applied alongside *outdoor* durability, *acoustic* comfort, and sustainability requirements. The optimal materials identified are 5000-series aluminium alloys for the structural frame, a multilayer sandwich configuration (treated softwood/bamboo outer layer, rigid polyurethane foam core, and cork inner layer) for enclosure panels, plywood and polypropylene for seating and desks, and treated bamboo/plywood with polycarbonate options for roofing. Quantitative comparisons demonstrate approximately 42% mass reduction for the aluminium frame versus stainless steel, and 3–5 times lower heat transfer for cork/PU foam panels compared with untreated wood. The selected materials balance structural performance, *thermal* and *acoustic* comfort, recyclability, and economic feasibility while supporting circular-economy principles. Limitations and future validation pathways, including finite element analysis and prototype testing, are discussed.

	Erratum				
11	**Corrigendum to "Environmental impacts of standard building renovations: City-level extrapolation from a real case study in Spain"(Energy and Buildings, (2026), 354, C, (116938), (S0378778825016688), 10.1016/j.enbuild.2025.116938)**	Monzon-Chavarrias M., Sierra-Perez J., Garcia-Perez S., Guillen-Lambea S.	Energy and Buildings , 367, 117701	2026	0

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Document title	Authors	Source	Year	Citations

The authors regret the omission of Appendix 1: Inventory of materials and processes from the published article. Appendice 1. Inventory of materials and processes Table A1 shows the inventory of materials and processes carried out in the rehabilitation of the case study, based on the actual execution project and discussions with the developers.PackageWork packageMeasurementMaterials and processDemolitionDemolition of tiled flooring with pneumatic hammer44 m^2Ceramic pavement 1110 kgmortar 2220 kgpneumatic hammer 2,2 kwDemolition of some partitions inside homes and to create the systems room100,51 m^2Ceramic Brick 4020 kgplaster coating 2412 kgmanual window removal156 m^2metallic frame 5080 kgmanual removal of drip edges93,87 mceramic 1877 kgdemolition of ceramic tiles with electric hammer26,11 m^2ceramic tiles 992 kgcement mortar 783 kgelectric hammer 2,2 kwdemolition of false ceilings26,11 m^2gymsum 939 kgCutting wooden doors18 udsTable saw 2 kwDisassembly and assembly of balcony railings27 udsCircular saw 1,2 kwPost-welding 4.3 kWFaçadeETICs695,7 m^2aluminum profile 18,9 kgPolymer mortar 6957 kgMineral wool, thickness 60 mm, density 150 kg/m^3 6574 kgMineral wool, thickness 60 mm, density 80 kg/m^3 3506 kgPolypropylene anchor plug 34,97 kgpotassium silicate primer 18,1 kgsilicate finishing coating 262 kgPlastic paint for facades373 m^2Acrylic primer 50,1 kgpaint 56 kghydrochloric acid solution (solucion acido clorhidrico) 0,0015 kgRepainting terrace railings161 m^2paint 32, 55 kgEnamel paint12,58 m^2Enamel paint 2,54 kgPrefabricated gutter4,11 mSheet metal 18,9 kgGalvanized steel screw 0,85 kgPVC downpipe57,9 mPVC 94,7 kgRoofBricks to build the new utility room32,64 m^2concrete brick 7703 kgcement mortar 1109 kgwater 196 kgContinuous mixer 2.2 kWFlooring for utility room24,2 m^2Ready-mix concrete 5844 kgQuartz aggregate topping mortar 72,6 kgFront-loading dumper 2 tons (1.69 kW)3 m vibrating screed (0.85 kW) Concrete mechanical trowel (7.5 hp)Cement grout 33,88 kgConcrete floor polisher, 1,200 WInsulation utility room floor24,2 m^2XPS 152 kgInsulation utility room ceiling40,3 m^3Blown mineral wool 847 kgcement mortar 24,2 kgInjection machinery, 3.6 kWTerrace waterproofing24,62 m^2Modified bitumen (APP) membrane 81,25 kgNon-woven geotextile made from polyester fibers 5,17 kgTerrace pavement24,62 m^2ceramic tiles 246 kgcement mortar 74 kgriver sand 73,86 kgContinuous mixer for mortar 30 MJ/hMetal walkway, galvanized profiles, tramex 30x30x24 mGalvanized steel 1370 kgconcrete 193,2 kgwaterproofing 8,4 kgRailingSolid steel 304 kgEnamel paint 2,95 kgMasonryinsulation floor and terrace174,36 m^2mineral wood 431 kgPolyethylene film 35,29 kgDrip edges52,85 martificial stone 1573 kgElastic adhesive 133 kgElastic single-component polyurethane putty 0,116 kgLaminated flooring for homes158,64 m^2laminated floor 297 kgHigh-density polyethylene Foam sheet 10,47 kgMDF baseboard 132,3 kgGalvanized steel nail 0,18 kgFiller wax 2,4 kgDoor frames111,78 m^2Plaster filler 671 kgRepairing holes in blinds103,96 m^2Spray polyurethane foam insulation 124 kgPlasterboard 524 kgFinishing of carpentry openings and painting133,77 m^2Plaster filler 1518primer 966paint 929,2Finishing of partitions and painting15 m^2Plaster filler 73,08 kgprimer 165,4 kgpaint 159 kgPaint2683,37 m^2Paint 542 kgVolume restoration17,7 mepoxy mortar formulation 176,55 kgFlexible nylon rods 0,0014 kgThixotropic epoxy adhesive 4673 kgDrilling machineLoading dock26 mSteel loading dock 326 kgMinium painting 70,2 kgFalse ceiling with plasterboard159,08 m^2Galvanized steel profile 1,34 kgSteel anchor and screw 0,99 kgGalvanized steel sheet 10,7 kgPlasterboard 1861 kgPolyethylene foam *acoustic* strip 127,26 kgEnamel paint on carpentry346,92Primer formulated with alkyd resins 40,43 kgEnamel 63,67 kgPulishing and polishing456,79 m^2Floor polisher 3 kW Polisher 2.2 kWTile tiling11,33 m^2Cementitious adhesive34 kgPVC corner trim 1,04 kgCeramic tile 204 kgCement grout 0,000011 kgPlaster coat49,83 m^2Cement mortarVinyl flooring13,4 m^2Acrylic resin adhesive 3,35 kgPVC tiles 87,1 kgstair flooring6,8 m^2artificial stone 183,6 kgadhesive 20,4 kgcement mortar 13,6 kgBaseboard7,3 martificial stone 9,26 kgadhesive 22,05 kgcement mortar 14,7 kgMicrograin terrazzo step17 martificial stone 183 kgcement mortar 0,17 kgTerrazzo repair1,62 m^2Epoxi primer 0,65 kgEpoxi mortar 20,25 kgAcrylic resin 1,13 kgDrip edges12,6 mAnodized aluminum sheet 54,34 kgWaterproofingRubber paint 0,87 kgPlaster ceilingGalvanized steel metal profile 12,84 kgPlasterboard 309,4 kgWooden interior door8 udsPine wood door frame 13,1 kgHollow melamine door 16,61 kgMDF casing 3,78 kgBrass hinge (60,76 kgBrass lock 3.6 kgWindowsAluminumDouble glazed 44/16argon/44.1BE128,4 m^2Units 86Aluminum Drip edges97,4 mAluminum 293 kgEPS 73 kgadhesive 9,25 kgCor-ten steel door3,53 m^2Cor-ten steel 135 kgFire door3,78 m^2Fire door 82,87 kgSystems: Heating and coolingHeat pump *outdoor* unit1 unitHeat pump indoor units8 units4,0 kW heating power and y 3,6 kW cooling power,refrigerant R410AControl system and wiring1Includes copper cables and a control unitPipingCopper and includes *thermal* insulationSystems: DHWHeat pumps for DHW2 unitshigh-efficiency circulating pump1 unitwith stainless steel casingExpansion vessel1 unit100 lPolypropylene pipessystems: PVPV system28 units28 PV panelsMounting systemTable A1. Inventory of the renovation. The authors would like to apologise for any inconvenience caused.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
12	**Dynamic turbulence modeling with experimental validation over a 13-km free-space optical link**	Ge H.,Song Y.,an Y., Wang T.,Dong K.	Optics Express, 34(18)	2026	0

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To evaluate atmospheric turbulence impacts on free-space optical (FSO) communications, this paper generates dynamic phase screens via the covariance matrix method (CMM) and the time-delay-induced autoregressive (TDI-AR) model to simulate the turbulence channel. Concurrently, a log-frequency weighted least squares (LF-WLS) inversion method is constructed to extract simulation parameters (transverse wind speed v and decorrelation time τ_{boil}) from temporal power spectra. Numerical simulations verify the spatiotemporal statistics of the generated phase screens and the accuracy of parameter inversion, after which the system reliability and channel fading characteristics are analyzed. Comparisons with a 13-km *outdoor* FSO experiment show good agreement in the statistical properties of the received power. For the 1.25-Gb/s link, the simulations employed a fixed equivalent SNR and captured the relative BER variation observed experimentally under different turbulence conditions. This research provides a theoretical and numerical basis for FSO channel modeling and performance evaluation under complex atmospheric conditions.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
13	**A Physical Environment Evaluation System for Outdoor Sports Spaces in Urban Residential Areas of a Severe Cold-Region City: A Case Study of Harbin, China**	Xia S.,Luo P.,Liu X.,Wang T.	Buildings, 16(18), 3762	2026	0

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Outdoor sports spaces in residential areas must provide safe and comfortable conditions for daily exercise, yet in severe cold regions their use is constrained by wind, solar exposure, *thermal* stress, *noise*, and air quality. Using six sites in Harbin, China, this study developed an Analytic Hierarchy Process (AHP)-based evaluation system covering the wind, light, *acoustic*, *thermal*, and air environments. Candidate indicators were screened using a user questionnaire, expert AHP judgments were used to derive weights, and site performance was quantified through field monitoring, site surveys, and ENVI-met simulation. The indicator weights, in descending order, were wet-bulb globe temperature (14.96%), pedestrian-level wind speed (13.67%), building layout conducive to air circulation (13.55%), pedestrian-level wind vortices and still-air zones (12.79%), sunlight exposure and lighting effectiveness (12.67%), shading rate of hard ground surfaces and structures (12.15%), reflectivity of pavement and covering materials (11.37%), and *outdoor noise* level (8.83%). Overall scores differed across the six sites. An AHP-weighted performance-shortfall analysis, calculated relative to the ideal Grade 5, identified *outdoor noise*, shading, building-layout-related air circulation, and sunlight/lighting performance as the four largest cross-site improvement needs. The resulting framework supports the diagnosis and renewal of *outdoor* sports spaces in cold-region residential areas. The empirical results represent the monitored summer and autumn transitional scenarios rather than year-round performance.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
14	**Hybrid Thermal-Compensated State Motion Detection Adaptive Kalman Filter for Error Suppression in Low-Cost MEMS IMU**	Zeng X.,Tong Q.,Gao F., ... Xing F.,You Z.	IEEE Sensors Journal, 26(18)	2026	0

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Low-cost microelectromechanical system (MEMS) inertial measurement units (IMUs) have gained widespread adoption across diverse application domains; however, their application in precision-demanding scenarios remains limited by stochastic errors and *thermal*-induced drift. To mitigate these limitations, this article first establishes a unified stochastic error model for accelerometers and gyroscopes and subsequently proposes a hybrid error suppression framework designated as the *thermal*-compensated state motion detection adaptive Kalman filter (TC-SMD AKF). The polynomial fitting-based temperature compensation eliminates drift-induced filtering misjudgment, while a direct gain-controlled AKF without *noise* matrix iteration is designed to dynamically adjust Kalman gain via SMD, balancing static *noise* reduction and dynamic tracking. Moreover, the integrated adaptive processing suits both gyroscopes and accelerometers without separate calibration. Taking the gyroscope of the IMU as an example, in the 24-h *outdoor* static test, the proposed method reduces the time-domain variance from 28.664◦/h to 2.297◦/h, optimizes the bias instability (BI) from 1.033◦/h to 0.493◦/h, and improves the angular random walk (ARW) from 0.032◦/

	Document title	Authors	Source	Year	Citations

√h to 0.017◦/ √h. Experimental results demonstrate the effectiveness of TC-SMD AKF in reducing the drift and random *noise* under static, dynamic, and long-term environmental conditions as compared with other existing algorithms.s.

Article • *Open access*

15 **Integration of BIM and Cloud-Based Tools for LEED Sustainable Building Design: A Case Study** — Chelaru B.,Ungureanu G.,Onutu C. — Buildings, 16(17), 3377 — 2026 — 0

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This research assesses the practical synthesis of Building Information Modeling (BIM), Autodesk Forma and Dalux to support LEED-oriented sustainable design for a higher education building. Autodesk Revit 2025 functioned as the central BIM platform, while Autodesk Forma enabled early-stage simulations of solar exposure, daylight potential, wind conditions, microclimate, *noise* and solar-energy potential. Dalux supported model coordination and information management in accordance with ISO 19650 principles. The workflow links simulation outputs to BIM elements through project-defined parameters, allowing performance evidence to inform design refinement. Quantitative indicators were consolidated for daylight exposure, wind comfort, *outdoor thermal* stress, *acoustic* exposure and photovoltaic potential. The solar-energy analysis considered an area of approximately 970 m^2, an annual potential of 1030 kWh/m^2 and a theoretical yield of approximately 999,100 kWh/year. Assuming 70% roof coverage and 18% panel efficiency, the estimated photovoltaic the expected output is approximately 125,113 kWh/year. These findings demonstrate the value of combining BIM, cloud-based analysis and CDE-based coordination for early-stage sustainable design, while LEED certification, operational energy modelling and lifecycle assessment require additional specialist validation. The proposed workflow provides a consistent approach for aligning design development with sustainability objectives and can be applied to similar building types.

Article • *Open access*

16 **Three-dimensional thermomechanical analysis and performance assessment of an aluminum-lined, glass-fiber overwrapped hydrogen tank under hot-climate conditions** — Tiam Kapen P. — Scientific African, 33, e03579 — 2026 — 0

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Hydrogen is increasingly central to decarbonizing transport and enabling long-duration energy storage, which places composite pressure vessels in harsh *outdoor* service. In sunbelt regions, tanks can face ambient temperatures approaching 50 °C together with intense solar irradiation, producing through-thickness *thermal* gradients that elevate liner temperature, modify composite response, and, when combined with internal pressure and cycling, reduce fatigue safety margins. However, many existing design and assessment studies assume near-isothermal behavior or consider *thermal* effects in isolation, limiting quantitative guidance for hot-climate deployment. This work presents a fully coupled three-dimensional *thermo*-mechanical finite-element analysis of a Type-III hydrogen tank (aluminum liner with glass-fiber/epoxy overwrap) using COMSOL Multiphysics 6.3. The model couples heat transfer in solids and gas with structural mechanics and imposes external convection-radiation, peak solar irradiance, and ambient temperatures up to 50 °C, together with internal hydrogen pressure and transient operating scenarios. Realistic solar exposure was achieved by computing the time-varying sun vector from the geographic coordinates of Maroua (Far North Cameroon) and applying it dynamically to the tank surfaces. Results showed pronounced vertical *thermal* stratification and crown-region heating, yielding an upper-dome *thermal* hotspot and circumferential stress bands peaking near 220 MPa (below liner yield), with predominantly radial displacements. Despite moderate stresses over most of the wall, fatigue damage localized near the lower outer shell/skirt junction, where geometric curvature and support constraints amplified the coupled *thermo*-mechanical stress range; the usage factor reached about 0.35 in this region while remaining below 0.1 elsewhere. Accordingly, the peak fatigue damage parameter increased from 0.291 under purely mechanical loading to approximately 0.35 ± 0.03 under coupled *thermo*-mechanical conditions. Importantly, this value represents the baseline environmental damage penalty per single 24-hour diurnal cycle rather than total cumulative lifetime damage, which will further accumulate under multi-cycle daily fast-refueling operations. To verify the framework's numerical robustness against operational uncertainties, model credibility was evaluated through a reproducible measurement-emulation benchmark: synthetic sensor time series for crown gas temperature, maximum von Mises stress, and maximum displacement were generated by sampling the simulated signals and adding sensor-like *noise* and occasional outliers. The reconstructed

	Document title	Authors	Source	Year	Citations

signals showed the lowest estimation errors (highest precision) for temperature and displacement, whereas peak-stress maxima exhibited higher dispersion, consistent with the sensitivity of extremum metrics to boundary-condition and property uncertainties. The study identifies the lower shell/skirt junction as the primary life-limiting region during hot-day cycling and motivates targeted mitigations such as local geometry smoothing, overwrap reinforcement, and *thermal* management to improve durability in hot climates.

Review

17 **Review of cross-modal and combined effects in real-world multi-domain outdoor environment on human responses** — Zhou H.,Wu Z. — Building and Environment , 302, 114881 — 2026 — 0

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Outdoor human comfort is a key indicator for evaluating urban environmental quality, public health, and sustainable development. While individual environmental domains are well-studied, human perception is powerfully shaped by the interaction of multi-domain stimuli (*thermal*, *acoustic*, visual, and air quality). However, existing reviews have mainly focused on single-domain *outdoor* comfort, while evidence on interactions among multiple *outdoor* environmental domains remains fragmented and inconsistent, highlighting the need for an integrated framework for systematic synthesis. This review therefore aims to synthesize empirical evidence on cross-modal and combined effects in real-world multi-domain *outdoor* environments, clarify their influence on human responses, and provide an interpretive basis for understanding inconsistent and context-dependent findings. Using the PRISMA framework, this review synthesized 49 peer-reviewed empirical studies. Results confirm the *thermal* (94%) and *acoustic* (76%) domains as primary cornerstones, with the *thermal*–visual–*acoustic* triad (n = 16) being most prevalent. Overall *outdoor* comfort is governed by dynamic, non-linear and context-dependent mechanisms: *thermal* stress often acts as a “one-vote veto” factor under extreme climatic conditions, whereas *acoustic* quality becomes more influential under non-extreme conditions. Visual quality can provide aesthetic compensation, particularly for psychological restoration, while air quality remains insufficiently integrated into multi-domain models. Future research should strengthen cross-study comparability and integrate subjective, physiological, and modelling evidence to support more generalizable *outdoor* comfort assessment. This review deepens the understanding of interaction patterns among different *outdoor* environmental domains and provides a conceptual foundation for *outdoor* environmental design and urban planning.

Article • *Open access*

18 **Mechanical, acoustic, and thermal assessment of mortar embedded with polycarbonate particles obtained from end-of-life roofing sheets** — Li T.,Proietti M.G.,Nobili A.,Moretti E. — Construction and Building Materials , 534, 146908 — 2026 — 0

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The study investigates natural hydraulic lime mortar specimens in which polycarbonate (PC) particles obtained from milling end-of-life roofing sheets are used as a partial replacement of natural sand. Compared to plastics such as PET, PS and PVC, PC offers exceptional toughness, excellent heat resistance and transparency, as well as good dimensional stability. Since PC is susceptible to weathering, two types of PC waste are considered, which differ by the environmental exposure conditions during the service of the roofing sheets from which they were milled: indoor and *outdoor*, respectively. PC waste replaces natural sand (NS) at two replacement levels, and specimens are assessed in terms of mechanical, *thermal*, and *acoustic* performance. To better highlight the role of PC waste, comparison with control is carried out at different levels of water reducer addition (PCE) so that the same workability is achieved for all admixtures. Besides, to rule out the role of grading, PC particles are replaced by NS that has been sieved to match the grading curve of the plastic particles. In contrast to the general finding associated with plastic waste embedment, it is found that PC addition offers significant mechanical enhancement, both in flexure and in compression, in terms of ultimate strength, which increases by 20% (30%) at low (high) dosage; in terms of dissipated energy, which multiplies by 3–4 times over control; and in terms of failure mode, which transitions from brittle to ductile. Such outcomes appear consistently and are little sensitive to the PC type and to the PCE amount. Indeed, they seem associated with the flake shape of PC particles as well as with the peculiarities of PC. Specimen density is also reduced, which entails an improvement in terms of *thermal* conductivity that decreased from 0.69 W/(m·K) for the control mortar to 0.48

	Document title	Authors	Source	Year	Citations

W/(m·K) at the highest PC content, corresponding to a reduction of about 30%. In contrast, *acoustic* absorption improved only for selected mixtures. Overall, results show promising potential for PC waste incorporation in mortar matrices, with significant benefits in the direction of waste reduction and reuse.

Article • *Open access*

19 **Differences in Multisensory Comfort Across Five Public Outdoor Functional Spaces: Evidence from an Autumn Field Study in Dalian, China**

Chen B.,Li X.,Deng S.,Li T., Zhang H.

Buildings, 16(16), 3340

2026

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Outdoor comfort is influenced by *thermal*, visual, and *acoustic* stimuli, whose relative roles may vary across functional spaces. This study compared multisensory comfort characteristics across five frequently used public *outdoor* functional spaces in the main built-up area of Dalian, China, during the autumn transition season and examined their associations with overall comfort. Rotational fixed-point monitoring and synchronous questionnaire surveys were conducted in residential, cultural, educational, commercial, and green spaces, yielding 1497 valid responses. The results showed that: (1) Among the five investigated public *outdoor* spaces, the residential space had the highest proportion of positive overall comfort votes, whereas the educational space showed more *thermal* discomfort and negative *acoustic* evaluations. (2) *Thermal* and *acoustic* sensations changed linearly with corresponding exposures, while visual comfort first increased and then decreased with illuminance. (3) *Thermal* comfort had the largest main effect on overall comfort, visual comfort was also significant, and *thermal–acoustic* interaction was observed. (4) SEM indicated distinct association pathways. The *thermal* pathway showed a significant serial indirect effect through *thermal* sensation and *thermal* comfort, while the visual pathway contained indirect effects in opposite directions and the *acoustic* pathway was characterized mainly by a direct negative association between LAeq and overall comfort. These findings support function-sensitive multisensory assessment and targeted *outdoor* environmental optimization.

Article • *Open access*

20 **Net heat-flux measurement method for passive radiative cooling materials with ambient self-tracking**

Bertiglia F.,Forte D., Lopardo G.,... Fasano M., Pattelli L.

Applied Thermal Engineering , 303, 132228

2026

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Passive Radiative Cooling (PRC) materials can efficiently dissipate *thermal* energy to outer space through the atmospheric window (8–14 μm), while reflecting most incident solar radiation, enabling sub-ambient cooling under direct sunlight. This work presents RAFT (Radiative Ambient-referenced Flux Testbed), a compact device for accurate *outdoor* PRC materials characterization. RAFT employs a heat flux meter to directly measure net heat flux, with the PRC sample mounted on the sensing surface and *thermally* coupled to an ambient-tracked reservoir via a compact fan-based convective heat exchanger integrated with heat pipes. *Outdoor* experiments demonstrated reliable ambient temperature tracking and accurate net heat flux measurements, which were further verified using an openly-released numerical model. Compared with conventional resistive heater-based *outdoor* setups, RAFT overcomes key limitations: it eliminates the need for feedback-control, shortens the measurement chain, reduces measurement *noise* when sample temperature approaches ambient, and enables quantification of both cooling and heating net fluxes with a total measurement uncertainty of ±1.8 W m^{-2}. Relative to liquid-flow-based systems, RAFT offers improved portability and avoids the complexities of fluid handling. The proposed setup provides a robust and practical framework for *outdoor* evaluation of PRC materials, addressing the current lack of standardized protocols for direct heat flux measurement.

Article

21 **Experimental performance analysis of a realistic FSO link in weak turbulent channels for 5G backhaul connectivity**

Saeed S.,Khan A.N., Younis U.

Journal of the Optical Society of America A Optics and Image Science and Vision , 43(8)

2026

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Document title	Authors	Source	Year	Citations

Free-space optical (FSO) communication links have emerged as a promising solution for future networks such as 5G, 6G, and beyond by enabling low-cost, power-efficient, and high-speed backhaul connectivity. However, atmospheric turbulence can degrade link performance by reducing the optical signal-to-*noise* ratio (OSNR), increasing the bit error rate (BER) and limiting the achievable data rate. In this work, we employ a weak-turbulence channel model based on measured temperature and pressure to estimate the refractive index structure parameter (C^2_n) and scintillation index (σ_I^2). These parameters are used to quantify turbulence strength and analyze its impact on optical beam propagation. The modeled OSNR and BER are analyzed for varying turbulence strengths. The measured OSNR of approximately 30 dB from the 50 m indoor channel corresponds to an achievable data rate of about 350 Mbps, yielding a BER of 10^{-8}, which is below the FEC threshold. The calculated OSNR is further validated using values derived from the measured received signal and *noise* optical powers. Finally, the (C^2_n) value obtained from the indoor measurements is used to estimate the *outdoor* transmission range that provides comparable performance. The analysis predicts an effective *outdoor* link range of approximately 800 m, demonstrating the potential of this approach for designing FSO backhaul links in beyond-5G/6G microcell networks.

Article

☐ 22 **Soundscape diversity governs thermo-physiological and psychological states in a sexually dimorphic manner** | Tang Z.,Huang W.,Zhang M.,Deng P.,Hong B. | Building and Environment, 301, 114742 | 2026 | 0

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Although soundscapes modulate *outdoor thermal* perception, sex differences in physiological and psychological mechanisms under *thermal* stress remain elusive. This study examined *thermo-acoustic* interactions across six *acoustic* conditions that varied in diversity (low/medium/high) and environmental context (nature-dominant versus anthropogenic-dominant) in male and female participants exposed to graded *thermal* stress. Physiological indicators (blood pressure, heart rate, and eardrum temperature), electroencephalographic (EEG) activity, and psychological metrics were continuously monitored alongside microclimatic parameters. Results demonstrate marked sexual dimorphism in *thermo-acoustic* perception, primarily within nature-dominant environments. Males exhibited cardiovascular attenuation under slight *thermal* stress, with systolic pressure declining 6.08 mmHg (medium diversity) and diastolic pressure 6.61 mmHg (high diversity) versus low diversity ($p < 0.05$), though physiological responsivity diminished as *thermal* stress escalated. Females conversely demonstrated intensified physiological reactions with increasing *thermal* load, including reduced heart rate under no *thermal* stress conditions in anthropogenic settings ($p < 0.05$). In nature-dominant *sound* environments, low soundscape diversity attenuated male δ power while potentiating α power, consistent with diminished somnolence and heightened relaxation; elevated female θ power, conversely, indexed enhanced positive affect. Psychologically, males demonstrated consistently higher sensitivity, while females exhibited a biphasic pattern of initial attenuation followed by enhancement. In anthropogenic-dominant environments, males displayed heightened physiological and psychological sensitivity to *acoustic* diversity, whereas females showed asynchronous patterns: physiological responses weakened then strengthened, while psychological responses progressively intensified with increasing *thermal* stress. Our findings inform sex-sensitive *acoustic* optimization frameworks for urban design, advancing evidence-based healthy built environment strategies.

Article

☐ 23 **Critical analysis of smart materials in adaptive transparent systems for building façade** | Jaffar H.,Javed N.,Riaz A.,... Zhou C.,Zhang J. | Energy and Environment, 37(5) | 2026 | 4

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Adaptive transparent system (ATS) is an aspiring and effective building system that can adapt itself through environmental changes in spontaneous and reversible ways. It proposes an automatic and real-time response to indoor and *outdoor* conditions by increasing building energy efficiency and user comfort. This research presents a critical analysis to provide a preference order of the smart materials being used in the adaptive transparent systems. The approach of using smart material technologies in the building façades is discussed as the promising research direction for the future. A comprehensive literature review is conducted to identify the smart materials used in adaptive transparent systems along with their properties. Seven types of smart materials that are aerogel (AG), phase

	Document title	Authors	Source	Year	Citations

change material (PCM), photovoltaic (PV), electrochromic (EC), *thermo*-chromic (TC), *thermo*-tropic (TT), and liquid crystal polymers (LCP) are selected based on evaluation criteria determined by the literature. The identified criteria are bulk density (BD), *thermal* conductivity (TCd), *sound* insolation (SI), fire retardation (FR), heat transfer coefficient (HTC), solar transmittance (ST), air purification (AP), annual energy saving (AES), cost saving (CS), ultraviolet (UV)/near infrared (NIR) control (UNC), and embodied carbon (EC). The identified seven smart materials are investigated with the help of two multi-criteria decision-making (MCDM) techniques named analytical hierarchical process (AHP) and technique for order preference by similarity to ideal solution (TOPSIS) for analysis and ranking of materials. Results of the analysis demonstrate that PV is the most optimum material for adaptive transparent systems while others are ranked according to their performance.

Article

24 **Experimental investigation of synthetic jet–based active cooling for photovoltaic modules: Thermal management, acoustic characterization, and field validation** — Ghaly H., Yakut K., Osta M.H. — Applied Thermal Engineering, 302, 132025 — 2026 — 1

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The electrical efficiency and long-term performance of photovoltaic (PV) modules are adversely affected by heat accumulation within solar cells. Synthetic jet (SJ) actuators have emerged as a promising active cooling technology for PV systems due to their low power consumption and operation without an external fluid supply. However, the influence of nozzle geometry and *outdoor* performance of synthetic jet cooling on PV modules remain insufficiently investigated. In this study, 57-orifice synthetic jet actuators with nozzle geometries of 3, 5, and 7 mm were experimentally evaluated for PV *thermal* management. The effects of excitation amplitude, frequency, and orifice diameter on *thermal* and electrical performance were investigated under 1000 W/m^2 solar irradiance. The optimal conditions were obtained at a 5 mm orifice diameter, 100 Hz frequency, and 3 V amplitude, resulting in a 16.5% reduction in average panel temperature and a 14.9% increase in maximum electrical power output. *Outdoor* experiments conducted in Erzurum demonstrated a reduction of up to 12 °C in surface temperature and a 7.5% increase in average electrical power output compared to the reference panel. Techno-economic analysis indicated a net annual energy gain of approximately 4.25%; however, the prototype-scale SJ system cost limited improvement in the levelized cost of electricity (LCOE). The obtained results are competitive with water-based active cooling systems, which typically achieve 6–22% power gains but require continuous water supply and maintenance. Overall, synthetic jet-based cooling provides a water-free, low-power, and scalable *thermal* management solution for photovoltaic systems under realistic operating conditions.

Article

25 **Compact High-Performance Sensing Platform for Environmental Trace-Gas Monitoring Using MEMS-Based Thermal IR Detectors** — Venditti A., Ayed F.B., Simeoni P., Qian Z., Rinaldi M. — IEEE Sensors Journal, 26(15) — 2026 — 0

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This article reports the demonstration of a compact high-performance sensing platform for atmospheric trace-gas detection based on laser absorption spectroscopy (LAS). The proposed system integrates an infrared emitter, a Herriott cell, a *thermal* infrared detector, and a low-*noise* readout circuit, providing narrowband selectivity, high sensitivity, fast *thermal* response, and a compact form factor. The detector combines a piezoelectric laterally vibrating resonator (LVR) with a spectrally selective metasurface, enabling a responsivity of 0.65 Hz/nW and a *thermal* time constant of 2.3 ms. Since the extremely responsive and selective detection relies on a plasmonic metasurface with a lithographically tunable absorption peak in the mid-infrared spectral range, the signature absorption bands of common trace gas molecules can be matched and their detection targeted by properly selecting the emitter wavelength. Among those, nitrous oxide is used for a system-level validation due to its increasingly concerning impact as a greenhouse gas. Using a laser with emission wavelength near 4.53μ m and an optical path length approaching 11 m, a sensitivity of 70.9 Hz/ppm is extracted. This corresponds to a limit of detection (LoD) of 14 ppb, considering the minimum Allan deviation of 1 Hz measured from the implemented readout circuit at an averaging time of 11 ms. The sensing platform is packed in a 3-D-printed housing with a total volume of 2 L and a weight of 800 g, suitable as a payload for integration with commercial drones. Preliminary robustness analyses and experimental

	Document title	Authors	Source	Year	Citations

validation under vibration conditions further support the feasibility of the proposed system for *outdoor* drone-based operation.

Article

26 **Thermally stable GaN RF HEMTs on proton irradiated GaN substrates for low-noise applications**

Fan K.,Xu C.,Han J., ... Mao J.,Liu X.

Applied Physics Letters , 129(4), 043502

2026

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In this study, proton irradiation is employed to fabricate semi-insulating GaN free-standing wafers with a sheet resistance of 5.9×10^7 Ω/sq. The mechanism for achieving semi-insulating GaN via proton irradiation is associated with the formation of a deep-level trap (E_C-0.54 eV), which is tentatively attributed to nitrogen antisite-related defects (N_{Ga}). GaN-on-GaN high electron mobility transistors fabricated on the irradiated substrates exhibit stable operation over 300–375 K, with a minimum *noise* figure as low as 0.25 dB, with suppressed low-frequency *noise* characteristics. Additionally, the proton irradiation treatment GaN substrate also exhibits high volume production potential. This device is ideal for installation in harsh operational environments, such as satellite communication systems or *outdoor* base stations.

Article

27 **Phase Change Microcapsules Loaded in Polyurethane Foam for Dual-Purpose Thermal and Acoustic Insulation**

Zhu H.,Tan J.,Qiang X., Yin Y.,Wang C.

ACS Applied Polymer Materials , 8(14)

2026

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Growing demand for lightweight and energy-efficient materials in building envelope systems has promoted the advancement of high-performance *thermal* insulation foams. Polyurethane foams (PUFs) are extensively adopted as porous insulation fillers owing to their low density, structural continuity, and suitability for scale construction deployment. However, insufficient skeleton strength and nonuniform pore size distribution limit the *thermal* insulation performance. In this study, n-tetradecane microcapsules were incorporated into a polyurethane raw material system, and phase change polyurethane composite foams (PCM-PUFs) were fabricated via a one-step foaming process. Microstructural characterization reveals that the composite foams exhibit a uniform closed-cell morphology, with the n-tetradecane microcapsules well dispersed and firmly embedded within the polyurethane cell walls; the average cell size is approximately 150 μm. At an optimal microcapsule loading of 15 wt % and a corresponding encapsulation efficiency of 89.7%, the resulting PCM_{15}-PUF foam demonstrates substantially enhanced resilience and *thermal* buffering capacity. The support factor reaches 5.45, indicating excellent load-bearing capability and comfortability. The *thermal* conductivity is reduced to 0.05 W/(m·K). Under *outdoor* solar exposure, the surface temperature of PCM_{15}-PUF is 6.2 °C lower than that of neat PUF. Furthermore, PCM_{15}-PUF achieves an average *sound* absorption coefficient (AAC) above 0.9 and a *noise* reduction coefficient (NRC) exceeding 0.7, confirming that the incorporation of microcapsules does not compromise the *sound* insulation performance. The developed composite foam holds strong promise for building applications, particularly for regulating temperature fluctuations in scenarios with significant indoor–*outdoor* temperature differences.

Article

28 **Do we all feel the city the same way? A proof-of-concept framework linking physiological wellbeing to neighborhood-scale resilience through multi-domain environmental sensing**

Grapas C.,Schweiker M., Pisello A.L.,Pigliautile I., Cureau R.J.

Building and Environment , 300, 114679

2026

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Outdoor comfort and climate adaptation studies usually focus on *thermal*-only investigations, typically relying on straightforward heat stress metrics. Human-centered validation frameworks to confirm whether built environment interventions reduce actual physiological stress versus predicted discomfort from population-averaged models still lack. To fill this gap, this proof-of-concept addresses three questions: (1) Do multi-domain environmental models explain physiological variance that *thermal* indices alone cannot capture? (2) Do individuals exhibit heterogeneous resilience to identical landscapes? (3) How can individual physiological data aggregate into neighborhood-scale

Document title	Authors	Source	Year	Citations

resilience zones? Using synchronized wearable sensors (heart rate, electrodermal activity) and multi-domain environmental monitoring (Universal *Thermal* Climate Index, Particulate Matter, CO_2, *Acoustic* Spectrum), we tested eight participants through contrasting Roman urban typologies such as exposed piazzas, tree-canopied riverbanks, enclosed gardens. *Thermal* comfort indices alone failed to predict stress, yet multi-domain models integrating *thermal*, air quality, and *acoustic* parameters explained up to 43% of physiological variance. Individuals with similar characteristics (age 28–38) and lifestyle exhibited markedly different physiological responses to identical microclimates, suggesting that the city is not felt the same way by everyone, and that population-averaged vulnerability models may not reliably predict individual-level stress outcomes. We operationalized three resilience metrics: Toughness, Recovery Capacity, Adaptation Duration, derived through generalized logistic growth modeling capturing nonlinear threshold dynamics in physiological stress responses. K-means clustering identified five statistically distinct adaptation zones corresponding to landscape typologies, demonstrating that individual wearable data enables neighborhood-scale resilience mapping. This framework establishes a preliminary methodological infrastructure for integrating physiological sensing with urban digital twins, enabling validation of whether landscape configurations reduce stress for inhabitants rather than relying on demographic prediction alone.

Article

29 **Research on Dual-Comb Ranging Technology Aiming at Space Applications (Invited) | 面向空间应用的双光频梳测距技术研究(特 邀)** Qin P., Wang S. Zhongguo Jiguang Chinese Journal of Lasers, 53(14), 1404022 2026 0

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Objective Space-based high-resolution optical and radio imaging plays an irreplaceable role in geographic information resource acquisition, environmental monitoring, terrestrial planet searches and early-universe observations. In order to overcome the size limitations of single-aperture antennas, deployable antennas and distributed synthetic aperture imaging have become an emerging trend, which are all on account of high-precision absolute distance measurement. Optical frequency combs (OFCs) inherently generate equally spaced ultrashort pulse trains, corresponding to equally spaced phase-coherent spectral lines in the frequency domain. Such unique output signal enables not only high-resolution time-of-flight information but also high-accuracy interferometric data, which is ideal to reconcile the trade-off between dynamic range and precision in conventional ranging methods. Since the very beginning of the 21st century, high-precision distance measurements utilizing OFCs have been fully demonstrated with various distance calculation methods, comb structures, and ranging scenarios. On the other hand, space flights of simple OFCs have been reported by Korean and German groups, which illustrates the potential for space applications. However, complete space-borne comb ranging systems have not been reported until now. This research carries out systematic work from theoretical design to engineering optimization, and eventually accomplish an environmentally-stabilized fiber comb ranging system, which would lay the foundation of future space missions. Methods Firstly, a numerical model based on the split-step Fourier algorithm was established to systematically calculate the influence of the intracavity dispersion distribution on the timing jitter of the OFC oscillator. Concurrently, an all-polarization-maintaining fiber-based figure-9 cavity comb oscillator and a timing jitter measurement system based on optical cross-correlation were constructed, and an optical comb oscillator with low timing jitter was developed. Secondly, by utilizing a specially designed mechanical structure and adhesive bonding process instead of traditional mechanical mounts, a compact fiber-based OFC engineering prototype was developed. The prototype underwent environmental tests including vibration, acceleration, shock, *thermal* cycling, and *thermal* vacuum to verify its environmental stability. Thirdly, a fiber dual-OFC ranging engineering prototype was developed based on the OFC prototype. The accuracy and precision of its measurement results were experimentally verified with the indoor standard facility and the *outdoor* standard baseline at the National Institute of Metrology. Results and Discussions Firstly, through systematic numerical simulations, we found that to reduce the sensitivity of the timing jitter level to the uncertainty of intracavity net dispersion, it is necessary to achieve low timing jitter over a relatively wide range of net dispersion. This requires controlling the intracavity net dispersion to a near-zero, slightly negative value. Guided by the numerical simulation conclusions and combined with precise timing jitter measurements based on optical cross-correlation, we experimentally realized an all-polarization-maintaining figure-9 cavity OFC oscillator with high-frequency timing jitter of 93 as. Secondly, by employing structural and process designs, we developed an engineering prototype of the fiber-based OFC. After various environmental tests, the central wavelength and bandwidth of the oscillator's output spectrum remained essentially unchanged, the signal-to-*noise* ratio of the radio

	Document title	Authors	Source	Year	Citations

frequency spectrum consistently remained above 75 dB, and the amplifier output power and spectral shape were also maintained. These achievements ensure the stability of the OFC output signal from both theoretical and engineering perspectives. Thirdly, the dual-OFC ranging prototype underwent systematic experimental verification with standard distance facilities. The effective measurement range covered 0.99 to 432 m. The difference between the measurement results and the standard distance remained within the uncertainty of the standard distance throughout the entire range. The measurement accuracy reached the micrometer level, with the Allan deviation at one-second averaging time reaching 42.7 nm at 72.82 m and 63.8 nm at 432 m. Conclusions This research is based on the fundamental principles of OFCs, and covers a complete development process including numerical simulation, experimental verification, engineering prototype development, environmental test, and performance verification. It has preliminarily achieved the integration of theoretical design and engineering optimization, laying a solid foundation for the development and deployment of fiber-based OFCs and their high-precision ranging systems as space-borne payloads.

Conference Paper • *Open access*

30 **Quantum Reinforcement Learning for Air Conditioning Control Based on Variational Quantum Circuits** — Gandhi H., Deng Z. — Proceedings of the 16th ACM International Green and Sustainable Computing Conference Igsc 2026 — 2026 — 0

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We present a novel approach integrating quantum neural networks with reinforcement learning (RL) for house air conditioning control. Our framework employs a quantum deep Q-network (QDQN) RL implemented with PennyLane, leveraging parameterized variational quantum circuits (VQC) as function approximators for Q-values. The agent interacts with simulated indoor *thermodynamic* modelling, incorporating *thermal* capacity, heat transfer, and *outdoor* temperature variations. The action space consists of discrete air conditioning power levels, and the reward function balances comfort, energy, and switching penalties. To stabilize training, we introduce a target network, experience replay, and epsilon-greedy exploration. QDQN encodes the state vector into quantum rotations and applies multi-layer entangling gates, providing a potentially richer representation of the environment compared to classical networks. The QDQN agent uses VQC architecture. It incorporates data re-uploading, parameterized rotations, and strongly entangled qubits to effectively capture inter-feature correlations. We demonstrate the algorithm on episodes with *outdoor* conditions in both winter and summer conditions, showing convergence to policies that maintain indoor temperatures within setpoint bounds with zero violation rate. To validate the framework in a real-world *noise* setting, the quantum pipeline is simulated on the IBM Marrakesh system, featuring a 156-qubit Heron r2 processor with a median two-qubit (CZ) error of approximately 2.59×10^{-3}. Our results highlight the promise of quantum RL for real-world controls.

Article • *Open access*

31 **Possibility of using immersion cooling technology with virgin coconut oil for split-type AC** — Wibolo A., Temaja I.W., Widiantara I.B.G., Arsana M.E., Yusuf M. — Sinergi Indonesia, 30(2) — 2026 — 0

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Conventional split-type air conditioners are limited by high electrical consumption, *noise* from cooling fans, and bulky condenser coils that restrict installation flexibility. However, the application of immersion cooling in household AC systems remains largely unexplored. This study investigates the feasibility of using Virgin Coconut Oil (VCO) as a dielectric fluid for immersion cooling to improve energy efficiency and *thermal* performance. An experimental setup was developed using 20 L of SNI-compliant VCO, with the *outdoor* unit modified for immersion operation. Key performance parameters, including condenser coil and evaporator coil temperatures and the Coefficient of Performance (COP), were measured. Results showed condenser coil temperatures ranging from 40 °C to 58 °C and evaporator coil temperatures between 6 °C and 11 °C. The system achieved an average COP of 4.65 and reduced electrical consumption by approximately 20 %, indicating significant energy savings without compromising cooling efficiency. VCO also demonstrated suitable dielectric properties, with a breakdown voltage of 29.17 kV, near the IEC

	Document title	Authors	Source	Year	Citations

156 threshold. These findings highlight immersion cooling with VCO as a promising, quiet, and energy-efficient alternative for household air conditioners. Further work is recommended to assess long-term reliability under varying operational conditions.

Article

32 **Interactive effects of noise and odor on construction workers' environmental perception and cognitive performance amid outdoor thermal stress**

Li Z.,Zhang X.,Peng Z., ... Li A.,Hong B.

Building and Environment , 297, 114576

2026

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Construction workers chronically exposed to dynamic *outdoor* environments constitute a critical cohort for investigating multisensory environmental stress responses and cognitive ergonomics. This study systematically deconstructs the interactive effects of *thermal* stress gradients and *noise*–olfactory multimodal stimuli on workers' subjective *thermal* perception and neurocognitive performance in a cold region of China. Using a controlled field experiment with synchronized micrometeorological monitoring, we simultaneously acquired subjective environmental evaluations, multidimensional cognitive performance metrics, and electroencephalogram (EEG) data from 72 construction workers under nine factorial combinations of *sound* pressure levels (50/70/85 dBA) and odor types (Mentha piperita, Citrus sinensis, and Rosmarinus officinalis). A multidimensional benefit assessment model quantitatively ranked intervention efficacy. The results indicate that: (1) *noise* significantly modulated *thermal* sensation votes ($p < 0.05$), while *thermal* stress drove auditory sensation votes ($p < 0.01$). Neurophysiologically, escalating *thermal* stress enhanced low frequency (δ) activity and suppressed high-frequency (α, β) oscillations, with *noise* further diminishing frontal α-wave relative power ($p < 0.05$). (2) C. sinensis elevated significantly frontal-occipital θ-wave power ($p < 0.05$). (3) M. piperita broadly potentiated β-band activity under “neutral” conditions, yet R. officinalis yielded significantly lower *thermal* comfort votes than C. sinensis under “warm” conditions ($p = 0.02$). Our findings offer actionable climate adaptation frameworks for occupational health, safeguarding vulnerable workers against intensifying extremes in *outdoor* working environments.

Article • *Open access*

33 **Auditory comfort and noise perception under thermal and visual conditions in urban parks**

Simsek O.

Scientific Review Engineering and Environmental Sciences , 35(2)

2026

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This study investigates auditory comfort in *outdoor* urban environments by examining the combined influence of visual and *thermal* factors. Field research was conducted in two major urban parks in Adana, Turkey, representing the Mediterranean climate, where simultaneous environmental measurements and surveys were performed during the autumn season. The collected data included air temperature, relative humidity, wind speed, mean radiant temperature, *sound* pressure level, and both horizontal and vertical illuminance. Correlation and regression analyses revealed that *sound* pressure level was the most influential parameter, showing a strong negative association with auditory comfort, while higher vertical illuminance slightly decreased comfort levels. *Thermal* variables exhibited indirect effects on *noise* perception rather than on comfort itself. Participants expressed a strong preference for natural *sounds* such as birdsong and flowing water, while human and mechanical *sounds* were generally disliked. The findings highlight that auditory comfort is a multisensory phenomenon shaped by both *acoustic* and contextual environmental qualities.

Article • *Open access*

34 **Theoretical and experimental study of a novel thermoacoustic engine for sound source application**

Zhang L.,Kang H.

Thermal Science and Engineering Progress , 74, 104699

2026

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Document title	Authors	Source	Year	Citations

This paper introduces a novel structures of standing-wave *thermoacoustic* engines designed for *outdoor sound* sources. The system is characterized by operational stability, enhanced SPL, and space-saving design. Compared to the conventional open-end system, the thin-plate *sound* source proposed is a closed-end one. The plate isolates the working gas from the external environment but remains *acoustically* transparent, enabling the use of alternative gases. Besides, it can effectively adjust the *acoustic* field distribution, enhance the *acoustic* power output. Experimental and theoretical research were conducted, the performance of systems using air or helium is analyzed and compared. The influence of thin plate properties on system behavior is investigated. Results show that with air as working gas, the system can achieve a higher SPL of 113.02 dB at 1 m away, marking a 5.4% increase in SPL compared to open-ended system. With helium adopted as the operational gas, the system can reach a highest SPL(1 m) of 123.49 dB with the condition of 0.3 mm SP and T_h = 675 K, indicating a 9.3% increase in SPL compared to system with air as working gas. The performance is significantly influenced by the thickness and Young's modulus. This system has potential for application as an *outdoor sound* source.

Article • *Open access*

35 **Quantifying outdoor thermal comfort in different street vegetation arrangements: Insights from a study in tropical and subtropical Australian cities**

Fu J.,Tavares S.,King D., Naylor S.

Urban Climate, 67, 102967

2026

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Street vegetation serves as a nature-based strategy for urban heat mitigation leading to improved *outdoor thermal* comfort (OTC). A *sound* understanding of spatial and temporal OTC variation in different street vegetation arrangements is essential to strategically develop urban heat mitigation planning and design. This study introduces a methodology framework that quantifies the *thermal* improvement in different street vegetation arrangements by using three indices - cooling intensity (CI), cooling area (CA) and cooling duration (CD). The framework is applied to four streets in tropical and subtropical Australian cities to quantify Physiological Equivalent Temperature (PET) over a 24-h period. Results indicate that the street with trees on the west-facing side experienced a longer duration of street *thermal* enhancement (up to 38% CD improvement). During daytime hours, the OTC improvement in the planting scenarios with additional trees in the street centre (median strip) incorporating multiple vertical layers of vegetation is more pronounced in the studied subtropical urban area (up to 2.02 °C CI in Ipswich) than that in the studied tropical urban area (up to 1.03 °C CI in Townsville). These findings can contribute to urban heat mitigation planning and design in tropical and subtropical cities worldwide. The methodology framework proposed in this study provides a foundation for quantifying spatial and temporal OTC variation in different street vegetation arrangements, and establishes an adaptive framework for landscape architects, urban designers and planners, and decision-makers worldwide to evaluate urban street *thermal* environments.

Article

36 **Gas leak region recognition method constrained by gray-level statistical characteristics | 灰度统计特性约束的气体泄漏区域识别方法**

Yang L.,Chen C.,Zhu T., ... Xiong W.,Lu J.

Infrared and Laser Engineering , 55(5), 20250566

2026

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Infrared *thermography* plays an important role in gas leakage detection, and efficient detection algorithms are essential for enhancing practical applicability. This paper proposes a gas identification algorithm based on gray-level statistical features, which identifies anomalous regions through background modeling and chi-square statistics, and further confirms gas leakage regions using optical flow and morphological constraints. The results show that the false alarm rate is only 2.2% under no-leak conditions, with a processing time of 0.016 s per frame. For gas flow rates of 1 L/min, 3 L/min, and 5 L/min, the identification rates are 87.5%, 86.11%, and 99.07%, respectively, with an average processing time of 0.14 s per frame. The overall detection performance is satisfactory, and false alarms are effectively suppressed. The proposed algorithm does not require large-scale training datasets or precise environmental parameter modeling, while maintaining good generalizability, real-time performance, and ease of engineering deployment, offering an efficient and practical solution for gas leakage detection. Objective Accurate detection of gas leaks is essential for maintaining safety and protecting the environment in industrial and environmental monitoring

Document title	Authors	Source	Year	Citations

contexts. This study presents a robust and real-time method for gas leak detection utilizing infrared *thermography*. The proposed approach operates independently of large-scale labeled datasets and does not require physical models based on radiative properties, offering a generalizable and practical solution suitable for a wide range of monitoring scenarios. Methods A statistical and dynamic analysis approach is proposed. First, a background model is constructed from infrared video frames captured without gas release to obtain reference gray-level distributions. The overall gray-level deviation is quantified using the chi-square statistic to identify potential abnormal frames (Fig.3). The area of gas leakage is identified through pixel-level analysis using localized chi-square statistics computed over 5x5 neighborhoods. To reduce false positives, optical flow analysis is applied to verify the dynamic behavior of detected pixels, ensuring that only moving gas clouds are considered. In addition, constraints based on connected regions, including minimum area, shape compactness, and aspect ratio, are used to remove *noise* and pseudo-targets. Results and discussions Experiments were conducted in an *outdoor* simulated sulfur hexafluoride gas leak scenario at flow rates of 0, 1, 3, and 5 L/min. The results show that the proposed method achieves high detection accuracy and efficiency (Fig.4). The false-positive rate in the absence of gas is reduced to 2.2%, compared with 11.37% when using only global chi-square statistics (Tab.1 and Tab.3). For flow rates of 1, 3, and 5 L/min, the detection rates for identifying gas leakage areas are 87.50%, 86.11%, and 99.07%, respectively (Tab.3). Higher flow rates lead to improved detection because of stronger absorption and larger coverage areas. The average processing time per frame ranges from 0.01 s for non-leak frames to approximately 0.14 s for leak frames, demonstrating good real-time performance (Tab.3). The method reliably distinguishes gas leak regions from complex backgrounds and shows advantages in simplicity and generalizability compared with traditional statistical or physically modeled approaches. Limitations include occasional false positives caused by system *noise*, frame jitter, and dynamic environmental interference, as well as dependence on the quality of background modeling data. Conclusions The proposed method combines gray-level statistical analysis, dynamic verification, and region-based constraints to detect gas leaks in infrared *thermal* images. It achieves high detection accuracy, low false-positive rates, and real-time performance without requiring large training datasets or environmental parameter collection. Future work may incorporate additional image features, optimize optical flow parameters, and enhance background modeling to further improve robustness and reduce misclassifications in complex or dynamic environments. The method provides a practical and deployable solution for industrial and environmental gas leak monitoring.

Article

37 **Uncooled infrared spaceborne imaging method based on stop-and-stare scanning | 基于摆停式扫描的非制冷红外航天成像方法** Shen J.,Zhao J.,Wu W., ... Zhang Y.,Qi H. Infrared and Laser Engineering , 55(5), 20250529 2026 0

Hide abstract Find It @ UCL View at Publisher Related documents

Objective Uncooled infrared focal plane arrays (UFPA) have attracted extensive attention for small satellite *thermal* imaging missions because of their advantages of low cost, low power consumption, and the ability to operate without deep cryogenic cooling. However, their inherently limited *thermal* response speed and residual image effects under motion prevent UFPA from meeting the performance expectations of high spatial resolution spaceborne imaging systems. The objective of this research is to establish a systematic imaging method that reconciles ultra-wide swath coverage and image sharpness for UFPA-based payloads. This is achieved by extending the effective integration time of the detector during wide-area scanning, mitigating torque disturbances at the payload level, and improving geometric consistency for accurate multi-strip mosaicking and reliable detection of small-scale *thermal* targets such as wildfires. Methods A lightweight stop-and-stare scanning architecture is proposed, in which the scanning mirror alternates between rapid transition and short-duration dwell exposure phases, referred to as “fast transit” and “dwell exposure.” This segmented control of line-of-sight angular velocity allows the focal plane to acquire near-static exposure conditions within each dwell, effectively increasing the integration duration without enlarging the detector. To suppress nonlinear disturbance torques generated during the mirror motion, a payload-level angular momentum compensation system is designed using a counter-rotating flywheel driven by a stepper motor. The system operates in strict temporal synchronization with the scanning mirror, and an adjustable compensation coefficient ensures real-time balance between the two opposite rotational motions. For geometric accuracy, an analytic coordinate transformation framework based on the relationship between the satellite, the Earth, and the imaging plane is established to correct the geometric deformation of non-nadir remote sensing imagery and ensure consistency during strip mosaicking. The method was verified through multiple experiments, including dynamic MTF measurements using a gimbal–collimator–slit setup, single-axis air-bearing torque compensation tests, geometric

Document title	Authors	Source	Year	Citations

correction using a blackbody calibration target, and *outdoor* wide-swath imaging and fire-point detection campaigns. Results and Discussions Experimental results demonstrate the efficiency and robustness of the proposed method. The dynamic modulation transfer function (MTF) remains stably above 0.23, maintaining image contrast comparable to that of static imaging. The angular momentum compensation system reduces residual angular momentum from the order of 10^{-2} N·m·s to below 10^{-3} N·m·s, significantly lowering the disturbance transmitted to the satellite attitude control system. Geometric correction results show spatial deviations confined within 0.07 pixels, indicating accurate restoration of image scale and orthogonality. Field experiments confirm seamless image mosaicking across swaths exceeding 1000 km, while artificial 10-meter-class *thermal* fire points are clearly detected with a signal-to-*noise* ratio of 5.2 and sharp, point-like image features without visible trailing. These outcomes confirm that the proposed stop-and-stare scheme effectively trades scanning time for image sharpness, while the compensation mechanism ensures imaging stability. Together, these improvements provide a unified optical–mechanical–control solution for enhancing the clarity and consistency of wide-swath *thermal* infrared imaging using uncooled detectors. Conclusions The developed stop-and-stare scanning strategy, combined with an active payload-level angular momentum compensation device and an analytic geometric correction framework, effectively overcomes the conventional trade-off between swath width and image sharpness in UFPA-based spaceborne imaging systems. Quantitative experiments verify that dynamic MTF is sustained at or above 0.23, residual angular momentum is suppressed to the 10^{-3} N·m·s range, and geometric accuracy reaches sub-pixel level. Moreover, 10-meter *thermal* sources can be detected with high signal-to-*noise* ratio and geometric fidelity, validating the feasibility of the proposed approach. This study provides a practical, lightweight, and cost-effective solution for small-satellite *thermal* infrared missions and offers a new technical path for wide-swath, high-resolution imaging under dynamic conditions. Beyond its immediate engineering value, the work contributes theoretical and methodological support for future constellation-based fire monitoring, atmospheric *thermal* research, and intelligent spaceborne infrared observation systems, promoting long-term advancement in space infrared imaging technology.

Article

38 **Multi-scenario assessment of living-wall effects on building air purification, noise attenuation, and thermal insulation using quantum machine learning**

Zhang H., Srinivasan R., Chen C., ... Ganesan V., Chen M.

Building and Environment, 296, 114492

2026

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Despite a surge in the deployment of building-integrated Living Wall Systems (LWS) to foster urban sustainability, the complexity of field measurements has led to disjointed and often insufficient quantification of contributions to air purification, *noise* abatement, and *thermal* stability. This study combines multi-scenario ecological assessment with a dedicated Quantum Machine Learning (QML) model to precisely extract features for complex environmental mitigation. Laboratory-scale LWS unit boxes with and without artificial pollution sources, and an in-situ test on an occupied LWS-equipped building were selected. An IoT-based monitoring system was developed to collect internal and external $PM_{2.5}$, PM_{10}, SO_2, NO_2, O_3, eTVOC, temperature, humidity, and *noise* with 1-minute intervals. A QML approach was developed, and accuracy was evaluated using R^2, RMSE, and ROC curves, compared with conventional random forest (RF) and artificial neural network (ANN) based models. A cross-scenario contribution analysis decomposed each indoor environmental variable into *outdoor* drivers to quantify the relative mitigation attributable to LWS. Under polluted conditions, SO_2 and NO_2 removal exceeded 60%, PM_{10} interception was about 40%, aggregated average *noise* attenuation in the building reached 25.4 dB, and sun-exposed façades exhibited a consistent cooling effect. The QML model outperformed RF and ANN, with R^2 ranging from 0.82 to 0.99, compared with 0.71 to 0.96 for RF; 0.35 to 0.90 for ANNs, and lower RMSE. This study provides quantitative evidence for the use of LWS in green buildings and introduces a quantum approach that redefines optimization methods for comparable indoor environments.

Conference Paper

39 **Thermal Drift Suppression Algorithm for Infrared Thermal Imaging Gas Detection Systems**

Wang X., Li S., Xu Z., Kang C., Xu J.

Proceedings of SPIE the International Society for Optical Engineering, 14177, 141778E

2026

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	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

Owing to their advantages of low cost, compact structure, and no requirement for cryogenic cooling, uncooled infrared *thermal* imagers have been extensively employed in fields such as infrared gas detection, industrial monitoring, and environmental remote sensing. However, their output is highly prone to *thermal* drift, leading to a significant degradation in stability under complex *outdoor* operating conditions. To address this issue, a *thermal* drift correction method for temperature measurement based on a grayscale response model is proposed in this paper. Derived from the physical response characteristics of infrared detector pixel units, the method establishes a three-stage coupled model linking grayscale, radiance, and temperature. Experimental results demonstrate that the proposed correction model can effectively suppress *thermal* drift-induced errors, with an average coefficient of determination R2 reaching 0.99996 and a mean square error (MSE) of approximately 5.85. These findings provide a robust technical foundation for the deployment of uncooled infrared *thermal* imagers in high signal-to-*noise* ratio (SNR) application scenarios, such as hazardous gas monitoring and industrial safety inspection.

Article • *Open access*

☐ 40 **Short-wave atomic sensing enhancement technology based on miniaturized multi-layer square spiral resonator | 基于小型化多层方形螺旋谐振器的短波原子传感增强技术**

Lin Y.,Zhou A.,Mao R., ... Yang K.,Fu Y.

Wuli Xuebao Acta Physica Sinica , 75(9), 090602

2026 0

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To address the long-standing bottlenecks of traditional short-wave sensing technologies, including the insurmountable sensitivity limit caused by Johnson-Nyquist *thermal noise* and the excessively bulky antenna size restricted by the Chu limit, as well as to further enhance the measurement sensitivity of existing Rydberg atom sensing systems, this study focuses on developing a miniaturized multi-layer square spiral resonator for short-wave atomic sensing enhancement. The proposed resonator is designed based on the traditional metamaterial split-ring resonator (SRR) structure and integrates the miniaturization principle of ceramic laminated wound inductors. It is composed of stacked multi-layer square spirals connected to parallel metal plates via epitaxial metal arms, which not only achieves device miniaturization but also enables responsive capability to multi-polarized waves, with the optimal enhancement effect on right-hand circularly polarized (RCP) waves. Systematic multi-parameter coupling simulations are conducted to clarify the influences of key structural parameters—such as square spiral side length, cross-sectional area, number of spiral layers, resonator height, and connecting arm distance—on the resonant frequency and electric field enhancement performance, thus forming a complete optimization design method. For experimental validation, a physical prototype of the resonator is fabricated using brass, with nylon dielectric pillars added for structural stability and tuning screws integrated to compensate for frequency deviations caused by manufacturing errors. Experimental validation uses two-photon transition to prepare the $72D_{5/2}$ Rydberg state of ^{133}Cs atoms, combined with electromagnetically induced transparency spectroscopy and heterodyne measurements. Results demonstrate that at 15.54 MHz, the resonator achieves an electric field enhancement factor of 3981 (corresponding to 72 dB), boosting the system's sensitivity from 10.39 $\mu V \cdot cm^{-1} \cdot Hz^{-1/2}$ to 2.60 $nV \cdot cm^{-1} \cdot Hz^{-1/2}$. *Outdoor* tests successfully receive the 13.85 MHz signal from China Radio International via the heterodyne method, with the demodulated signal producing clear audio output without preprocessing. This miniaturized resonator enhances atom-electric field coupling, breaking through the limitations of traditional short-wave sensing and providing technical support for short-wave communication and direction-finding applications. Future work will optimize the structure to balance sensitivity and bandwidth.

Article • *Open access*

☐ 41 **Multi-sensory campus: The impact of thermal-visual-acoustic interactions on emotional perception in campus environments**

Chen B.,Qu J.,Guo P.,Lu X., Zhang H.

Building and Environment , 295, 114444

2026 4

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Multi-sensory interactions represent a key pathway through which environmental interventions shape emotional states and perceptual experiences, and may play a regulatory role in supporting students' physical and mental well-being in university campus settings. This study proposes a "*thermal*-visual-*acoustic*" multi-sensory interaction analysis framework. Walking experiments were conducted during transitional seasons across five typical *outdoor*

Document title	Authors	Source	Year	Citations

space types on a university campus in Dalian, China, integrating mobile meteorological monitoring with subjective questionnaires. The total survey route covered 5.6 km and included 22 measurement points. The study aims to examine the differential effects of various *outdoor* spaces on emotional and perceptual responses and to identify the dominant influencing factors. The results indicate that: (1) The *acoustic* environment (LAeq) is the key driver of negative affect (NA). (2) Human emotional perception can be enhanced through multi-sensory compensation mechanisms. (3) The multi-sensory interaction effect is regulated with significant spatial heterogeneity across different campus areas. Lake adjacent area exhibited the greatest emotional restoration, while main teaching area showed the strongest emotional stress. (4) The interaction among *thermal*, visual, and *acoustic* factors significantly contributed to predicting both positive affect (PA) and negative affect (NA). This study systematically elucidates the interactive effects of multi-sensory environmental on emotion and perception, providing both a theoretical basis and practical implications for the multi-sensory optimization of emotion-supportive campus environments.

Article • *Open access*

☐ 42 **FRESH: An Autonomous IoT Platform for Multi-Parameter Environmental Sensing and Short-Term Forecasting** Pan F.,Covington J.A. Sensors, 26(10), 3015 2026 1

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Environmental monitoring systems are often constrained by high cost, limited portability, restricted pollutant coverage, and dependence on fixed infrastructure, which can limit their suitability for distributed real-time sensing. This study presents FRESH, an autonomous Internet of Things (IoT)-based platform for multi-parameter environmental monitoring and short-term forecasting. The system integrates sensors for air quality, *thermal* conditions, light, *acoustics*, and weather, together with GSM-based remote data transmission, onboard data logging, and hybrid battery–solar power management. FRESH was deployed across multiple indoor and *outdoor* locations in Coventry and at the University of Warwick, UK, and operated over a 10-month period to assess practical performance under varied environmental conditions. In addition to continuous environmental sensing, machine learning models were developed to predict short-term changes in selected environmental variables. Across the tested models, the best predictive performance was obtained for several key parameters, including particulate matter (R^2 = 0.93), volatile organic compounds (R^2 = 0.92), and ozone (R^2 = 0.98). The results suggest that FRESH has potential to support portable, multi-parameter environmental monitoring with integrated short-horizon forecasting, providing a basis for further development of distributed sensing and localised early-warning applications.

Article

☐ 43 **Wearable Flexible Acoustic Alarm Device Based on MXene/Laser-Induced Graphene@Polyimide Thermoacoustic Film** Zhang Y.,Han D.,Wang Y., ... Zhai W.,Wang J. ACS Sensors, 11(4) 2026 1

Hide abstract Find It @ UCL View at Publisher Related documents

Flexible *acoustic* devices have advantages of high flexibility and large *sound* output area, capable of adhering to the surface of the skin or clothing, which shows great potential in the field of wearable electronics. However, the design of *acoustic* material structures and their functional applications are still difficult. Here, we fabricated a porous laser-induced graphene (LIG) microarray layer on a polyimide (PI) film by laser direct writing. Subsequently, we uniformly coated the few-layer MXene nanosheets onto the LIG surface to obtain an MXene/LIG@PI film featuring low specific heat capacity per unit area, large interlayer spacing, excellent electrical conductivity, and high *thermal* conductivity. Meanwhile, the MXene/LIG@PI-based *acoustic* device composed of flexible copper electrodes exhibits excellent *acoustic* performance with a high *sound* pressure level of 77.6 dB and maintained a stable *acoustic* spectrum as the frequency increased from 20.0 Hz to 20.0 kHz. Finally, the portable *acoustic* alarm device composed of the piezoelectric module and the MXene/LIG@PI-based *acoustic* module was designed, which can be worn on various parts of the human body clothing, to provide audible alarm reminders in case of accidental falls or drops during *outdoor* activities. This study provides a novel approach for designing new flexible *acoustic* materials and devices.

Article

	Document title	Authors	Source	Year	Citations
44	**Investigation of the figure of merit to 256QAM FSO link for the next-generation terrestrial network**	Vyas A.K.,Kumar S.	Journal of Optical Communications , 47(2)	2026	1

Hide abstract Find It @ UCL View at Publisher Related documents

Free space optical communication (FSO) has become an emerging broadband network. Even with the potential advantages of FSO technology, atmospheric turbulence, cloud-induced fading, and other environmental conditions such as fog, aerosol, temperature changes, storms, heavy rain, aiming inaccuracy, and scintillation reduce the performance of the faithful transmission. This article presented the 256QAM-FSO link, investigated the effect of different weather conditions on the received signals, and summarized bit error rate, error vector magnitude, symbol error rate (SER), power spectral density, and signal-to-*noise* ratio, and constellation diagram with and without atmosphere turbulence scenarios. The power spectral density of the output signal is a maximum of 11.61»dBm in no turbulence and a minimum of -72.80»dBm for heavy rain conditions to *noise* ratio degraded in the worst case of heavy rain 0.01»dB and achieved 18.57»dB in the best case of no turbulence, whereas SNR is obtained 9.37»dB in case of fog. Error vector magnitude suffers more from the haze condition, and SER varies from 0.70 to 0.85. The maximum bit error rate obtained for heavy rain is 5,432. With the advanced modulation scheme, the 256QAM-FSO link will provide wide bandwidth, which is helpful for *outdoor* and indoor services, terrestrial transmission, mile solutions, and so on.

Article

	Document title	Authors	Source	Year	Citations
45	**A stretchable and scalable Janus coating film with a multiscale porous structure integrating passive radiative cooling and electromagnetic shielding**	Ma L.,Liu L.,Wang L., ... Wang C.,Huang W.	Progress in Organic Coatings , 213, 109968	2026	4

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The escalating global energy crisis and electromagnetic pollution pose dual threats to sustainability and human health. Developing multifunctional materials for zero-energy cooling and effective EMI shielding is therefore crucial. Herein, based on high-speed shear and ultrasonic dispersion, a stretchable Janus coating film was fabricated via a scalable method combining solvent evaporation and humidity-assisted phase separation. The film integrates passive radiative cooling, electromagnetic interference shielding, Joule heating, self-cleaning, and flame retardancy in a single system. The radiative cooling layer, comprising a *thermoplastic* polyurethane matrix with $BaSO_4$ micro-nanoparticles and SiO_2-HA fillers, forms a multiscale porous structure that leverages Mie scattering, multiple reflection, and phonon-polariton resonance to achieve high solar reflectance (90.46%) and emissivity (95.71%). It also exhibits superhydrophobicity (CA > 150°) and low *thermal* conductivity. The EMI layer, constructed with a graphene-based dense-porous gradient, resolves the recalcitrant problem of combining high shielding efficiency and excellent mechanical properties with graphene as filler material. It employs an absorption-reflection-absorption mechanism that yields an average shielding effectiveness of 36.07 dB (X-band) and fast electrothermal heating to 93.3 °C at 7 V. A halogen-free phosphorus additive provides self-extinguishing flame retardancy, and the film maintains excellent flexibility (strain 243%). *Outdoor* tests show an average daytime sub-ambient cooling of 5.5 °C and a theoretical cooling power of 84.27 W/m^2. Building energy simulations further demonstrate Janus film's significant cooling energy savings across global climates. With superior mechanical robustness, environmental stability, and adaptive *thermal* management, this Janus coating film offers strong potential for building energy conservation, wearable electronics, and *outdoor* applications.

Article

	Document title	Authors	Source	Year	Citations
46	**Thermal and power generation performance of photovoltaic noise barriers with a heat sink using a deep neural network**	Sim J.,Yun S.,Lee D.	Applied Thermal Engineering , 288, 129575	2026	3

Hide abstract Find It @ UCL View at Publisher Related documents

Photovoltaic *noise* barriers (PVNBs) enable space-efficient solar energy utilization but are confined to elevating module temperature and reducing efficiency and lifetime. Despite its importance, intensive analysis on the direct correlation among irradiation, surface temperature, and power generation in PVNB has not yet been conducted. This study experimentally evaluated the *thermal* and electrical performance of three photovoltaic (PV) configurations with indoor and *outdoor* experiments: PV-only, PVNB, and PVNB with heat sink. Indoor experiments measured averaged

	Document title	Authors	Source	Year	Citations

surface temperatures of PV for various configurations, while *outdoor* experiments collected power-generation data which were used to train a deep neural network (DNN) that predicts power output using solar irradiation and averaged surface temperature of module. The PVNB exhibited the highest temperature rise, producing power reductions of 1.3 % at 700 W m^{-2} and 0.8 % at 800 W m^{-2} relative to PV-only, whereas PVNB with heat sink enhanced natural convection and recovered the barrier-induced losses, fully at 700 W m^{-2} and by approximately 85 % at 800 W m^{-2}. The developed DNN model captured physical trend, achieving coefficients of determination of 0.8887 for training and 0.9217 for validation, and most predictions fell within a ±20 % error. Economic analysis reported that the heat sink retrofit of PVNB provided similar annual revenue but resulted in a 29.5 % higher net present value owing to the enhanced lifespan of the PV module caused by the reduction in averaged surface temperature. Despite the limitations of narrow climate conditions and using only single crystalline silicon module with one built-on PVNB design, this study provides essential insight and a practical tool for assessing PVNB retrofit and predicting *thermal* and electrical performances.

Article

☐ 47 **A highly resilient and large-strain wearable hydrogel sensor based on acrylamide/gelatin/Zr4+ and its application in human motion monitoring and information transmission**

Zhang Y., Liang M., Yan Z., Zhang S.

Polymer, 346, 129635 2026 5

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The application potential of flexible electronic technologies across diverse interdisciplinary domains has become increasingly evident. Owing to their unique combination of flexibility, biocompatibility, and skin-mimetic mechanical properties, hydrogels have emerged as a central material for wearable flexible sensors. Nevertheless, conventional hydrogel-based sensors often suffer from inadequate mechanical performance under high stress and large strain conditions, along with limited toughness. Gelatin, a natural protein, possesses a molecular structure abundant in hydroxyl and amino functional groups, enabling the formation of a highly extensible, hydrogen-bonded flexible network and exhibiting excellent biocompatibility. In this study, a *thermally* induced free-radical polymerization approach was employed to construct a covalently cross-linked rigid framework using polyacrylamide (PAM), while incorporating gelatin to establish a hydrogen-bond-reinforced flexible network. Concurrently, Zr^{4+}were introduced to coordinate with anionic functional groups on both gelatin and PAM chains, resulting in the fabrication of a polyacrylamide/gelatin/zirconium ion composite hydrogel sensor (PMG_xZ_y). The PMG_xZ_y hydrogel sensor demonstrates exceptional toughness, rapid response dynamics, high sensitivity, and a broad sensing range. It achieves a strain of up to 924 % under a stress of 0.26 MPa, with a fracture energy of 1.12 MJ m^{-3}. The sensor not only enables real-time monitoring of large-amplitude human motions but also facilitates *sound* recognition, handwritten pattern detection, and information transmission via integration with Morse code encoding. This work effectively overcomes the mechanical limitations of traditional hydrogel sensors, offering a promising new material platform for wearable sensing applications in fields such as *outdoor* sports monitoring and rehabilitation training.

Article • *Open access*

☐ 48 **Acoustic Wave-Powered Durable Icephobic Duplex Coating Design with Superior Deicing Performance**

del Moral J., Haworth L., Montes L., ... Gonzalez-Elipe A.R., Fu Y.

ACS Applied Materials and Interfaces, 18(5) 2026 5

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Piezoelectric thin film-based surface *acoustic* wave (SAW) deicing technology has recently emerged as an attractive and energy-efficient alternative with direct applications across multiple industrial sectors. However, the generation of SAWs on piezoelectric thin films, such as ZnO, faces diverse challenges, including its low long-term stability and variable wetting properties upon exposure to UV radiation and other environmental hazards. To overcome these challenges, we propose a bilayer coating design that integrates a diamond-like carbon (DLC) thin film with an atop CF_x layer (DLC-CF_x). This design is intended to serve as both an anti-icing and a protective coating for ZnO SAW devices built on aluminum substrates, which are specifically selected for critical ice-exposed applications in the aeronautics or wind turbine industries. We demonstrate that, unlike the implementation of single fluorinated polymer

	Document title	Authors	Source	Year	Citations

layers, such as commercial CYTOP, the DLC-CF_x hydrophobic duplex coating effectively protects the ZnO surfaces while maintaining optimal SAW transmission and wave propagation and reducing the fluorine content. The SAW-induced deicing on these devices is achieved through a highly effective mechanism involving the interfacial ice melting, followed by a rapid ice sliding detachment for both small ice droplets and large ice aggregates. Experiments at laboratory scale and in an icing wind tunnel facility reveal that deicing involves SAW activation of the interface between the ice and the DLC-CF_x bilayer, as well as an effective *thermal* contribution resulting from the rapid heat transmission through the aluminum substrate. Our studies demonstrate that the highly conformal deposition of DLC-CF_x through a room temperature plasma-assisted method ensures reliability and long-term stability of thin-film-based *acoustic* wave devices in harsh *outdoor* conditions.

Article

49 **On the applicability of knowledge sources in climate-based design: Thermal-spatial micro-spaces in historical gardens versus contemporary outdoor design strategies** — Nouri-Horzvili N., Suzanchi K., Haghighat Bin M. — Urban Climate, 65, 102713 — 2026 — 0

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Quantitative research on *outdoor thermal* comfort often produces strategies that are technically *sound* but limited in systemic integration and practical design applicability. This study reveals the limitations of existing strategies in capturing the complexity of *thermal*-spatial mechanisms within historical gardens' micro-spaces. Using Alexander's pattern language as a shared framework, we translated pure science-based strategies and design-informed interpretation into a hierarchical *thermo*-spatial syntax to decode two comparable representations. We analyzed each representation using five systemic complexity indicators: four; Betweenness Centrality, Katz Centrality, Strongly Connected Components (SCCs), and feedback loops, capture horizontal integration and connectivity, while Hierarchy Depth captures vertical, multi-level dependencies. Accordingly, scientific decoding often reflects static, component-based logic, whereas designer-based decoding reveals dynamic, cyclic systems rich in integration. Additionally, bidirectional mapping shows how designer insights can enrich scientific structures, highlighting evolutionary micro-spaces as reservoirs of human-scale *thermal* strategies and revealing system flows that reduce the complexity of *outdoor*.

Article

50 **Acoustic enhancement and weathering durability of rice husk/PVC composites via microperforated design and nanocoating** — Ibrahim N., Jaafar M., Che Hassan N.H., Garcia Sanz R. — Journal of Building Engineering, 119, 115253 — 2026 — 3

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Rice husk-reinforced polyvinyl chloride (rice husk/PVC) composites valorize agro-waste but suffer from poor durability and weak *acoustic* absorption, limiting their use in construction. This study adopts a dual approach, a microperforated panel (MPP) design was introduced to enhance *acoustic* performance, and a dual-layer nanocoating of multi-walled carbon nanotubes (MWCNTs) and silicon dioxide (SiO_2) was applied to improve weathering resistance. Rice husk/PVC composites were fabricated via extrusion and compression molding. For *acoustic* evaluation, samples with different perforation patterns were tested using an impedance tube, where MPP panels achieved a maximum *sound* absorption coefficient (SAC) of 1.0 at 1045 Hz, with resonance tunable by varying rear air gaps (10–30 mm). For durability evaluation, coated and uncoated composites were exposed to four weeks of accelerated UV-moisture weathering. Coated samples showed superior retention of tensile strength (only 18.5 % reduction compared to 56 % in uncoated composites) and maintained high hydrophobicity (contact angle 144° vs. 75° for uncoated). Fourier Transform Infrared Spectroscopy and Scanning Electron Microscope confirmed reduced oxidation and microcracking, while *thermal* conductivity values (0.312–0.327 W/m·K) remained comparable to other agro-waste-based insulation materials. Overall, the findings demonstrate that microperforation effectively addresses *acoustic* limitations, while nanocoating significantly enhances environmental durability. Together, these strategies highlight a sustainable pathway to transform rice husk into multifunctional composites for building engineering applications, particularly *acoustic* cladding, facade panels, and *outdoor noise* barriers.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
51	**Prefabricated Co-Working Spaces' Window Design: Emotional Salience Scale-Based Optimisation**	Ciervo A.,Masullo M., Maffei L.,... Morelli M.D., Scorpio M.	Buildings, 16(4), 875	2026	1

Hide abstract Find It @ UCL View at Publisher Related documents

Windows are key elements of the building's system; they connect workers with the *outdoor* environment, influence daylight penetration, *sound* insulation, and *thermal* exchanges of façades, but they also moderate the workers' well-being and productivity. This research investigates how the window-to-wall ratio, as well as the position and orientation of mullions, in movable offices affect the combination of workers' perceptual and emotional responses. A smart co-working prefabricated movable office was modelled in virtual reality to include dynamic visual elements and *acoustic* stimuli. Experiments were performed in a laboratory under controlled *thermal* conditions involving 32 volunteers. The Igroup Presence and Emotional Salience Questionnaires were used to collect subjective responses. ANOVA analysis and post hoc test with the Bonferroni correction were used for data elaboration. Results revealed that window design affects emotional salience. High window-to-wall ratio and no mullions achieved the highest scores. Increasing the number of mullions, particularly when they obstruct key visual elements, reduced the positive emotional salience rating. Horizontal mullions diminish the outdoors' spatial perception, interrupting visual continuity and restricting users' capacity to recognise variations in the views. Finally, the results suggest some valuable insights and suggestions that can help designers improve window design and people's well-being and satisfaction.

Article • *Open access*

52	**Prediction of outdoor ground effect**	Attenborough K., Taherzadeh S.	Applied Acoustics, 242, 111067	2026	3

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A classification of *outdoor* ground surfaces for calculating the attenuation due to destructive interference between direct and ground-reflected *sound* is proposed which uses a physically admissible ground impedance model with parameters of porosity, flow resistivity, and, for some ground surfaces, layer depth. This impedance model is shown to enable good fits to short range level difference measurements over a wide range of ground surfaces. Ranges of parameter values are suggested in each class to allow for variations in soil depth profiles, moisture content and surface roughness. Also, methods are proposed to account for mean ground roughness heights of less than 0.1 m and for waves on a water surface. The influence of atmospheric turbulence on ground effect is included through a coherence factor which assumes a Kolmogorov turbulence spectrum with parameters for which values can be calculated if heat flux and friction velocity are known. To account for changes in impedance along the propagation path, a Fresnel zone method is proposed which weights pressures squared since this method has been shown to compare better with 2D Boundary Element Method predictions than a method that weights excess attenuation.

Article

53	**A Flexible Temperature Sensor with Ultrafast Response Speed and High Stability Achieved by Improving Substrate Thermal Conductivity and Radiative Cooling**	Sun G.,Wang D.,Wang P., ... Yan H.,Meng C.	Advanced Functional Materials , 36(2), e12296	2026	67

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Flexible temperature sensors have great potential for application in human-computer interaction and disaster prevention systems. However, the slow response rate makes it difficult to quickly respond to human health conditions, and its stability is difficult to maintain in various environments, limiting its range of use. In this study, a fast-response, highly stable temperature sensor based on a breathable nanofiber platform is prepared. The response speed (0.32 s) is improved by introducing oriented boron nitride nanosheets (BNNS) with high *thermal* conductivity into TPU (*thermoplastic* polyurethane elastomer) nanofibers; the high sensitivity (0.077 $°C^{-1}$) is realized by adding polyaniline (PANI) to graphene (G). High stability and signal-to-*noise* ratio (SNR) are achieved by covering the sensing layer with the TPU/silica (SiO_2) nanofiber and using the direct printing preparation method. B Besides, the TPU/SiO_2 nanofiber, with its high visible light reflectance, and the TPU/BNNS nanofiber enable the sensor to exhibit excellent human *thermal* management capabilities, lowering body temperature by up to 5 °C in *outdoor* conditions compared to standard clothing. For application verification, the sensors are used for long-term temperature monitoring of the

	Document title	Authors	Source	Year	Citations

human body in various environments, as well as for respiratory and battery temperature monitoring. This study opens a promising path for improving the response speed and reliability of temperature sensors.

Article

54 **Hollow glass beads/polyurethane multifunctional foamed membranes based on structure-function co-design and their extreme environmental applications.** Liu Y.,Ye H. Colloids and Surfaces A Physicochemical and Engineering Aspects , 728, 138512 2026 6

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Outdoor adventures in extreme environments place stringent requirements on the multifunctional integration of materials, and the existing research has made it difficult to synergize the UV resistance, infrared stealth, and *thermal* insulation properties due to the limitations of structural design, let alone expanding the additional functions. In this study, based on the concept of "structure-function synergistic design", the HGB/polyurethane foamed film was prepared by the scraping method through the synergistic effect of the sodium dodecyl sulfate (NXZ) and hollow glass beads (HGB), which realizes the synergistic enhancement of multi-functionality and expansion of functions for the first time. Experiments show that NXZ (0.2 wt%) and HGB (24 wt%) synergistically enhance the UPF of the film by 12 times compared with that of the NXZ-free film through the "scattering-barrier" synergistic effect, reduce the infrared emissivity by 10.19 %, and the *thermal* conductivity is as low as 0.03995 W/(m·K), and at the same time, realize the *acoustic* absorption factor of 0.227 (2450 Hz). With the introduction of iron oxide yellow 313 dye, the UPF was further increased by 11.77 times, the IR emissivity was reduced by 20.96 %, and the self-cleaning function was given through the interfacial synergistic effect. The synergistic mechanism of "porous structure-functional unit" established in this study provides a new paradigm for the design of materials for extreme environments, which has significant potential for application.

Article • Article in Press

55 **Outdoor comfort in historic spaces under thermoacoustic interactions** Zhang R.,Cui L.,Liu J., ... Wang Z.,Zhen M. Building Simulation 2026 0

Hide abstract Find It @ UCL View at Publisher Related documents

Outdoor comfort in historic spaces has received increasing attention under the pressures of urban heat and environmental *noise*, yet *thermoacoustic* research has focused mainly on modern urban settings. This study investigated Jinci Temple, a historic site in China's cold region. Field measurements and questionnaire surveys were conducted in four open-space types to examine the relationships among *thermal* conditions, *acoustic* conditions, and *outdoor* comfort. Representative optimization scenarios were simulated using ENVI-met and CadnaA and prioritized through an exploratory assessment informed by Heritage Impact Assessment (HIA). Respondents exhibited relatively high summer heat tolerance. The neutral physiological equivalent temperature (NPET) was 29.4 °C, the neutral PET range was 23.7–35.1 °C, and the upper acceptable PET threshold was 34.3 °C. *Acoustic* evaluation varied markedly with perceived *sound*-source type. At a neutral mean *acoustic* sensation vote, *sound* pressure levels were 72.2, 68.8, and 60.9 dB for positive, neutral, and negative *sound* sources, respectively, indicating greater *acoustic* tolerance under positive *sound* conditions. *Thermoacoustic* associations were context-dependent: *thermally* comfortable conditions were associated with lower sensitivity to SPL variation, whereas moderate *acoustic* conditions were associated with weaker *thermal* discomfort. Scenario simulations predicted that reversible shading with a solar transmittance of 0.3 would reduce mean radiant temperature and PET by 12.5 and 5.3 °C, respectively, whereas the predicted *sound* pressure level reductions for the 3-m and 5-m multilayer green buffers were only 0.2 and 0.4 dB, respectively. HIA identified vegetation shading and greening as priority strategies, water–vegetation synergy as a localized conditional strategy, and visitor *noise* management as a supplementary measure.

Conference Paper

56 **Review on Soundproof Ventilation Techniques: Study on Ventilation and Acoustic Performance** Baskara S.A., Trihamdani A.R. Lecture Notes in Civil Engineering , 875 LNCE 2026 0

Hide abstract Find It @ UCL View at Publisher Related documents

	Document title	Authors	Source	Year	Citations

Natural ventilation has been known as a cost-effective and sustainable solution to provide good indoor air quality (IAQ) and *thermal* comfort in the hot-humid climate. Nevertheless, in the urban setting, opening windows also allows *outdoor noises* to penetrate the indoor spaces. This problem forces occupants to close the window, which will lead to poor IAQ and *thermal* comfort. This research aims to provide a solution to the aforementioned conflicting issues, which are ventilation performance and *outdoor noise* reduction. Firstly, several *noise* resonator techniques is compared. Then, a *noise* silencer based on the best resonator performance is developed. Finally, the detailed assessments of ventilation and *noise* reduction performances of selected (Soundproof Ventilation) SPV techniques, i.e. *noise* resonator, window plenum, and *acoustic* louver were conducted through simulation. The simulation results show that the soundproof ventilation is 24.1 dBA, 9.1 dBA, 2.7 dBA, 2.6 dBA SPL reduction and 0, 0.036, 0.209, and 0.362 discharge coefficients for *noise* silencer, plenum window, chevron *acoustic* louver, and single bank *acoustic* louver respectively. Discussion to improve the SPV technique was also conducted.

Article • *Open access*

57 **Environmental barriers meet functional limitations: an exploratory qualitative study on the usage and experience of outdoor gym among older adults**

Zhou T.-Y.,Lu H.-].,Li C., Lyu H.

Frontiers in Public Health , 14, 1861748

2026

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Introduction – *Outdoor* gyms are widely used in urban public spaces, yet there is a lack of specialized research for older adults in urban parks from the perspective of functional limitation. Methods – This study conducted in-depth interviews with 30 older adults via purposive sampling and used thematic analysis to explore their usage and experience of *outdoor* gym in urban park and related environmental and individual influencing factors. Results – Results show that *outdoor* gym have positive impacts on the physical, psychological and social health of older adults. The main motivations include pursuing physical health and satisfying social and emotional needs. Individual constraints involve declines in lower limb function, balance, flexibility, as well as visual and cognitive impairments. Environmental support factors include pleasant natural settings, low *noise*, comfortable *thermal* conditions and diverse equipment, while environmental pressure factors include bad weather, insufficient or poorly maintained equipment, *noise* and inadequate lighting. Older adults proposed optimization suggestions covering safety, information accessibility, comfort and equipment suitability. Conclusion – This study deepens the understanding of environment-behavior interactions in older adults with functional limitations, offering a valuable empirical baseline for rehabilitation-oriented design for healthy aging.

Conference Paper • *Open access*

58 **Optimization Analysis of Thermal Distribution in a Confined Balcony Space Under Air-Conditioning Outdoor Unit Operation Using Taguchi-Based CFD Simulation**

Dang H.-S.,Nguyen T.-A.-T., Lai H.-N.

2026 11th International Conference on Applying New Technology in Green Buildings Atigb 2026

2026

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This study analyzes the *thermal* behavior and airflow characteristics within a confined balcony space under typical hot and humid climatic conditions in Ho Chi Minh City. A computational fluid dynamics (CFD) model was developed and numerically assessed through mesh-independence and convergence analyses, incorporating an external air domain to ensure realistic boundary conditions. The Taguchi method, combined with analysis of variance (ANOVA), was employed to evaluate the influence of four key factors: number of *outdoor* units, installation position, ventilation opening ratio, and ambient wind velocity. The results indicate that the number of *outdoor* units is the dominant factor affecting the average balcony temperature, contributing the largest variation. Ambient wind velocity has a moderate influence, while installation position and ventilation opening ratio exhibit relatively minor effects. The consistency between Taguchi and ANOVA analyses confirms the reliability of the findings. Based on the signal-to-*noise* (S/N) ratio analysis using the 'smaller-thebetter' criterion, the optimal configuration was identified as A1-B2-C1-D1. A verification simulation showed that the model converged and achieved an average temperature of 304.24 K, significantly lower than in the initial simulation cases. The findings highlight the importance of controlling internal heat sources and

	Document title	Authors	Source	Year	Citations

optimizing airflow pathways to improve *thermal* conditions in confined balcony spaces. This study provides practical insights into the design and installation of airconditioning systems in urban residential buildings.

Article • Article in Press

59 **Threshold-based multi-scale analysis of thermal-acoustic comfort in urban public spaces: identifying critical thresholds and individual heterogeneity**

Xu L.,Jing W.,Ding W., ... Ge Z.,Ai Y.

Environment Development and Sustainability

2026

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Outdoor users encounter *thermal* and *acoustic* conditions simultaneously, and pooled comfort scores may obscure differences among respondents. We analysed 975 valid questionnaires paired with concurrent environmental measurements at a university plaza in Xi'an to examine response thresholds, conditional associations and heterogeneity. The analysis combined established approaches: piecewise regression, regression trees, K-means clustering, random forest interpretation using Shapley additive explanations (SHAP), and partial dependence analysis. The *thermal*-sensation curve changed slope near 30 ℃. A *thermal*-comfort breakpoint was estimated at 29.95 ℃, but the comparison with a linear model did not meet the 0.05 significance criterion. *Acoustic* sensation had an estimated breakpoint of 64.8 dB, while *acoustic* comfort declined without a significant breakpoint. Clustering yielded three response profiles. Under the combined high-temperature and high-*sound*-level condition, *thermal* comfort was 0.63 units below the additive prediction. Temperature accounted for the largest reported SHAP contribution to *thermal* comfort. The findings motivate local tests of coordinated shading and *sound* management, together with alternative *outdoor* settings for users with different preferences. Validation at other sites is needed before applying the thresholds or profile-based recommendations more widely.

Article • *Open access*

60 **Assessment of Psychological Effects of the Built Environment Based on TFN–Prospect–Regret Theory–VIKOR: A Case Study of Open-Plan Offices**

Cheng X.,Zhao G.,Xie M.

Sustainability Switzerland , 18(2), 1104

2026

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As people spend more time indoors, the impact of the built environment on psychological health has attracted growing attention. Yet existing studies often have difficulty capturing decision-makers' reference dependence and loss aversion under uncertainty. To bridge this gap, we propose an evaluation framework comprising three first-level criteria—*Outdoor* Environment, Physical Comfort (including *thermal*, lighting, and color environments), and *Acoustic* Comfort—and determine combined weights by integrating subjective analytic hierarchy process (AHP) judgments with objective entropy weighting based on triangular fuzzy numbers (TFNs). We further incorporate prospect–regret theory to represent loss aversion, expectation-based reference points, and counterfactual regret/rejoicing, and couple it with the VIKOR compromise ranking method, forming an integrated "TFN + Prospect–Regret + VIKOR" approach. The proposed method is applied to four retrofit alternatives for an open-plan office floor (approximately 1200 m^2), each emphasizing *outdoor* environment, physical comfort, *acoustic* comfort, or no single priority. Experts assessed the schemes using fuzzy linguistic variables. The results show that lighting conditions, *thermal* comfort, color scheme, and internal *noise* control receive the highest comprehensive weights. Extensive sensitivity analyses across value/weighting functions and regret-aversion parameters indicate that the ranking of alternatives remains stable while exhibiting clearer separation. Comparative analyses further suggest that, although the overall ordering is consistent with baseline methods, the proposed model increases score dispersion and improves discriminative power. Overall, by explicitly accounting for decision-makers' psychological behavior and information uncertainty, the framework enables robust and interpretable selection of retrofit schemes for existing office spaces.

Article • Article in Press

	Document title	Authors	Source	Year	Citations
61	**Photovoltaic Rapid Shut-off Device Health Status Assessment Model Based on Multi-Modal Feature Fusion and Transfer Learning**	Wang Q.,Li K.,Nie C., Man J.,Wang W.	International Journal of Computational Intelligence and Applications , 2650023	2026	0

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Photovoltaic rapid shut-off devices operate for a long time in *outdoor* high-temperature, high-irradiation and humid-heat environments. Their early degradation characteristics are often weak and vary widely across sites. Traditional diagnostic methods based on a single electrical or *thermal* imaging signal are difficult to meet actual operation and maintenance needs. Therefore, this study proposes a photovoltaic rapid shutdown device health status assessment model based on multi-modal feature fusion and transfer learning. The model improves the complementary expression between different modalities through weak mode enhancement and cross-modal attention mechanisms, and it also uses a distribution alignment strategy to alleviate cross-site feature drift. Experiments showed that this method significantly improved recognition performance in multi-modal fusion experiments. The F1 value of the early fusion model increased from 82.21% to 89.54%, and the accuracy of the weak mode complete enhancement scheme increased from about 74% to nearly 86%, and remained the best in the four types of ablation comparisons. Furthermore, in the *noise* perturbation experiment, when the *noise* intensity reached 0.4, it still maintained an AUC of 0.89, which was 5-8% higher than the comparative method on average. At the same time, the performance retention rate still exceeded 89% when the mode was missing 30%, showing more stable robustness. The study verifies the stability of the proposed method under the conditions of weak degradation identification, *noise* immunity, and mode loss, and provides an effective technical approach for photovoltaic equipment health monitoring.

Book Chapter

	Document title	Authors	Source	Year	Citations
62	**Physics-Informed Kalman Filtering for Multi-Step Bias Correction in Indoor Temperature Forecasting**	Daoutis G., Tomoutzoglou O., Kornaros G.,Coppola M.	Pervasive Intelligence from Architectures to Sustainable Edge AI Systems of Systems	2026	0

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Indoor temperature prediction is often utilized to maintain the right environment in which wine can be fermented or stored. However, forecasting these indoor microclimates is challenging due to inaccuracies in *outdoor* temperature predictions combined with the difficulty of modelling a building's *thermal* inertia, which changes due to unpredictable environmental factors. This paper proposes a two-stage machine learning and state-estimation pipeline to predict temperatures inside a winery up to an 18-hour horizon. First, feature extraction is performed to train an Autoregressive with eXogenous inputs (ARX) model. Then, to mitigate the drift caused by factors such as *noise* in the API predictions or errors caused by the model, a Two-State Kinematic Kalman Filter (KKF) is introduced. The proposed KKF lies in its dual-state formulation which tracks both the magnitude of the prediction error and its latent velocity. In this way, the filter effectively models the building's *thermal* inertia and identifies the steady, low-frequency velocity of the *thermodynamic* drift, resulting in a stable and physically realistic forecast.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
63	**Techno-economic evaluation of PVT-sourced heat pump systems: can solar compete with air and ground in cold climates?**	Beltran F.,Sommerfeldt N.	Solar Energy Advances , 6, 100133	2026	2

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There are various ways to integrate solar photovoltaic-*thermal* (PVT) collectors with heat pumps, but using the PVT array as the sole source offers unique advantages: eliminating *noise* from air-source units and avoiding the cost and land use of boreholes. This work investigates whether PVT collectors can operate as the sole source for a brine-water heat pump in cold-climate multi-family houses (MFH), and how absorber design and system sizing influence techno-economic performance. Five unglazed PVT absorbers with different design characteristics, are simulated in a validated TRNSYS model of a 2000 m^2 Swedish MFH. Collector area is varied from 1.1 to 4.5 m^2/kW_{th} of nominal heat pump capacity, equivalent to a rooftop coverage of 25 to 100 %, while brine flow rate is varied from 24 to 96 $l/h\text{-}m^2$.

	Document title	Authors	Source	Year	Citations

Finned designs achieve the highest seasonal performance factor (SPF) of 2.4 to 2.6 at moderate rooftop coverage (2.3–3.4 m^2/kW_{th}), with 15–20 % auxiliary heating share, while unfinned concepts converge at an SPF of approximately 2.2 and 25–35 % auxiliary heating share at full rooftop coverage (4.5 m^2/kW_{th}). Arrays below ~2 m^2/kW_{th} are non-viable (SPF ' 1.8; 40–70 % auxiliary heating). The economic optimum occurs at an SPF of around 2.2, not at the highest efficiency, yielding two optimal pathways: fin-enhanced absorbers at ~2.3–3.4 m^2/kW_{th}, or low-cost polymeric absorbers at 4.5 m^2/kW_{th}. The best PVT-only configurations achieve cost parity with air source heat pumps (±5 %) and remain ~10–15 % below district heating. They remain more expensive than GSHP systems, but avoid both *outdoor noise* and drilling requirements.

Conference Paper

64 **Field Investigation on the Window-Opening Behavior and Indoor Environment Performance in Naturally Ventilated Residential Building: A Case Study of Student Dormitory in ITB Jatinangor Campus**

Nurjannah A., Trihamdani A.R., Alfata M.N.F., Nurdini A. | Lecture Notes in Civil Engineering , 875 LNCE | 2026 | 0

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Window-opening behavior is one of the critical factors to ensure the effectiveness of natural ventilation in improving indoor air quality, *thermal* comfort, and energy-saving in buildings. This research aims to investigate the window-opening behavior and its effects on indoor *thermal* condition in a public apartment in Jatinangor, Indonesia. A questionnaire survey and field measurement were carried out to analyse the window-opening behavior and *thermal* environment, respectively. The result shows that most occupants prefer to open the balcony door than the window. Fresh air intake and heat removal were the main reasons for opening the window; on the other hand, the *outdoor* environment (e.g., dust and *noise*) was the reason for closing the window. This study also revealed that opening the window at night could lower indoor temperature up to 2.7 °C. Meanwhile, opening windows during the daytime may lead to excessive heating in indoor spaces. Furthermore, opening the window/door could elevate the indoor airspeed up to 0.5 m/s higher compared to the unit where all windows/doors are closed. This research provides insights into the window/door design for low-energy buildings.

Article • Article in Press

65 **Embodied eco-consciousness in paragliding: a multi-method phenomenological approach to a cosmic sport**

Schirrer M., Aubin N., Schnitzler C., Lefevre L. | Sport Ethics and Philosophy | 2026 | 0

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Anchored in the philosophy of sport, this article adopts a phenomenological approach centred on the embodied experience of free flight. It explores paragliding as a cosmic sport, in which the body enters into a continuous dialogue with the invisible forces of air, gravity, and the rhythms of the sky. Departing from performance or control-oriented conceptions of sport, this research makes visible the sensory and pre-reflective experience of pilots—both novice and expert—in their ways of inhabiting the air. Three qualitative field studies were conducted: the accompaniment of high-school students during initial slope training, a longitudinal exploration of novice high-school students' first major paragliding flights and explicitation interviews with experienced cross-country pilots. The methodology combines multisensory experience narratives, eco-consciousness drawings and interviews supported by GPS flight traces, providing access to the fine textures of lived experience, like tensions in the lines, wind pressure on the skin and the *sound* of rising *thermals*. Our findings reveal a progressive development of embodied eco-consciousness, from an anthropocentric perspective towards a more cosmic attunement, in which body, wing and elements become experientially intertwined. This research contributes to the philosophy of sport by offering an empirical phenomenological account of how nature-based practices cultivate embodied eco-consciousness, challenging performance-oriented models of sport and demonstrating how cosmic sports can foster ecological awareness through sensory attunement rather than environmental mastery. These experiences invite a rethinking of *outdoor* sports ethics: rather than seeking to dominate the environment, the aim becomes one of the attunement and transformation through it.

Article

	Document title	Authors	Source	Year	Citations
66	**Performance Study of Thermal Control for Uncooled IRFPA Cores in Quantitative Detection in Complex Environments | 复杂环境下定量检测的非制冷IRFPA机芯温控性能研究**	Yuan P.,Cui H.,Wang M., ... Qiu S.,Wang X.	Guangxue Xuebao Acta Optica Sinica , 46(15), 1512002	2026	0

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Objective: Uncooled infrared focal plane arrays (IRFPAs) have been widely applied due to their advantages of operation without cryogenic cooling, compact size, low power consumption, and long lifespan. However, their high sensitivity to ambient temperature constrains the accuracy and stability of quantitative imaging detection in complex environments, such as industrial gas leakage infrared imaging detection and division-of-focal-plane polarization imaging. Traditional temperature compensation and non-uniformity correction (NUC) algorithms also struggle to completely suppress residual *thermal* drift and *noise*. Therefore, in-depth research on the variation patterns of uncooled IRFPA imaging performance with temperature changes and investigation of high-precision temperature control technology and the corresponding NUC algorithms are of significant importance for enhancing infrared quantitative imaging detection capabilities in complex environments. Methods: This study presents the design and implementation of a high-precision active temperature control system based on a *thermoelectric* cooler (TEC) that enables precise adjustment of uncooled IRFPA assembly temperature within a wide operating range of 15–42 °C and achieves stable temperature control with ±0.01 °C accuracy. A series of experiments were conducted under both *thermally* stabilized and free-running conditions to characterize the effect of temperature on the IRFPA's response parameters (gain, offset) and 3D *noise* profile, including system performance validation, multi-temperature response analysis, and assessments of repeatability and stability. Furthermore, through experiments with filter-wheel multispectral IRFPA imaging systems and division-of-focal-plane polarization *thermal* imaging systems, we validated the effectiveness of the stray radiation model-based NUC algorithm (SRM-TUIRF) considering temperature control conditions and the comprehensive correction algorithm for *thermal* drift in division-of-focal-plane polarization *thermal* imaging (CCMTD), respectively. Results and Discussions: The TEC-based temperature control system achieved high-precision temperature stabilization of ±0.01 °C across a wide temperature range of 15–42 °C, providing a technical guarantee for stable operation of uncooled IRFPA in complex environments. Experiments revealed that IRFPA gain exhibited a negative correlation with focal plane temperature, offset showed a positive correlation with core temperature, and temperature transition-triggered shutter correction caused synchronous updates of response parameters. The temperature control system significantly improved IRFPA *noise* performance and response stability. Under TEC-stabilized 24 °C operation, temporal *noise* was reduced from 14.55 to 11.82, while spatial *noise* was drastically suppressed from 469.2 to 8.93, demonstrating a substantial reduction in 3D *noise*. The relative errors of the repeated tests did not exceed 0.42%, and 24-hour long-term stability errors remained below 1.51%, indicating excellent operational reliability. For the filter-wheel multispectral IRFPA imaging systems, the temperature-controlled SRM-TUIRF algorithm effectively overcame internal stray radiation interference caused by environmental temperature fluctuations. The root mean square error (RMSE) for all six spectral channels was significantly reduced, with correction performance superior to traditional two-point correction and fixed pattern *noise* correction methods. This approach is particularly suitable for low-frequency stray radiation suppression in *outdoor* environments with large temperature variations. It significantly enhances multispectral image uniformity and detail representation. For the division-of-focal-plane polarization *thermal* imaging systems, combining the temperature control system with CCMTD reduced the maximum measurement errors of degree of linear polarization (DoLP) and angle of polarization (AoP) by 33.89% and 72.97%, respectively, achieving accurate and stable measurement of polarization parameters across a wide range of operating temperatures. Conclusions: By employing a high-performance uncooled IRFPA active temperature control system, this study systematically analyzed the variation patterns of uncooled IRFPA system performance with core temperature changes, demonstrating that active temperature control is an effective means to suppress *thermal* drift and reduce *noise* in uncooled IRFPA imaging systems. It provides a stable and reliable physical foundation for advanced NUC algorithms that depend on precise temperature parameters, serving as a key enabling technology for unleashing advanced algorithm performance. The combination of temperature control systems and novel correction algorithms can provide technical support for uncooled IRFPA applications in complex environments such as industrial gas leakage multispectral infrared imaging detection, epidemic mobile population imaging temperature screening, and division-of-focal-plane polarization imaging for quantitative imaging applications. This research provides reliable experimental evidence and technical pathways for promoting the development of uncooled IRFPA imaging technology from traditional qualitative observation to high-precision, high-reliability quantitative detection applications.

	Document title	Authors	Source	Year	Citations

Article • *Open access*

67 **Over-the-Air Federated Learning for Urban Cellular Networks: Demonstrations and Scalability**

Pradhan S.,Koc A., Alemdar K.,... Hudon M., Chowdhury K.

IEEE Transactions on Networking , 34

2026 0

Hide abstract Find It @ UCL View at Publisher Related documents

Over-the-air federated learning (OTA-FL) offers an exciting advancement over classical FL using the physics of analog signal propagation for model update aggregation. Since each participant broadcasts its model weights concurrently in time and frequency, this paradigm conserves communication bandwidth and model upload latency. Despite its potential, no prior large-scale real-world demonstration exists. This paper is the first to demonstrate OTA-FL deployment in a cellular network within the constraints of a 5G-compliant OFDM (Orthogonal Frequency Division Multiplexing) waveform. We identify key challenges, including multi-path fading, *thermal noise*, and the need for precise client synchronization for coherent OTA aggregation. To address these, we propose a unified real-world implementation incorporating pre-equalization, model update-to-OFDM mapping, and a dual-layer synchronization interface for OTA-FL training. Additionally, we develop a simulation framework in NVIDIA Sionna for large-scale evaluations in an urban microcell (UMi) environment, integrating an adaptive power control scheme to manage complex *outdoor* propagation. We experimentally validate OTA-FL for Channel Estimation and Object Classification using the ORBIT testbed with users at varying distances, along with an indoor testbed featuring a mobile client for dynamic channel conditions. Furthermore, we analyze model generalization and adaptive power control across a wide-range emulated urban cell, along with additional operator-level benefits. Under specific conditions, OTA-FL achieves comparable model performance while improving spectrum utilization by 43× and energy efficiency by 7× over classical FL with five participants, with the benefits growing proportionally as more participants are added.

Conference Paper

68 **Investigating Relationships Between Windows' Opening and Closing Sessions and Indoor Environmental Conditions in Buildings**

Adekunle T.O.

Lecture Notes in Civil Engineering , 844 LNCE

2026 0

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Windows in buildings primarily enhance occupants' views of the *outdoor* surroundings, improve ventilation, mitigate pollution sources, add visual appeal, and provide natural light. They are crucial for maintaining indoor *thermal* conditions and minimizing overall energy consumption. The study aims to assess the connections between the frequency of window openings and closings and the indoor environmental conditions. This research examines how these conditions impact window operation across various seasons. The case study is in Buckinghamshire, England, and features innovative buildings constructed with structural insulated panels. The research examines three residences within this development, utilizing HOBO sensors, including state loggers, to monitor indoor conditions and track window operations at intervals during summer and winter. The results indicate that windows in lower and upper floor spaces facing main streets were closed over 50% of the time due to *noise* pollution, safety concerns, and other factors, with limited openings for only a few hours each day. In contrast, windows in inner street locations remained open for more than half of the monitoring period. Patterns were observed between higher temperatures and window status throughout the buildings. During winter, windows were kept closed to improve indoor conditions and reduce heat loss. This study demonstrates that indoor environmental conditions have a significant influence on window operations. Although the preferred temperatures for residents to open windows varied, this research provides valuable insights into the temperature range occupants prefer for opening and closing windows. These findings could significantly inform future studies on smart window regulations and the implementation of intelligent controls for indoor environmental conditions in buildings.

Conference Paper

69 **Design and Performance Evaluation of a Wearable Cooling Device Utilizing Peltier Technology and Water-Cooling Mechanism**

Basak A.,Hasan M., Shourov A.,Ahmed J., Tahmid A.

Proceedings of the Thermal and Fluids Engineering Summer Conference

2026 0

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Document title	Authors	Source	Year	Citations

Thermal comfort is essential in human productivity, wellbeing, and safety, making comfort engineering a fundamental aspect in modern building and product design. While conventional air conditioning remains the dominant method for maintaining indoor *thermal* comfort, wearable coolers offer a promising alternative because of their energy efficiency, localized temperature control, portability, and reduced global warming risks. The wearable cooler is an innovative *thermal* management solution, especially beneficial for athletes, medical patients, and *outdoor* workers. As global warming and related risks like heatstroke increase, personal body cooling technologies are becoming increasingly important. This study presents an experimental and simulation-based approach for designing a portable, ergonomic neck-band cooling system that utilizes a *thermoelectric* Peltier module for effective heat absorption from human body. The design integrates a *thermoelectric* module, a water circulation system, and an aluminum fin for efficient heat dissipation unlike traditional air blowers or ice-based solutions. The prototype achieves a stable cooling performance reaching by up to 12°C without body contact and can be maintained user comfort around 24.7°C while consuming minimal power. The cooling performance is adjustable by modifying the voltage supplied by a rechargeable battery. Furthermore, using water as the working fluid instead of refrigerants provides a greener environment-friendly alternative. Additionally, lower vibration and *noise* enhance user comfort. This system demonstrates significant potential for practical applications in variable temperature conditions and medical conditions, offering an effective solution for personal *thermal* management.

Conference Paper

70 **Indoor Temperature Prediction Model Based on SVMD-CNN- Informer** — Han Z.,Yang F. — 2026 5th International Conference on Smart Grid and Green Energy Icsgge 2026 — 2026 — 0

Hide abstract Find It @ UCL View at Publisher Related documents

Accurate indoor temperature prediction can effectively optimize the control strategies of heating systems, enhance users' *thermal* comfort, and achieve the goals of energy conservation and carbon reduction. To further improve the accuracy of indoor temperature prediction, this paper proposes a hybrid time-series prediction model based on SVMD-CNN-Informer and applies it to indoor temperature prediction for central heating in northern China in winter. The model first uses SVMD to adaptively decompose the time-series data of indoor temperature and related influencing factors, effectively separating *noise* components and multi-scale fluctuation features. It then employs a convolutional neural network to accurately extract local abrupt features such as sudden changes in *outdoor* temperature and short-term strong winds. Combined with the ProbSparse self-attention mechanism of the Informer model, it captures long-period temperature variation patterns, and realizes the deep fusion of bimodal features through an adaptive weight assignment mechanism. To verify the model performance, relevant experiments were conducted. The results show that the MAE, MSE, RMSE, and R^2 of the proposed model reach 0.37°, 0.03°C, 0.59° C, and 0.962, respectively. Compared with the traditional LSTM, CNN-LSTM, and original Informer models, the prediction accuracy is improved by 10.3%-35.7%, This study provides a basis for the refined regulation of central heating systems, and can serve as an engineering application reference for reducing heating energy consumption, cutting carbon emissions, and ensuring winter livelihood heating.

Conference Paper

71 **Urban Oases in the City: A Human-Centric Assessment of Multi-dimensional Comfort in Different Park Types** — De Quadros B.M., Pigliautile I.,Pisello A.L., ... Mizgier M.O.,Cureau R.J. — Lecture Notes in Civil Engineering , 745 LNCE — 2026 — 1

Hide abstract Find It @ UCL View at Publisher Related documents

The rapid urbanization of cities has increased exposure to environmental stressors such as *noise*, air pollution, and excessive heat while reducing access to nature, impacting public health. Urban parks serve as “urban oases” that mitigate these issues, but their design faces challenges in dense urban areas. Despite various strategies, the influence of different park types on multi-dimensional comfort—*thermal*, *acoustic*, and air quality—remains underexplored. This study evaluates pedestrian comfort in two distinct parks—an elevated urban park and an artificial island—compared to a high-traffic street. Using environmental measurements, walking paths were selected to ensure comparable conditions. Findings indicate that both parks enhance comfort, each excelling in different aspects: the

	Document title	Authors	Source	Year	Citations

elevated park is praised for its air quality, while the artificial island is noted for its *acoustic* conditions. Both parks significantly improve air quality, reducing PM2.5 concentrations relative to the street, and provide more stable *thermal* conditions. While the street exhibited a warming effect of 1.2 °C compared to meteorological stations, the parks offered cooler environments. The artificial island also proved quieter than the elevated park. This integrated approach highlights how urban design influences comfort, demonstrating the benefits and trade-offs of different park types in mitigating environmental stressors.

Article • *Open access*

72 **Quality Assessment of Privately Managed Public Space: Āgenskalns Market Exploratory Case Study** — Braslins M.,Tisenkopfs T. — Urban Science, 10(1), 33 — 2026 — 0

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This exploratory study addresses the problem of limited research on quality assessments of newly emerging multi-use market formats that function as social hubs and their management as privately managed public spaces. Using Āgenskalns Market, a revitalised multi-use market hall in Riga, as a case study, the authors apply an assessment framework based on Yuri Impens' study on covered food halls, incorporating quality criteria from Vikas Mehta's Public Space Index and the UN-Habitat's Site-Specific assessment methodology. Leclercq et al.'s works on privatisation of public spaces are integrated in the analysis of "publicness". This framework evaluates user and observer perceptions across four dimensions: environmental quality and comfort, accessibility and amenities, social experience, and market offer. Data comprised an online survey of 318 respondents and 21 structured observations conducted during summer in 2024 and 2025. The preliminary results suggest users perceive the market as a well-maintained, aesthetically pleasing, accessible space, while identifying room for improvement regarding restroom facilities, indoor *thermal* regulation, *noise* mitigation, *outdoor* weather protection and parking arrangements. As for meaningful use and promoting sociability, findings highlight that flexible seating areas that allow high degrees of temporary personalisation and appropriation, alongside tailored programming and diverse activities beyond retail and dining, play an important role in attracting and retaining diverse audiences. While pricing concerns were noted for specific product groups, exclusionary effects appear to be counterbalanced by openness and inclusivity of cultural programmes and free events. The findings contribute to broader urban scholarship discussions calling for new typologies that better capture the changing character of public space use. This research suggests that private-public partnerships involving multiple stakeholders can enhance "publicness" by promoting inclusivity and social life through accessible infrastructure, diverse activities and free events, as well as enabling opportunities for temporary appropriation by users.

Article • *Open access*

73 **Multisensory Interactions in Greenway Plazas of Differing Openness and Effects on User Behaviors** — Peng Z.,Liu W.,Teng M., ... Keyhani A.B.,Wang P. — Urban Science, 10(1), 60 — 2026 — 3

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Spatial openness affects the subjective evaluation of soundscape, landscape, and *thermal* perceptions, leading to various restoration effects and recreational behaviors. However, the literature lacks studies investigating the effects of multisensory interactions under different levels of spatial openness in plazas on users' behaviors in urban greenways. Thus, this study contributes to the enhancement of recreational experiences and the environmental design of urban greenways by examining the interaction between multisensory evaluations and recreational behaviors in greenway plazas with different levels of spatial openness. Three types of plazas (enclosed, semi-enclosed, open) were selected along an urban greenway to analyze interactions through in situ measurements, questionnaires, and behavior observation. The results showed that people rated the environment as the quietest and coolest in enclosed plazas, although the *sound* pressure level of these plazas was the highest. Furthermore, the visual evaluation (VE) was mostly correlated with *acoustic* evaluation (AE) in plazas with high openness, while the correlation effect between AE and *thermal* evaluation (TE) was only significant in enclosed plazas. In other words, AE was the key factor targeting the improvement in comfort in greenway plazas. Secondly, improving AE was more effective for stimulating the frequency of interactive activities in enclosed plazas, compared to improving TE. However, AE had a negative effect on the time that people were willing to spend on interactive activities in semi-

	Document title	Authors	Source	Year	Citations

enclosed plazas. Finally, these findings provide corresponding strategies for creating comfortable audio, visual, and *thermal* environments in greenway plazas with different levels of openness, as well as strategies for enhancing the recreational experiences of visitors.

Book Chapter • *Open access*

74 **A Case Study on the Thermal Performance of Materials in an Earthen House in the Anoual Region, Morocco: Towards Smart Buildings and Durability**

Laaouar B.,Horma O., Drissi M., ... Moussaoui M.A., Mezrhab A.

Green Energy and Technology , Part F2003

2026

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Raw earth, an ancient and sustainable building material, has demonstrated its durability and strength through enduring structures, many of which are designated as World Heritage sites. This eco-friendly material not only offers excellent *thermal* and *acoustic* performance but also requires significantly less energy to produce compared to modern construction materials. Furthermore, it is fully recyclable, positioning it as a key material in the context of sustainable development. In Morocco, traditional earthen construction techniques such as rammed earth, adobe, and compressed earth blocks are still in use; however, these practices rely on disappearing artisanal knowledge and lack formal technical standards. This chapter examines an earthen house in the Anoual region of Morocco constructed with compacted earth walls. Indoor temperature measurements were compared with *outdoor* temperatures using a Testo device to evaluate the building's *thermal* performance. Additionally, the *thermal* properties of the soil samples—including *thermal* conductivity, *thermal* diffusivity, and specific heat capacity—were characterized to assess the material's behavior under various conditions. Computational Fluid Dynamics (CFD) simulations using ANSYS software were also conducted to validate the experimental results, providing a comprehensive analysis of the *thermal* efficiency of earthen materials in construction. This chapter highlights the contribution of earthen materials to the development of smart and sustainable buildings.

Article • *Open access*

75 **Thermal Comfort Assessment in Faculty Offices of a High-Performance Educational Building in Canada: A Mixed-Methods Approach**

Mazandarani M., Ting D.S.K.,Stagner J.A.

Indoor Air, 2026(1), 9639599

2026

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Single-occupant faculty offices remain underexplored in indoor environmental quality research, despite extensive studies on classrooms and open-plan offices. This study provides case-specific exploratory evidence on how spatial orientation and *acoustic* conditions influence *thermal* satisfaction in a LEED-Gold academic building. Offices were classified as street-oriented, corridor-oriented, or void-oriented based on their exposure to *outdoor* streets, internal corridors, or the central atrium. Over a 40-day summer period, temperature, relative humidity, and airspeed were continuously monitored, and daily surveys captured occupants' *thermal* and *acoustic* perceptions. Three offices representing the three orientations were instrumented, and 12 faculty members participated. Corridor-oriented offices showed the warmest and most humid conditions, with mean values of 29°C and 74.6% RH, exceeding ASHRAE 55-2020 thresholds. Void- and street-oriented offices maintained more moderate conditions. A significant association between *acoustic* satisfaction and *thermal* comfort ($\rho > 0.60$, $p < 0.05$) was observed, suggesting that sensory dimensions may reinforce one another. Because of the small sample size ($n = 12$), all findings should be interpreted as exploratory. Even so, the results provide empirical evidence that spatial microclimates and *acoustic* perception shape comfort in high-performance educational offices. Practical implications include refining HVAC zoning, improving *acoustic* control, and strengthening occupant-centered post-occupancy evaluation strategies.

Article • *Open access*

76 **ENVIRONMENTAL QUALITY OF TRANSITIONAL SPACES AND STUDENT BEHAVIOUR ON A UNIVERSITY CAMPUS**

Ziyi Z.,Marzbali M.H.

Planning Malaysia, 24(4)

2026

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Transitional spaces (*outdoor* and semi-open areas) are increasingly used by university students for informal learning and social interaction; however, these spaces are often overlooked in campus planning, resulting in environmental

Document title	Authors	Source	Year	Citations

conditions that may not support learning-related behaviours and user satisfaction. This study aims to evaluate the environmental qualities of campus transitional spaces and examine how these qualities relate to students' behaviours and satisfaction in a tropical university setting. A case study was conducted at the Universiti Sains Malaysia (USM) main campus using on-site observation and a questionnaire survey (n = 300). Environmental quality was assessed across six dimensions (colour, size, temperature, ventilation, *acoustics*, and lighting), while students' behaviours were measured through learning outcomes and interactions, together with satisfaction regarding environmental quality. Descriptive and inferential analyses show that temperature and lighting are the most problematic aspects across the study areas and are associated with students' behaviours and satisfaction. Higher temperatures, unstable *thermal* conditions, and insufficient lighting are linked to reduced learning outcomes and lower satisfaction in transitional spaces. These findings suggest that improving transitional spaces requires a system-level approach rather than isolated upgrades, and they support planning actions to integrate transitional spaces into campus public realm optimisation strategies for tropical university campuses.

Article • *Open access*

77 **Research on AE method for thermal fatigue damage in heat transfer oil pipes** — Hu Z.,Luo X.,Zhang J., ... Lu C.,Li X. — Frontiers in Physics, 14, 1783743 — 2026 — 0

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Thermal oil pipes in concentrated solar power systems face prevalent risks of *thermal* fatigue damage during long-term *outdoor* service, while the dynamic monitoring of such structures remains an unaddressed gap. This study develops a dynamic monitoring and evaluation method for *thermal* fatigue damage using *acoustic* emission (AE) technology. Firstly, three types of U-shaped specimens are designed and fabricated from 20# steel, considering the material and structural characteristics of actual heat transfer oil pipes. Secondly, an inner-wall *thermal* cycle loading device is developed to simulate the real service environment, aligning with the actual *thermal* fatigue stress mode of the pipes. Finally, an AE detection experimental system is established; *thermal* fatigue loads are applied to the specimens via the loading device, and AE detection experiments are conducted. The study focuses on analyzing AE characteristic parameters during two stages of *thermal* fatigue, establishing the intrinsic correlation between signals and the damage process, with results validated against X-ray detection. Findings indicate distinct differences in AE characteristics across damage stages: The initiation stage is dominated by low-energy signals, while energy significantly increase during the propagation stage, the average energy increment is approximately 4, 200 kpJ. X-ray detection confirms the effectiveness and accuracy of AE technology in characterizing *thermal* fatigue damage. This research demonstrates the promising application of AE technology for monitoring *thermal* fatigue damage in heat transfer oil pipes, providing experimental and theoretical support for online monitoring and service life prediction of such critical components.

Article • *Open access*

78 **Advanced drone detection via ground-based RGB surveillance using efficient deep learning architectures** — Nguyen H.,Van V.D., Ka Duong M. — Journal of Information and Telecommunication, 10(3) — 2026 — 3

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The rapid proliferation of unmanned aerial vehicles (UAVs) in civilian airspace has introduced serious security challenges. Unauthorized drones can disrupt airport operations and may also be exploited for illicit activities, creating an urgent need for reliable detection systems. Sensor-based counter-drone solutions such as radar, radio frequency, *acoustic*, and *thermal* sensing have been widely studied, but each modality can degrade under specific conditions. Vision-based detection with standard RGB cameras is attractive due to low cost and rich visual cues. However, detecting small drones in complex *outdoor* scenes is difficult because distant micro-drones occupy only a few pixels and are often confused with birds or background clutter, causing missed detections and false alarms. This paper proposes the Hybrid Efficient Drone Detector (HEDD), an efficient deep learning framework for robust small-drone detection in RGB video. HEDD introduces an edge-aware feature extraction module that enhances object contours from single-camera input, combined with a lightweight convolutional backbone and multi-scale feature fusion. This design improves sensitivity to tiny targets across distances while preserving real-time throughput.

	Document title	Authors	Source	Year	Citations

Experiments on the DUT Anti-UAV benchmark and a custom surveillance dataset show that HEDD achieves high accuracy, low false-alarm rates, and competitive or superior performance compared with state-of-the-art methods.

Conference Paper

79 **Properties of the Mortar Composites Containing Waste Polycarbonate Particles: A Preliminary Study**

Moretti E.,Proietti M.G., Li T.,Nobili A.

Lecture Notes in Civil Engineering , 778 LNCE

2026

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Sustainability is a key issue in the building and construction sector, which is credited as the largest emitter of greenhouse gases, accounting for around 37% of global emissions. Among many pathways to sustainability, the possibility of adopting end-of-life materials as lightweight/insulating agents appears very promising given that it contributes to reducing the pressure on virgin resources and provides a new market for waste materials, that are usually disposed of by incineration or landfilling. This is particularly true for plastic waste, which is harmful in many ways, some of which have only recently been studied and appreciated. In this work, we present the preliminary results of *thermal*, *acoustic*, and mechanical testing of specimens of mortar-based composite materials that incorporate end-of-life polycarbonate. Polycarbonate is a *thermoplastic* polymer that finds many applications considering its optical properties. In this study, polycarbonate is obtained from grinding end-of-life clear sheets that are used for roofing. As a result, the polycarbonate is severely degraded by atmospheric agents and is unsuitable for remixing with virgin polycarbonate. Two degraded polycarbonate qualities are considered, which differ by the amount of exposure to UV rays (indoor vs *outdoor* applications). We show that the resulting material retains significant mechanical properties while acquiring interesting *thermal* insulation capabilities, which makes it a sustainable alternative for lightweight flooring solutions.

Book Chapter

80 **Differential Pressure Effect on Air Flow Through the "Takhtabush"**

Mohamed M.

Sustainability Resilience and Inclusiveness Through Human Centred Architecture and Design

2026

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People have tried from ancient times to adapt their buildings to the harsh climate in the hot–dry zone by reducing heat impacts. They used to open their houses onto a private internal open space that was visually and *acoustically* separated from the outside, called Sahn "The courtyard." The courtyard helps maintain cooled indoor temperatures by employing the stack effect phenomenon. Clean, oxygen-rich air for breathing is human life's most pressing environmental requirement. The Takhtabush, "A covered *outdoor* sitting area at ground level," was introduced to the traditional courtyard to ensure the airflow. It is located between the courtyard and the back garden, opening completely onto the courtyard and through a Mashrabiya onto the back garden, which ensures a steady flow of air by convection. Since the back garden is bigger and has more sunlight, it will heat up faster than the courtyard. The sun's radiation will make air movement and create a cool air flow in the Takhtabush, leading to the courtyard. However, the orientation of the building is also a big matter when using Takhtabush. Knowing the good wind and the sunlight orientation will surely affect the way it works. This chapter aims to investigate the effectiveness of the size of the backyard and its exposure to solar radiation on the *thermal* performance of the courtyard on the urban scale. The expected results confirm quantitatively and qualitatively the importance of employing the Takhtabush with the exposed-to-sun backyard to enhance the *thermal* performance of the open courtyard.

Article • *Open access*

81 **Dynamic objective prioritization-driven optimization method for infrared adversarial patches | 动态目标优先级驱动的红外对抗贴片优化方法**

Wang J.,Miao Z.,Hou Y., ... Wang J.,Li Y.

Journal of Image and Graphics , 31(10)

2026

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Document title	Authors	Source	Year	Citations

Objective With the growing adoption of deep learning in infrared object detection, such detectors have become core technical support for key applications including nighttime warning in autonomous driving, nocturnal security monitoring, and long-range body-temperature screening in epidemic prevention. Their detection accuracy and resistance to interference are therefore closely tied to operational safety and system reliability. Infrared object detectors are widely used because they are not limited by visible-light conditions and can penetrate certain partial occlusions. Nevertheless, like other deep-learning-based vision systems, they remain vulnerable to security threats. In particular, the emergence of adversarial examples has posed serious challenges to the robustness and security of infrared object-detection models. As an important tool for evaluating and improving model security, research on adversarial examples is therefore of great significance. Existing work on infrared adversarial examples mainly falls into two categories: digital adversarial attacks and physical adversarial attacks. Although digital attacks can achieve high success rates, the perturbations exist only in the digital domain and cannot be directly transferred to the physical world. Current research on physical attacks is concentrated mainly in the visible-light domain; for infrared imagery, the unique imaging mechanism limits high-level semantic cues such as texture and color, which makes physical adversarial optimization substantially more difficult. Although existing studies have explored this area actively, white-box methods remain the most effective attack setting because they can exploit model structure and parameter information. However, current approaches still suffer from two major limitations. The first is imbalance in multi-objective optimization: most methods use fixed weights to balance attack effectiveness and physical realizability, and when the optimization difficulty of achieving strong adversarial effects greatly exceeds that of maintaining physical feasibility, the process can easily lead to insufficient attack strength or poor real-world applicability. The second is inadequate physical constraint design: existing methods typically consider only basic constraints such as patch size and shape, while neglecting pixel-level consistency in the digital domain, which often causes the resulting patches to fail under real infrared imaging conditions. Method We propose a dynamic objective prioritization-driven optimization method for infrared adversarial patches. First, we introduce two regularizers: aggregation regularization and sparsity-binarization regularization. Aggregation regularization uses a graph-theoretic approximation of the local clustering coefficient (LCC) to force mask pixels into a compact and continuous geometry, while sparsity-binarization regularization drives mask values toward a bimodal distribution with sharp boundaries, thereby facilitating material cutting and fabrication. Next, to avoid the optimization imbalance caused by fixed loss weights, we devise a dynamic objective prioritization strategy. Key performance indicators (KPIs) are defined to quantify the current optimization difficulty of the attack loss, aggregation loss, and binarization loss. Focal loss (FL) is then used to reallocate computational resources: difficult objectives receive larger gradients, whereas easier ones are down-weighted, ensuring that the solver focuses on the most challenging constraints at each stage. Finally, we design a dual-projection optimization algorithm that combines projected gradient descent (PGD), tailored for adversarial tasks, with a momentum-based weighted update. The first projection constrains the updated mask to the feasible set immediately after each gradient step, preventing wasted gradients; the second projection recalibrates the mask after momentum fusion, smoothing local gradient *noise* and promoting global convergence under strict physical constraints. Result Experiments were conducted primarily on the FLIR ADAS v1.3 dataset, with additional validation on FLIR ADAS v2.0 and LLVIP. YOLOv3 was used as the target detector, and attack effectiveness was evaluated in both digital and physical settings. In the digital domain, the average attack success rate (ASR) reached 75.08%, and the average precision (AP) of YOLOv3 across datasets dropped to 36.18%, significantly outperforming state-of-the-art methods. Moreover, the proposed method achieved the best ASR and AP across datasets, demonstrating strong generalization. To verify cross-model robustness, we further tested the method on detectors with different architectures, including single-stage models (YOLOv5 and YOLOv8), a two-stage detector (Faster R-CNN), and the Transformer-based DETR. Except for DETR, the ASR exceeded 74% on all models, and the ASR against Faster R-CNN reached 78.90%, confirming strong transferability across detector architectures. In physical experiments, heat-insulating gel was used as the *thermal*-insulation material. The position and shape of the adversarial mask were optimized digitally, and the material was then cut into physical patches for testing in both indoor controlled environments and *outdoor* scenes. Indoors, the ASR reached 98.5% in the frontal view and remained above 84.5% at ±30° under changes in angle, distance, and posture. Notably, even a single patch maintained a high success rate at left and right viewing angles of 30°. In complex *outdoor* environments, the ASR still reached 90.5%, indicating that the proposed method can retain high attack effectiveness under diverse physical interference using only a single patch. Conclusion By integrating dynamic objective prioritization with a dual-projection optimizer, the proposed infrared physical adversarial patch method resolves the core conflict between attack aggressiveness and physical realizability. The dynamic strategy balances multiple objectives, the dual-projection optimizer ensures strict constraint satisfaction and stable convergence, and the dual-regularization design reduces fabrication difficulty. The

	Document title	Authors	Source	Year	Citations

method achieves strong digital attack effectiveness and cross-model robustness while maintaining stable performance in complex physical scenarios, thereby striking an effective balance between attack strength and deployment cost. It provides a reliable red-team tool for security evaluation of infrared detection systems and offers important insights for the design of countermeasures, especially for safeguarding infrared perception modules in autonomous driving and security applications.

Article • *Open access*

82 **High-quality fusion of infrared and visible images by integrating frequency information with atmospheric scattering models | 融合频率信息与大气散射模型的高质量红外与可见光图像融合**

Song C.,Hu J.,Jin Q. | Journal of Image and Graphics , 31(2) | 2026 | 0

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Objective In the domain of image processing, infrared and visible image fusion (IVIF)holds a pivotal position. Its main aim is to synthesize a single image that can optimally integrate the complementary aspects of infrared and visible light images. Visible light images are rich in texture details and color information, while infrared images offer valuable *thermal* radiation characteristics. By fusing them, we can create a comprehensive and useful visual representation that is applicable in numerous scenarios such as surveillance, autonomous driving, and remote sensing. However, real-world atmospheric conditions pose significant challenges. During the transmission of light from the source to the sensor, it undergoes various alterations due to atmospheric scattering and energy attenuation. Such alterations lead to issues including light energy loss, scattering effects that blur images, and inaccurate color contrast. In complex *outdoor* or dynamic environments where the atmosphere has a strong influence, these problems become pronounced, resulting in a significant degradation of image quality and making it difficult to obtain accurate and useful fusion results. Hence, developing a method that can effectively overcome these atmospheric-induced limitations is crucial to enhance the overall quality and practicality of fused images. Method To address these challenges, we propose an innovative IVIF framework. This framework combines the physical model of atmospheric scattering with frequency-domain feature components. First, within the atmospheric scattering model, we focus on accurately estimating and predicting two critical parameters: transmission map and atmospheric light. The transmission map reflects how light propagates through the atmosphere and gets attenuated, while the atmospheric light represents the ambient light conditions in the scene. By precisely determining these parameters, we can deeply understand the impact of the atmosphere on images and use this knowledge to enhance visible light images. This enhancement helps in reducing the negative effects of energy loss and scattering. Additionally, to counteract the artifacts and texture loss that might occur as a result of applying the scattering model, we integrate a Fourier transform and a spatialchannel attention mechanism. The Fourier transform allows us to analyze the images in the frequency domain, where we can selectively amplify the amplitude and phase features. Meanwhile, the spatial-channel attention mechanism helps in focusing on the most relevant regions and features in the spatial and channel dimensions, ensuring that the texture fidelity of the fused images is improved and the fine details are well preserved. Result We conducted extensive experiments on four publicly available datasets, namely, RoadScene, TNO, M3FD, and VT5000, along with an extreme foggy scene dataset (AWMM-100K). The outcomes of these experiments were quite compelling and provided solid evidence of the superiority of our proposed method. In the qualitative comparison, our method outperformed the existing state-of-the-art fusion techniques, especially when dealing with complex scenes. The fused images generated by our approach boasted clear object contours, which is of great significance for subsequent object recognition and detection tasks. Moreover, these fused images had a high visual quality, presenting a visually appealing and information-rich appearance. They were able to vividly showcase the details that mattered, making them conducive to in-depth analysis and practical applications. Quantitatively, on the RoadScene and TNO datasets, we carried out a meticulous comparison with other deep learning methods in terms of several key metrics that are crucial for evaluating image quality and fusion performance. Specifically, in the aspects of information entropy, standard deviation, spatial frequency, mean gradient, peak signal-to-*noise* ratio, and correlation coefficient, our method demonstrated remarkable advantages. Detailed calculations and comparisons revealed that compared with other deep learning methods, our method showed increased mean values in these six important indicators by 7. 44%, 44. 22%, 97. 89%, 91. 01%, 59. 15% and 83. 68%. Such significant improvements in these metrics clearly indicated that our method had a strong ability to enhance image quality from multiple dimensions and was highly capable of supporting high-level visual tasks such as object detection. Notably, in the

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extreme foggy scenes of the AWMM-100K dataset, which are particularly challenging because of the heavy influence of fog, our method shone brightly. It effectively mitigated the adverse effects of fog, significantly reducing the blurriness that is commonly associated with foggy conditions. By doing so, it enhanced the clarity of the fused images and managed to produce higher-quality fused images that were more suitable for further analysis and application in various fields. Overall, the experimental results comprehensively demonstrated the effectiveness and superiority of our proposed IVIF framework in different scenarios and laid a solid foundation for its potential real-world applications. Conclusion Our study has introduced a novel approach to IVIF by integrating traditional atmospheric scattering models with deep learning techniques. Through the combination of Fourier transforms and spatial-channel attention mechanisms, we have achieved enhanced image fusion accuracy and improved preservation of critical image details. This approach offers a promising solution for real-world applications in the field of computer vision. The ability to mitigate the effects of atmospheric scattering, along with advanced frequency-domain processing, enables our frame work to deliver superior fusion results. Consequently, it is highly suitable for various high-level tasks such as object recognition, segmentation, autonomous driving, remote sensing, and security surveillance in dynamic environments. The potential of our proposed approach for practical use across diverse fields is truly significant. It sets a new benchmark for future research in this area, inspiring researchers to conduct further investigations and improvements in IVIF. This breakthrough will likely lead to the development of more effective and efficient methods for dealing with complex atmospheric conditions and attaining enhanced visual representations, thus advancing the entire domain.

Article • *Open access*

☐ 83 **The role of campus green spaces in improving microclimate, reducing noise, and achieving thermal comfort | ةرارحلا ةحارلا ةيقحتو ءاضوضلا ليلقتو ير غصلأا خانملا نيسحت يف يعماجلا مرحلل ءارضخلا تاءاضفلا رود**

Alojany Z.O., Al-Zurfi M.T.H., Abdulrazzaq Z.M.

Basrah Journal of Agricultural Sciences , 38(2)

2025

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The importance of campus green spaces lies in their planting with various plants (trees, shrubs, green spaces, etc.) which work to purify the air, reduce the intensity of solar radiation and temperatures, increase humidity in the atmosphere, and decrease wind speed, which results in achieving *thermal* comfort for campus occupants. This is what the results of the study proved, as the temperature, amount of solar radiation, relative humidity, and wind speed under the canopy of trees in the gardens planted with grass in the morning and afternoon were 36.7 and 37.1 C°, 1031 and 1551 candelas, 43.3%, 34.3%, 0.9, and 1.2 m s^{-1}respectively, compared to what was measured in the paved streets recorded41.6 and 45.3 °C, 5587 and 6933 candelas, 23%, 20.6%, 2.9, and 2.8 m s^{-1}respectively, At midday, the microclimatic contrast between tree-lined areas and paved streets manifested in mean differences of up to 8 C° in air temperature, 5382 candelas in solar radiation intensity, 13.7% in relative humidity, and 1.6 m s^{-1}in wind velocity. The results also showed that plants had a significant impact on reducing *sound* intensity (*noise*) under the tree canopy in turf-planted gardens in the morning and afternoon, with levels of 52.7 and 51.8 dB, compared to levels of 66.7 and 65.6 dB, respectively. Plants were more effective at absorbing *sounds* at higher frequencies. The results thus demonstrate the significant impact of plants in improving the microclimate and reducing *noise* in *outdoor* environments.

Article

☐ 84 **Comfort assessment in outdoor spaces supported by dynamic environmental monitoring: A comparison between different travel conditions**

Kruger E.,Cerutti S., Rosa L.,Boeno M.

Science of the Total Environment , 1006, 180860

2025

3

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This study investigated *outdoor* comfort during climate walks by comparing subjective and physiological responses of 24 participants at designated stopping points and while moving. In accordance with the study objectives, two participant groups were evaluated: active participants (walkers) and sedentary participants (using an adapted wheelchair). Mobile data collection and thematic mapping were employed to evaluate objective environmental parameters. *Thermal* maps indicated that both natural and artificial shading reduced ambient temperatures. Elevated *noise* and carbon dioxide (CO2) concentrations were detected near road junctions, where traffic contributed

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to environmental disturbances. Confounding variables affecting physiological data included minor differences in body mass and clothing among participants. Skin temperature differences reached statistical significance ($p < .01$), although the relationship was counterintuitive, suggesting that a within-subjects design is more appropriate than between-subjects or pairwise comparisons. Analysis of variance (ANOVA) revealed statistically significant differences between subjects at $p < .01$ for responses to four of seven questionnaire items. The effect sizes, measured by Cohen's omega squared, ranged from weak to medium. Walkers reported greater *thermal* discomfort and dissatisfaction. These results underscore the necessity of accounting for physical activity levels when evaluating environmental comfort, as metabolic rate influences environmental perception.

Article

85 **Research on summer microenvironment and human comfort in the space under viaducts in humid and hot regions | 湿热地区高架桥下空间夏季微环境及人体舒适度研究**

Wu J.,Liu Y.,Zhong Z., ... Guo L.,Ding G.

Xi an Jianzhu Keji Daxue Xuebao Journal of Xi an University of Architecture and Technology , 57(6)

2025

1

Hide abstract Find It @ UCL View at Publisher Related documents

In order to explore the microclimate conditions of the space under viaducts in high-density urban areas in hot and humid areas in summer and its response to human *thermal* comfort, the space under viaducts on urban expressways within the second and third rings of Fuzhou City was used as the research object, based on a comprehensive investigation of the spaace under the bridge, the environmental types were divided into different surface types, greening rates and plant community structures. By conducting targeted measurements of objective microenvironment factors at various observation points under the bridge, combined with three human comfort evaluation indicators of *outdoor thermal* comfort, *noise* pollution, and air quality, this study explores their correlation with various microenvironment factors. The results show that : (1) The microenvironment of the space under the bridge reflects the characteristics of heat and humidity, and the air temperature, black ball temperature, and relative humidity are significantly affected by the form of the underlying surface; plants play an obvious cooling and humidifying role in the space under the viaduct, and the greening rate increases the higher and richer the community structure is, the stronger the cooling and humidifying effect;(2) The *thermal* neutral temperature of the space under the Fuzhou bridge in summer is 27. 7 °C, and the *thermal* comfort range is 25.4℃~30.2℃; (3) The human comfort under the bridge at each measuring point shows the characteristics of morning and evening comfort and noon heat, and the more single the vegetation. *Thermal* comfort of the human body is lower if the type of chemical and underlying surface is poorer; the *thermal* comfort of the human body has a very significant positive correlation with the solar radiation intensity, air temperature and black ball temperature among the microenvironmental factors, and has a very significant negative correlation with the relative humidity and *noise*; (4) The *noise* index under the bridge in summer has a very significant negative correlation with the air temperature and black ball temperature. The higher the temperature under the bridge, the faster the *noise* propagates; (5) There is an extremely significant positive correlation between the inhalable particulate matter PM2. 5 and PM10 in the air, and relative humidity and wind speed have a great impact on air quality. Finally, based on the research results, optimization suggestions are put forward to improve the micro-environment under the bridge in terms of greening, paving materials, etc. This study will help to understand the microenvironmental profile of the space under viaducts of different types of underbridge spaces in hot and humid areas, and provide a theoretical basis for studying the relationship between various microenvironment factors and human comfort, *noise* index, and air quality. It will also provide a basis for more. It provides a good reference for optimizing the micro-environment of the viaduct public space and improving *thermal* comfort.

Article • *Open access*

86 **Lessons learned from developing the worldwide IEQ guidelines database**

Clinchard S.,Maula H., Jacobs P.,... Toyinbo O., Haverinen-Shaughnessy U.

Indoor Environments , 2(4), 100124

2025

4

Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

Indoor environmental quality (IEQ) guidelines and standards aim to ensure consistent, safe and healthy indoor spaces by providing clear benchmarks for air quality, lighting, *thermal* comfort and *acoustics*. Many countries and organizations have established their own IEQ regulations and standards or guidelines, but discrepancies in parameters, definitions, methods, and updating frequency often cause them to lag behind the latest scientific and technological advancements, potentially limiting their effectiveness. Researchers and practitioners advocate for the simplification and unification of IEQ guidelines, though risking overlooking geographical and cultural specificities. To address these challenges, and to help preserve and use current knowledge, an open database compiling worldwide IEQ guidelines was developed, ensuring equitable access to up-to-date information. The database covers four key IEQ domains, indoor air quality (IAQ), *thermal* comfort, *acoustics*, and lighting, as well as two related domains: ventilation and *outdoor* air quality (OAQ). Ventilation is a critical factor influencing most of the other IEQ domains, while OAQ guidelines are sometimes used in the absence of indoor-specific guidelines. Each of the six resulting tables includes both mandatory governmental regulations and voluntary guidelines, along with internationally recognized standards for broader relevance. This paper summarizes key lessons learned from database development and data collection efforts over the past five years. Practical IEQ guidelines and standards balance health and wellbeing outcomes with environmental, economic, and comfort considerations. Guidelines and standards can work best when science-based, adaptable to regional contexts, and structured with clear, measurable parameters, such as exposure limits, timeframes, and validated measurement methods. Collaboration between scientific research, technological advancements, and policy development can ensure regulations remain effective and up to date. To increase compliance and raise public awareness, the database consolidates and shares existing international guidelines and standards, supporting global alignment with best practices while accommodating resource constraints and regional specificities.

Conference Paper • *Open access*

87 **Experimental and Simulation Analysis of IEQ and Energy Demand in an Office Building in Northern Greece** — Papadopoulos G., Panaras G. — E3s Web of Conferences , 669, 01001 — 2025 — 1

Hide abstract Find It @ UCL View at Publisher Related documents

The present study aims to assess both the indoor environmental quality (IEQ) and the energy demand of an office building in Kozani, Western Macedonia, Greece. *Thermal* comfort was evaluated by analysing questionnaire responses and by applying both the Fanger and adaptive models, while CO2 concentration served as an index of indoor air quality (IAQ). Illuminance (lux) readings and *acoustic* measurements were also collected to assess the lighting and *acoustic* conditions, respectively. The IEQ assessment indicates that the Fanger model has limited applicability in naturally ventilated or mixed-mode spaces; CO2 concentrations fluctuated at low levels due to natural ventilation; and the *acoustic* and lighting indicators were within acceptable limits. A simulation study was conducted to assess the building' s energy consumption under different cooling and ventilation scenarios, as well as the impact of climate zone on cooling loads. The analysis shows that natural ventilation achieves greater reductions in cooling loads in climates with lower *outdoor* temperatures, and that the ventilation period is also critical for energy savings. Overall, the proposed approach combines experimental and computational tools and can be further developed into an integrated methodology to ensure occupant comfort and health at the lowest possible energy cost.

Article

88 **Optimization and field evaluation of a detachable thermal insulation fabric for rail temperature mitigation** — Park J.,Kang D.,Kim H.-S. — Construction and Building Materials , 500, 144205 — 2025 — 1

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Rail temperature control is a critical factor in ensuring the safety and reliability of railway systems, especially under extreme summer heat conditions. Conventional countermeasures such as water spraying systems and solar-reflective coatings have demonstrated limited applicability due to high installation and maintenance costs, short duration of cooling effects, and practical constraints in field deployment. As a novel solution, this study introduces and validates a detachable *thermal* insulation fabric designed for steel rails, consisting of a heat-reflective fluoropolymer-coated glass fiber fabric and embedded magnets for easy installation and reuse. Through controlled laboratory tests, the

	Document title	Authors	Source	Year	Citations

optimal coating thickness (100 μm) and effective application area (rail web only) were determined. Field application over two summer seasons on an actual railway track demonstrated an average temperature reduction of 4–5℃ and a maximum of 9.7℃, particularly effective during peak solar radiation periods. Compared to conventional coatings, the fabric exhibited similar or superior *thermal* mitigation performance, with significantly improved reusability and maintenance convenience. A one-year *outdoor* exposure test confirmed minimal performance degradation (~3 %) primarily due to UV-induced discoloration. This study is the first to provide a comprehensive and quantitative validation of detachable *thermal* insulation fabric for rail applications. The results suggest that this fabric-based approach offers a practical, durable, and scalable alternative to existing methods, with broader applicability to urban infrastructure, *noise* barriers, and steel bridge decks under *thermal* stress.

Article

89 **Comprehensive effects of thermal and acoustic environments on overall comfort in urban parks of cold regions in China** — Wu X.,Yuan C.,Xue H.,Yan Y. — Energy and Buildings, 347, 116329 — 2025 — 9

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Urban parks play a crucial role in enhancing residents' quality of life; however, research on the combined effects of *thermal* and *acoustic* environments on subjective evaluation remains limited. This study explores these interactions in cold-region urban parks, focusing on People's Park in Zhangjiakou, China. A combination of field measurements and questionnaire surveys was conducted across three representative spaces characterized by three common *sound* types: human-generated, artificial, and natural *sounds*. The results indicate that, in terms of *thermal* perception, natural *sounds* significantly reduce heat stress by lowering the *thermal* sensation vote and improving *thermal* comfort, whereas artificial *sounds* exacerbate it. Additionally, high *sound* pressure levels (L_{Aeq} ≥ 60 dB(A)) exacerbate *thermal* discomfort, making individuals feel hotter in summer and colder in winter, thereby reducing *thermal* comfort. Regarding *acoustic* perception, a *thermally* comfortable environment enhances the *acoustic* sensation vote and improves *acoustic* comfort. For overall comfort, during summer and transitional seasons, the overall comfort vote is negatively correlated with both the Universal *Thermal* Climate Index (UTCI) and *sound* pressure levels, while in winter, overall comfort vote is positively correlated with UTCI but negatively correlated with *sound* pressure levels. The analysis of equivalent overall comfort vote further reveals that in summer and transitional seasons, rising temperatures under extreme heat conditions (UTCI ≥ 35°C) have a more pronounced negative impact on overall comfort vote than decreasing temperatures in winter. Conversely, in winter, variations in *sound* pressure levels exert a stronger influence on overall comfort vote compared to those in summer and transitional seasons. This study addresses a research gap in understanding the interaction between *thermal* and *acoustic* environments in cold-region urban parks and provides valuable insights for the future planning and optimization of urban green spaces in such climates.

Article • *Open access*

90 **The Nexus Between Indoor and Outdoor Environmental Conditions and Teacher Perceptions in Naturally Ventilated Primary School Classrooms, in Ireland** — Honan D.,Garvey J.,Littlewood J.,Horrigan M.,Gallagher J. — Sustainability Switzerland, 17(21), 9873 — 2025 — 3

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Indoor air quality (IAQ) and *thermal* comfort influence the health and cognitive performance of school occupants. This study investigated carbon dioxide (CO_2), nitrogen dioxide (NO_2), *thermal* comfort, and ventilation rates (VRs) in eight naturally ventilated (NV) primary school classrooms in Ireland during October 2024, combining environmental monitoring with teacher surveys. Mean CO_2 concentrations ranged from 796 ppm to 2469 ppm, exceeding national guidelines in seven of the eight classrooms. NO_2 levels ranged from 3.4 μg/m^3 to 7.2 μg/m^3, with indoor/*outdoor* ratios increasing with VRs and influenced by window orientation and road proximity. Indoor temperatures remained within recommended limits, while relative humidity ranged from 53% to 78% mirroring CO_2 trends and exceeding guideline levels in classrooms with lower VRs and temperatures. Occupied VRs ranged from 1.2 L/p/s to 4.1 L/p/s with window opening behaviours, reliant on teachers' perceptions of *thermal* comfort, accounting for 84% to 96% of VRs. Ventilation in NV classrooms is often insufficient, yet increasing VRs can compromise *thermal* comfort and increase ingress of *outdoor* pollutants and *noise*. The findings highlight the ineffectiveness of current school ventilation

	Document title	Authors	Source	Year	Citations

standards, which rely heavily on user operation. Integrating occupant-led strategies, including scheduled purging, awareness campaigns, and pre-emptive air quality alerting, into policy offers practical, immediate pathways to improving IAQ, fostering healthy, sustainable learning environments.

Article • *Open access*

91 **Design of low-noise high-frame-rate short-wave infrared imaging system | 低噪声高帧频短波红外成像系统设计**

Ren T.,Jia J.,Yu Y.,... Ma S., Chen S.

Guangxue Jingmi Gongcheng Optics and Precision Engineering , 33(21)

2025

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To meet the demands for low *noise* and high linearity in ground-based,large-aperture shortwave infrared (SWIR) imaging systems and to circumvent international constraints on scientific-grade SWIR imaging while expediting project timelines,a cooled SWIR imaging system based on an InGaAs focal plane array(FPA)was developed. The system architecture was first established,followed by the design of a low-*noise* hardware circuit incorporating *thermoelectric* cooling(TEC)to minimize readout *noise* and dark current. An FPGA-based sensor drive circuit and an optimized image readout and data-transmission program were implemented to realize a prototype. Experimental results indicate stable operation at - 10 ℃ with a maximum full-frame output rate of 116 frames per second. In low-,medium-,and high-gain modes,*noise* levels of 364. 87 e^- ,40. 66 e^- ,and 15. 73 e^- were measured,respectively,together with excellent linearity and response uniformity. The system demonstrates robust performance and produces high-quality images in both indoor and *outdoor* environments,satisfying specifications for scientific-grade detectors.

Conference Paper • *Open access*

92 **Research on Multi-dimensional Evaluation and Low-carbon Update Strategy of Outdoor Environment in Existing Residential Areas from the Perspective of Green Building Standards ——An Integrated Optimization Method Combining BIM Technology**

Yu Q.,Guo J.

Proceedings of 2025 2nd International Conference on Digital Society and Artificial Intelligence Dsai 2025

2025

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The environmental performance enhancement of existing residential areas, as the primary component of urban building stock, represents a pivotal strategy for realizing the dual-carbon targets. Conventional retrofitting methodologies, however, are constrained by over-reliance on empirical practices, resulting in fragmented data ecosystems and suboptimal multidisciplinary coordination. This research addresses these challenges by pioneering an integrated framework that synergizes Building Information Modeling (BIM) technology with green building evaluation standards, specifically tailored for residential communities in severe cold climates. The proposed methodology establishes a holistic workflow encompassing digital twin creation, multi-physics simulation, precision diagnostics, and iterative optimization.Leveraging Revit's parametric modeling capabilities, the study constructed a BIM platform that harmonizes geometric, material, and operational data, transcending conventional 2D design limitations. Coupled simulations utilizing SVE software enabled the concurrent analysis of wind flow patterns, *acoustic* propagation, *thermal* dynamics, and solar irradiance, creating an interactive digital analysis matrix. Quantitative assessments revealed critical environmental deficits: 12% of the residential area was identified as summer windless zones, with banking facilities exceeding *noise* standards by 9 dB(A), and a complete absence of green roof infrastructure.Targeted interventions comprised three-dimensional greening systems integrated with architectural reconfiguration strategies. Computational fluid dynamics (CFD) simulations optimized aerodynamic tree formations, achieving an 80% reduction in windless areas. Stratified *acoustic* vegetation buffers demonstrated 14 dB(A) *noise* attenuation, while comprehensive roof greening systems incorporating vertical gardens and sky farms increased vegetation coverage to 57%. Post-optimization analyses indicated a 42% improvement in overall environmental performance, coupled with 28% reduction in building energy consumption and 35% decrease in carbon emission intensity.This study demonstrates BIM's transformative potential in orchestrating data-driven retrofitting processes, validating its integrated advantages in complex urban ecosystems. The research delivers a

	Document title	Authors	Source	Year	Citations

scalable technical paradigm for enhancing residential environments in cold climates, offering quantifiable pathways toward sustainable urban transformation and carbon neutrality.

Conference Paper

93 **MFB-YOLO: A high-precision algorithm for moving fluid boundary gas cloud detection**

Yan B.,Chen H.,Zeng D., Li S.

Proceedings of SPIE the International Society for Optical Engineering , 13672, 1367205

2025

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To address the challenges in infrared remote sensing for gas leakage detection in complex industrial environments - such as irregular gas cloud shapes, unstable boundaries, difficulties in achieving high-precision detection, and high false alarm rates caused by multisource interferences (e.g., personnel and vehicles) in gas cloud recognition within infrared images - this paper integrates infrared-visible fused images with the well-developed deep learning technology for visible object detection. Targeting moving fluid boundary objects, we propose a high-precision gas cloud recognition algorithm based on the YOLO architecture: Moving Fluid Boundary YOLO (MFB-YOLO). Built upon the YOLOv8 framework, MFB-YOLO optimizes feature learning mechanisms to address the highly irregular and unstable nature of gas cloud boundaries. First, in gas detection scenarios, stable boundary interferences (e.g., personnel and vehicles) are eliminated from fused images to achieve differentiated recognition of gas clouds. To fully capture spatial distribution features of targets, a Global Attention Mechanism (GAM) is introduced during the feature fusion stage. This employs coordinated channel-spatial attention and dynamic weight allocation strategies, establishing two convolutional submodules: a channel attention submodule and a spatial attention submodule. The channel attention submodule reduces computational load through channel-wise recalibration, enhances feature channels relevant to gas targets, and suppresses *noise* from irrelevant channels. The spatial attention submodule applies spatial recalibration to cover regions where gas clouds predominantly exist, preventing overfitting to shape features by small convolutional kernels. Together, these mechanisms significantly improve gas cloud detection precision. First, ablation experiments were conducted to analyze the contributions of each module while validating the algorithm. *Outdoor* field experiments for HFC-152a gas detection were also performed. The experimental results demonstrate that the MFB-YOLO algorithm achieves approximately 10% higher precision than YOLOv8 with comparable recall rates. The mAP:50 metric improved by about 55%, reaching 84%. Although the single-sample response time increased by roughly 30%, the total processing time remains merely 0.08s - considerably less than the acquisition time of a single-frame infrared image (3s). Compared to traditional YOLO methods, MFB-YOLO enhances gas cloud recognition precision through infrared-visible image fusion and Global Attention Mechanism. This advancement increases the practicality of infrared remote sensing in complex environments and holds significant scientific and engineering value for applications such as industrial gas leak alarming.

Article • *Open access*

94 **Indoor air quality of academia-related workshops based on health complaints**

Awad A.H.,El-Gendy S., Saeed Y.,Kamal S.

Trakya University Journal of Natural Sciences , 26(2)

2025

3

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Indoor Air Quality (IAQ) is a result of the interaction between micro-environmental conditions, location, and building characteristics. IAQ directly affects human health, comfort, productivity, and performance. However, very little attention has been paid to the IAQ of nonindustrial workshops. This cross-sectional survey aimed to determine the IAQ of academia-related workshops based on the factors such as the microbial load (including bacteria, fungi, and actinomycetes), particulate matter (PM) content, presence of chemical pollutants (such as ammonia [NH_3], volatile organic compounds [VOCs], and formaldehyde [HCHO]), and physical conditions (such as temperature [T°C], relative humidity [RH%], light intensity, *noise*, dewpoint and air speed). Moreover, the perception weights of IAQ factors affecting the indoor comfort condition were also examined. A two-stage viable, Andersen cascade impactor, was used by suctioning air onto the selective culture media. The PM content was determined by using a preweighted membrane filter. Portable air quality monitors were used to estimate the chemical and physical factors. A questionnaire survey

	Document title	Authors	Source	Year	Citations

was employed to assess the health complaints and the participants' perception weights on the indoor environmental parameters (such as *thermal*, *acoustic*, visual environment, and air quality). The concentrations of mesophilic bacteria, fungi, and actinomycetes were found to be higher indoors than outdoors, with indoor/*outdoor* (I/O) values of 3.13, 1.56, and 1.53, respectively. The Global Index of Microbial Contamination/m^3 exceeded 7,000 colony forming units/m^3 in approximately 46% of the workshop areas. The I/O ratios of PM, VOCs, HCHO, and NH_3 were 1.69, 1.52, 0.65, and 0.6, respectively. T°C and RH% values ranged 18–35°C and 40–56%, respectively. *Noise* values exceeded 70 dBA in both the indoor and *outdoor* environments. Light intensity was also unacceptable (≤ 300 lux) at 84.6% of the workshop areas. VOCs and dewpoint revealed significant positive and negative effects on microbial viability, differing with regard to the microbial type. Fatigue (45.5%), allergies (38.6%), and headache (35.2%) were the common complaints of the occupants. All of the tested IAQ parameters influenced the workplace environment, with *noise* ranking as the main factor (40.9%). Microbial air quality is differently associated with the indoor environmental factors. The IAQ in the workshops was poor and potentially affected the occupant's well-being. The perception of comfort varied among the occupants under the same IAQ factors. Thus, corrective actions based on comparative analysis should be implemented to promote the indoor quality of even nonindustrial and academia-related workplaces.

Article

95 **Optimizing urban block morphology for photovoltaic power and thermal comfort in hot and humid regions** — Bai B.,Li T.,Wang S.,Yan H.,Dong J. — Engineering Applications of Artificial Intelligence, 158, 111377 — 2025 — 7

Hide abstract Find It @ UCL View at Publisher Related documents

This study explores the influence of urban block shape on solar energy potential and *outdoor thermal* comfort. A multi-objective optimization model was developed to improve both photovoltaic efficiency and *thermal* comfort in hot and humid regions. Using Rhino/Grasshopper parametric modeling, a digital prototype of a typical block in Foshan Shishan Town was created. Performance was evaluated through coupled simulations with Ladybug Tools. Results show that the Non-dominated Sorting Genetic Algorithm II (NSGA-II) produced Pareto front solutions. The north-south high-low bench layout increased photovoltaic efficiency by 18 % and improved *thermal* comfort by lowering summer temperatures by 0.8–1 °C. Additionally, placing water bodies in the southern and central parts of the block reduced ambient temperatures by 1.2–2.0 °C. Based on these findings, the study proposes guidelines for low-carbon community planning in hot and humid areas. The suggested layout—"multi-layer (south) to high (middle), multi-layer (north)"—can achieve 12–15 % energy-carbon sink synergy and a 20–25 % reduction in air conditioning load. Density-Based Spatial Clustering of Applications with *Noise* (DBSCAN) analysis shows that this framework has an 84 % applicability to planning schemes in similar climates. The study offers both algorithmic innovations and practical solutions for low-carbon community development.

Article • *Open access*

96 **A Combined Weighting Method to Assess Indoor Environmental Sub-Factors for Human Comfort in Offices in China's Severe Cold Regions** — Li Z.,Song G.,Zhang Q.,Yu J.,Liu Y. — Buildings, 15(19), 3529 — 2025 — 3

Hide abstract Find It @ UCL View at Publisher Related documents

Indoor environmental quality in offices, comprising *thermal*, *acoustic*, lighting, and air quality domains, is known to influence human comfort, yet the relative importance of their sub-factors—particularly in severe cold regions—remains unclear. This study addresses this gap by integrating objective (Criteria Importance Through Intercriteria Correlation, CRITIC) and subjective (Analytic Hierarchy Process, AHP) weighting methods, supported by field measurements and questionnaire surveys in open-plan offices in three provinces in northeastern China. Cluster analysis categorized *acoustic* sub-factors into *outdoor* traffic, *outdoor* entertainment, people conversation, burst *sound*, and people movement. Results show that temperature is the dominant *thermal* comfort driver (39.7% CRITIC; 45.5% AHP), exceeding air velocity and humidity, which had nearly equal influence. Indoor *sound* exerted greater impact than *outdoor sound*, with people conversation ranked highest among indoor *noise* sources, and burst *sound* and movement showing similar but slightly lower weights. Natural light outweighed artificial light in importance

	Document title	Authors	Source	Year	Citations

(54.2% CRITIC; 61.0% AHP), while air freshness and pollution were nearly equally influential. Compared to CRITIC, AHP produced more dispersed weights, reflecting subjective bias toward pronounced differences. These findings provide a quantitative basis for prioritizing environmental design interventions—such as controlling indoor conversational *noise*, optimizing natural lighting, and stabilizing temperature—to enhance comfort in offices in severe cold regions.

Article • *Open access*

97 **Outdoor Comfort Optimization in Historic Urban Quarters: From Multisensory Approaches to Operational Strategies Under Resource Constraints** — Su H.,Ma H.,Liu K. — Buildings, 15(19), 3616 — 2025 — 1

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During the transition from urban expansion to renewal, optimizing environmental comfort under resource constraints presents critical challenges. While existing research confirms that multisensory interactions critically shape environmental comfort, these insights are rarely operationalized into protocols for resource-constrained contexts. Focusing on historic urban quarters that need to balance modification and preservation, this study quantifies multisensory (*acoustic*, visual, *thermal*) interactions and integrations to establish operational resource-optimization strategies. Through laboratory reproduction of 144 field-based experimental conditions (4 *sound* sources × 3 *sound* pressure levels × 4 green view indexes × 3 air temperatures), systematic subjective evaluations of *acoustic*, visual, *thermal*, and overall comfort were obtained. Key findings demonstrate: (1) Eliminating extreme comfort evaluations (e.g., “very uncomfortable”) within any single sensory domain stabilizes cross-sensory contributions to overall comfort, ensuring predictable cross-domain compensations and safeguarding resource efficacy; (2) Accumulating modest improvements across ≥2 sensory domains reduces per-domain performance threshold for satisfactory overall comfort, enabling constraint resolution (e.g., visual modification limits in historic districts); (3) Cross-domain optimization of environmental factors (e.g., *sound* source and air temperature) generates mutual enhancement effects, maximizing resource economy, whereas intra-domain optimization (e.g., *sound* source and *sound* pressure level) induces competitive inefficiencies. Collectively, these principles establish operational strategies for resource-constrained environmental improvements, advancing sustainable design and governance through evidence-based multisensory approaches.

Article • *Open access*

98 **Mitigating overheating in renovated Danish residential buildings: The potential of implementing louvred openings** — Bugenings L.A., Andersen K.P.,Rong L., Kamari A. — Journal of Building Engineering, 111, 113034 — 2025 — 4

Hide abstract Find It @ UCL View at Publisher Related documents

As *outdoor* temperatures rise and heat waves become more frequent, the absence of overheating prevention measures and impending renovations pose challenges to summer *thermal* comfort in residential buildings. Manual window opening, as the preferred method to address high temperatures, comes with concerns such as increased safety risks, potential *noise* disturbances, and vulnerability to adverse weather conditions. To address this, a combined system of louvre and ventilation hatches (named louvred opening hereafter) is proposed as an alternative to energy-intensive air conditioning and traditional window ventilation. This system enables ventilation even during occupants' absence, adverse weather conditions, and potential *noise* disturbances. The study investigates the implementation of louvred openings in existing residential buildings and their potential to mitigate overheating during the warm season. The systems’ impact on load-bearing structure, architectural value, daylight availability, and overheating is analysed using the Danish building stock and two Danish case study buildings. Additionally, influential factors such as occupant behaviour and opening characteristics are scrutinised in the overheating study. The findings revealed that the load-bearing structure and the architectural value of post-1950 Danish buildings are well-suited for incorporating louvred openings. However, placing louvred openings in existing window spaces significantly reduces daylight availability, indicating that buildings from earlier construction periods with smaller glazed areas may necessitate adding new external wall openings. Nevertheless, louvred openings demonstrate superior performance in mitigating overheating compared to conventional window ventilation, especially when occupants utilise night ventilation effectively.

	Document title	Authors	Source	Year	Citations
	Article • *Open access*				
99	**A repeated cross-sectional study of a biophilic building's effects on workers' perception of indoor environmental quality, health, well-being and productivity**	Jones R.V.,Fuertes A., Bilverstone P.A.	Journal of Building Engineering , 111, 113476	2025	5

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The study aimed to investigate the sustained effects of a case study biophilic designed building, The Enterprise Centre (TEC) in the UK, on its permanent workers' satisfaction with the indoor environmental quality (IEQ) and their health, well-being and productivity. A repeated cross-sectional research design, following the Building Use Studies (BUS) Methodology, was used with two data collection periods 12 and 24 months after building occupation with participation rates of 86 % and 97 % respectively. Results showed high and sustained satisfaction with IEQ, including *thermal* comfort, *acoustics*, lighting, and air quality. Workers consistently reported improved health and well-being, with productivity gains of 12.98 % amongst the 12-months cohort and 9.17 % for the 24-months cohort. This study contributes novel insights by: (1) providing evidence of biophilic design's sustained impact over time, an area with limited existing research; (2) evaluating combined biophilic strategies at full-building scale (including natural materials and colours, connection with nature, *outdoor* views, natural light and air, and climate-aligned indoor conditions) rather than isolated biophilic interventions; and (3) establishing new insights into the effects of biophilic design on occupant health, well-being, and productivity. The positive outcomes suggest that biophilic design could be successfully integrated into future buildings and spaces to improve satisfaction with IEQ and enhance workplace health and performance.

	Article • *Open access*				
100	**Operational Performance of an MVHR System in a Retrofitted Heritage Dwelling: Indoor Air Quality, Efficiency and Duct Constraints**	Giraldo-Soto C., Azkorra-Larrinaga Z., Uriarte A., Romero-Anton N., Odriozola-Maritorena M.	Sustainability Switzerland , 17(18), 8493	2025	4

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The integration of Mechanical Ventilation with Heat Recovery (MVHR) systems into heritage buildings poses a series of challenges, largely attributable to architectural constraints and conservation requirements. The present study offers an operational campaign of an MVHR system installed during the energy retrofit of a protected residential heritage dwelling in Vitoria-Gasteiz, Spain. Although environmental monitoring was carried out throughout the year, representative spring, autumn and winter days of continuous operation were analysed, as the occupants frequently avoided using the system due to *noise* perception. This limitation highlights the importance of considering *acoustic* comfort and user acceptance as critical factors in the long-term viability of MVHR in heritage contexts. The system was assessed under real-life conditions using continuous environmental monitoring, with a focus on indoor air quality (IAQ), *thermal* efficiency, airflow balance, and pressure losses. Despite the acceptable mean apparent *thermal* effectiveness (0.74) and total useful efficiency (0.96), the system's performance was found to be constrained by significant flow imbalance (up to 106%) and elevated pressure drops, which were attributed to the legacy of the duct geometry. The results obtained demonstrate IAQ improved overall, with mean CO_2 concentrations below ~650 ppm across the analysed dataset; however, daily means occasionally exceeded 900–1000 ppm during high-occupancy periods and in the absence of spatially distributed demand control. These exceedances are consistent with the measured *outdoor* baseline (~400–450 ppm) and reflect the need for post-commissioning balancing and room-level sensing to sustain Category II performance in heritage dwellings. This study provides empirical evidence on the limitations and opportunities of MVHR deployment in historic retrofits, thus informing future guidelines for sustainable interventions in heritage contexts.

	Article • *Open access*				
101	**Study protocol for an observational panel study of heat strain in the general adult population in Basse Santa Su, The Gambia**	Tadiri E.,Saucy A.,Bonell A., ... Minor K., Vicedo-Cabrera A.M.	Plos One, 20(9 September), e0332238	2025	0

Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

Heat is among the most hazardous environmental factors for human health, but humidity's role in heat-related health effects remains unclear. This study will assess the effect of humid heat and other environmental conditions on health in a representative population in Basse Santa Su, The Gambia, a region at high risk of humid heat exposure. We will examine the association between humid heat exposure and physiological heat strain, identify vulnerable sub-groups, and evaluate adaptive behaviours. We will recruit 60–90 healthy adults from Basse Santa Su and surrounding areas. Participants will be monitored for four non-consecutive weeks across dry (November–May) and rainy (June–October) seasons. Daily questionnaires will assess activities, *thermal* comfort, adaptation behaviours, heat strain symptoms, mood, and sleep quality. Wearables will collect time-resolved personal and indoor exposure (temperature and humidity), heat strain, and further physiological covariates. A fixed monitoring network will measure *outdoor* air temperature, humidity, air quality, and environmental *noise*. Descriptive analyses will assess baseline characteristics, heat stress and heat strain. Case-time series analysis with distributed non-linear lagged models will estimate immediate and delayed associations between exposure to humid heat and physiological heat strain. Stratified analyses by individual characteristics will explore possible vulnerability groups. Multiple exposure models and interaction terms will explore cumulative effects of multiple environmental factors. Multilinear land use regression modelling will develop high-resolution maps of temperature, humidity, and heat stress. This study will provide new insights into humid heat's effect on health, particularly in low-income, high-exposure settings. This study addresses limitations in prior epidemiological research on heat, humidity, and health, including lack of high-resolution and individual-level data, and limited focus on humidity as a heat-health driver, on non-mortality outcomes and on climate-vulnerable populations. This study combines high-resolution microclimate mapping and individual-level measurements which may inform future epidemiological studies and heat-health interventions.

Article

☐ 102 **Seeking comfort in the urban heat: Unraveling the effects of thermal-acoustic environments on psychological restoration in urban greenways** — Gao M.,Zhu X. — Building and Environment , 282, 113269 — 2025 — 19

Hide abstract Find It @ UCL View at Publisher Related documents

As environmental pressures from high-density urbanization increase, the comfort of *outdoor* spaces and their restorative benefits for urban residents have become increasingly important. Although *thermal* and *acoustic* environments are key factors influencing *outdoor* comfort, evidence linking *thermal* and *acoustic* comfort with restorative benefits is scarce. This study explores the impact of the *thermal-acoustic* environment on psychological restorative benefits in urban greenways through on-site meteorological measurements and questionnaire surveys. The results indicate that, during the hot season, compared to densely vegetated greenways, sparsely vegetated greenways are perceived as hotter and noisier. Furthermore, the relationship between the *thermal-acoustic* environment and restorative benefits is more strongly correlated in sparsely vegetated greenways than in densely vegetated ones. In other words, the environmental comfort in different types of greenways can be targeted for improvement. Secondly, the interaction between *thermal* and *acoustic* environments influences restorative benefits. At the same *sound* pressure levels, restorative benefits decrease as the Universal *Thermal* Climate Index (UTCI) increases. However, on sparsely vegetated greenways, under conditions of 70–75 dB(A), the increase in UTCI does not significantly affect restoration. Finally, to identify the factors hindering psychological restoration, the study compares the veto power of *thermal-acoustic* comfort voting (TCV, ACV). This veto power varies according to greenway type. In densely vegetated greenways, ACV has greater veto power than TCV, while in sparsely vegetated greenways, the reverse is true. The findings offer valuable insights for urban planners and policymakers, suggesting that creating comfortable *thermal-acoustic* environments in *outdoor* spaces can enhance restorative benefits and contribute to the sustainability of high-density urban areas.

Article

☐ 103 **A multi-domain simulation approach for enhanced indoor environmental quality in offices: Challenges and opportunities** — Loeser B.D.C., Bavaresco M.,Melo A.P. — Energy and Buildings , 341, 115829 — 2025 — 10

Hide abstract Find It @ UCL View at Publisher Related documents

	Document title	Authors	Source	Year	Citations

As occupants experience simultaneous stimuli from multiple environmental domains, integrating multi-domain simulation practices can significantly enhance decision-making in the building design phase. This study introduces a multi-domain simulation workflow, merging *thermal*, visual, *acoustic*, and air quality domains to evaluate IEQ in offices. A case study uncovered significant challenges in integrating the various domains, underscoring the need for better interoperability between simulation tools. Systems, boundaries, and fluid dynamics shape the interplay between *thermal*, *acoustic*, and air quality. Besides, the *thermal* and visual domains emphasise the need for a holistic approach that also considers the influence of *outdoor* climate on indoor environments. This comprehensive approach supports initial design processes while highlighting the relevance of detailed multi-domain interactions.

Article • *Open access*

☐ 104 **Impact of atmospheric turbulence on shock detection using background-oriented Schlieren** — Liu S.,Ren Y.,Mei H., ... Ma X.,Rao R. — Optics Express, 33(16) — 2025 — 2

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Background-Oriented Schlieren is a widely adopted optical technique for flow visualization, with increasing applications in aero-optical diagnostics. However, its *outdoor* performance is often affected by atmospheric turbulence, which distorts flow-induced features. To quantify this influence, we present a simulation framework based on anisoplanatic image degradation, which not only models the coupled effects of turbulence and shocks, but also supports the extraction of flow field information. Simulation results reveal that even moderate turbulence induces nonlinear degradation in shock visibility, with a threshold beyond which reliable detection fails. Contrary to traditional assumptions, smoother natural backgrounds enhance shock detectability under turbulence due to lower gradient *noise*. These findings provide guidance for BOS deployment in airborne and remote sensing scenarios.

Review • *Open access*

☐ 105 **Enhancing Outdoor Environmental Comfort: A Review of Façade-Surface Strategies and Microclimate Impacts** — Khan Z.,Ghiai M. — Buildings, 15(16), 2829 — 2025 — 5

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Building façades traditionally focus on enhancing indoor environmental quality and improving energy performance, but undermine their influence on *Outdoor* Environmental Comfort (OEC), including *thermal*, *acoustic*, and visual conditions. With technological advancements in envelope design, research on new materials and green systems has been introduced in the last few decades. This review examines the role of two key elements—façade materials and green façades—in shaping OEC. A total of 41 peer-reviewed studies (24 on urban scale and 17 on building scale) were categorized into three focus areas: (1) *outdoor thermal* comfort; (2) *outdoor acoustic* comfort; and (3) *outdoor* visual comfort. The analysis was structured across three levels: (a) Performance Determinants; (b) Metrics/Models; and (c) Material or Façade Types. We proposed this analytical structure to highlight the interactions between building façades and OEC domains (*thermal*, *acoustic*, and visual comfort). Our results showed façade treatment can impact all three comfort factors related to OEC, but trade-offs must be evaluated. Moreover, the findings highlighted that additional research is required to cover variations in both climate and context conditions, due to their close association with the OEC. Finally, the conceptual framework is presented to synthesize the three comfort domains for sustainable *outdoor* environments.

Article • *Open access*

☐ 106 **Impact of personal and contextual factors on multi-domain sensation in classrooms: Results from a field study** — Morandi F.,Gasparella A., Tzempelikos A.,Pittana I., Cappelletti F. — Building and Environment, 280, 112892 — 2025 — 7

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In indoor spaces shared by many people, such as school classrooms, the contextual and individual characteristics can affect the perceived indoor environmental quality and explain the distribution of the occupants' subjective comfort evaluations. Understanding the impact of these factors on the subjective assessment of the indoor environmental quality (IEQ) and multi-domain comfort is crucial for reliable post-occupancy evaluations and retrofit solutions. This work presents the impact of personal (i.e., gender, clothing level, body mass index, age, use of glasses) and contextual (i.e., sitting position inside the classroom, facade orientation/*outdoor* view) factors on the students' *thermal*/air

Document title	Authors	Source	Year	Citations

quality/visual/*acoustic* sensation votes collected in 50 classes during regular lectures, through the administration of 825 questionnaires. Students' sensation votes regarding the four perceptual domains were collected and correlated with environmental parameters (i.e., indoor and *outdoor* air temperature and relative humidity, indoor mean radiant temperature, indoor CO_2 concentration and TVOC, indoor horizontal illuminance, indoor *sound* pressure level). Multi-linear regression analyses were performed to identify environmental variables significantly correlated with sensation votes, exploring cross-domain interactions. Then, the sensation votes were grouped according to different levels of personal and contextual factors to study correlations between votes and related binned environmental parameters for each set, by means of covariance analysis and post-hoc tests. Among personal factors, age and gender were proven to impact on the sensation votes in all the four domains. Furthermore, Body Mass Index and clothing level influence *thermal* sensation while the use of glasses do not affect visual sensation. The sitting position in the classroom impacts *thermal*, IAQ, visual sensation votes in relation to the proximity to the windows and the visual and *acoustic* sensation in relation to the proximity to the teacher and blackboard. Finally, groups identified by classroom orientation and occupant density significantly differ in terms of sensation votes.

Article • *Open access*

107 **Predicting the student's perceptions of multi-domain environmental factors in a Norwegian school building: Machine learning approach** — Alam A.G.,Bartonova A., Hoiskar B.A.K., ... Fredriksen T.,Cao G. — Building and Environment, 280, 113144 — 2025 — 9

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Poor Indoor Environmental Quality (IEQ) in schools significantly impacts students' well-being, learning capabilities, and health. Perceived dissatisfaction rates (PD%) among students often remain high, even when indoor environmental variables appear well-controlled. This study aims to predict perceived dissatisfaction rates (PD%) across multi-domain environmental factors—*thermal*, *acoustic*, visual, and indoor air quality (IAQ)—using machine learning (ML) models. The research integrates sensor-based environmental measurements, *outdoor* weather data, building parameters, and 1437 student survey responses collected from three classrooms in a Norwegian school across multiple seasons. Statistical tests were used to pre-select relevant input variables, followed by the development and evaluation of multiple ML algorithms. Among the tested ML models, Random Forest (RF) demonstrated the highest predictive accuracy for PD%, outperforming multi-linear regression (MLR) and decision trees (DT), with R^2 values up to 0.91 for overall IEQ dissatisfaction (PDIEQ%). SHAP analysis revealed key predictors: CO_2 levels, VOCs, humidity, temperature, solar radiation, and room window orientation. IAQ, *thermal* comfort, and *acoustic* environment were the most influential factors affecting students' perceived well-being. Despite limitations as implementation in building level scale, the study demonstrates the feasibility of deploying predictive ML models under real-world constraints for improving IEQ monitoring system. The findings support practical strategies for adaptive indoor environmental management, particularly in educational settings, and provide a replicable framework for future research. Future research can expand to other climates, buildings, measurements, occupant levels, and ML training optimization.

Article

108 **Temperature behaviour Analysis of a GaN LNA under Harsh Environmental Conditions for Telecommunications** — Sharma S.,Sharma S.S., Longhi P.E., ... Serino W.C.A.,Limiti E. — International Journal of Microwave and Optical Technology, 20(4) — 2025 — 0

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This paper illustrates the design of a two-stage low-*noise* amplifier using 0.12 µm GaN HEMT technology for X-band applications in telecommunications. The first stage of the LNA circuit is chosen to have higher gate fingers to handle a large amount of input power. The LNA is biased with a gate voltage of -1 V and a drain voltage of 8 V. To achieve the *noise* figure of less than 1.5 dB, source degeneration feedback topology has been used with an appropriate input/output matching network. LNA survivability is also examined in terms of temperature stress and non-linearity. The measured MMIC low-*noise* amplifier shows the gain greater than 20 dB, Input Return Loss (IRL) / Output Return Loss (ORL) greater than -11 dB and *noise* figure of 1.5 dB in the frequency band from 9 to 11 GHz. The size of the chip is 3.65mm * 2mm. In contrast to traditional designs documented in the literature, this work puts a strong emphasis on high-performance operation, *thermal* reliability, and environmental longevity. The suggested LNA is put through

	Document title	Authors	Source	Year	Citations

extensive stress testing, which includes: temperature tolerance between 25°C and 125°C and resilience of input power up to 40 dBm. The results of these investigations validate the amplifier's strong performance in the midst of severe electrical and environmental stress, which makes it ideal for use in hazardous *outdoor* communications, aerospace, and defense applications.

Article • *Open access*

109 **Performance Evaluation and Simulation Optimization of Outdoor Environmental Space in Communities Based on Subjective Comfort: A Case Study of Minhe Community in Qian'an City**

Rong Y.,Song J.,Xu Z., ... Yang B.,Guo S.

Buildings, 15(12), 2078 2025 2

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With the continual expansion of global urbanization and population growth, urban energy demands have intensified, and anthropogenic activities have precipitated profound shifts in the global climate. These climatic changes directly alter urban environmental conditions, which in turn exert indirect effects on human physiological function. Consequently, the comfort of *outdoor* community environments has emerged as a critical metric for assessing the quality of human habitation. Although existing studies have focused on improving singular environmental factors—such as wind or *thermal* comfort—they often lack an integrated, multi-factor coupling mechanism, and adaptive strategy systems tailored to hot-summer, cold-winter regions remain underdeveloped. This study examines the Minhe Community in Qian'an City to develop a performance evaluation framework for *outdoor* spaces grounded in subjective comfort and to close the loop from theoretical formulation to empirical validation via an interdisciplinary approach. We first synthesized 25 environmental factors across eight categories—including wind, *thermal*, and lighting parameters—and applied the Analytic Hierarchy Process (AHP) to establish factor weights, thereby constructing a comprehensive model that encompasses both physiological and psychological requirements. Field surveys, meteorological data collection, and ENVI-met (V5.1.1) microclimate simulations revealed pronounced issues in the community's wind distribution, *thermal* comfort, and *acoustic* environment. In response, we proposed adaptive interventions—such as stratified vegetation design and permeable pavement installations—and validated their efficacy through further simulation. Post-optimization, the community's overall comfort score increased from 4.64 to 5.62, corresponding to an efficiency improvement of 21.3%. The innovative contributions of this research are threefold: (1) transcending the limitations of single-factor analyses by establishing a multi-dimensional, coupled evaluation framework; (2) integrating AHP with ENVI-met simulation to realize a fully quantified "evaluation–simulation–optimization" workflow; and (3) proposing adaptive strategies with broad applicability for the retrofit of communities in hot-summer, cold-winter climates, thereby offering a practical technical pathway for urban microclimate enhancement.

Article • *Open access*

110 **Infrared Super-Resolution Imaging Using Minimalist Optical Systems | 极简光学系统红外超分辨率成像**

Wang T.,Wang C.,Wu X., Sun H.,Liu J.

Guangxue Xuebao Acta Optica Sinica , 45(11), 1111002 2025 1

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Objective With the rapid advancement of optical imaging technologies, there is an increasing demand for higher resolution, better image quality, and more compact, cost-effective optical systems. In recent years, super-resolution imaging based on digital micromirror devices (DMDs) has attracted considerable attention for its ability to surpass the resolution limits of traditional optical systems. However, the integration of DMDs introduces challenges such as optical eccentricity and tilt-induced aberrations. To mitigate these issues, the design of DMD-based super-resolution imaging systems often requires additional optical lenses and complex surface types to correct aberrations and enhance imaging quality. While this improves accuracy, it reduces energy transmittance and significantly raises manufacturing costs. In this paper, we focus on the simplified design and image quality optimization of such systems, aiming to enhance system performance and imaging accuracy. Methods To meet the needs of simplified design and improved imaging quality in DMD-based infrared super-resolution systems, we examine the computational super-resolution imaging theory based on DMDs, analyze their working mechanisms and applications, and propose a simplified design approach. The original projection lens is then simplified using Zemax software, and the point spread function (PSF) of the modified system is obtained. Building on this, we develop a super-resolution restoration and

Document title	Authors	Source	Year	Citations

image quality optimization model using T-L (TVAL3-Lucy Richardson), along with an algorithm that combines super-resolution reconstruction and image deblurring restoration. This algorithm improves both resolution and clarity in degraded images captured by the simplified system. Simulation, indoor target, and *outdoor* scene imaging experiments are conducted to validate the method's accuracy. The findings offer new insights for designing cost-effective DMD-based optical systems. Results and Discussions To validate the method, two images with a resolution of 512 pixelx512 pixel and two images with a resolution of 144 pixelx144 pixel are selected for simulation experiments (Fig. 8). Based on evaluation metrics (Table 3), the average peak signal-to-*noise* ratio (PSNR) and structural similarity index measure (SSIM) outperform other restoration methods, with PSNR exceeding 27 and SSIM exceeding 0.85. Compared to blurred images captured by the simplified system, PSNR improves by an average of 78%, and SSIM by 71%. The root mean square error (RMSE) decreases by 36% compared to blurred images captured by the simplified system. Indoor target imaging experiments (Fig. 11, Table 4) demonstrate significant improvements after restoration, with PSNR reaching 20.37 and SSIM rising to 0.77, representing improvements of 58.5% and 57.1%, respectively, over the original images. Meanwhile, RMSE drops to 27.46, a 27% reduction. Finally, *outdoor* scene imaging experiments (Fig. 13) demonstrate excellent image enhancement under various distance conditions. Compared to direct super-resolution methods, the proposed approach significantly enhances detail reproduction while maintaining overall imaging stability. The natural image quality evaluator (NIQE) values (Table 5) for two *outdoor* scenes decreases from 8.10 to 5.13 and from 7.75 to 4.09, representing reductions of 36.7% and 47.2%, respectively. Conclusions The proposed method effectively simplifies the structure of DMD-based infrared super-resolution imaging systems, reduces alignment complexity, improves infrared transmittance, and lowers production costs. By iteratively combining super-resolution reconstruction and deblurring, and analyzing convergence, optimal image enhancement is achieved. Theoretically, the TVAL3 algorithm uses total variation (TV) regularization to enable sparse reconstruction while preserving edges and textures, thus reducing artifacts. The Lucy-Richardson algorithm iteratively corrects blur and enhances image sharpness. In simulations, PSNR improves by an average of 78% and SSIM by 71%, compared to directly captured images. In indoor experiments, PSNR improves by 58.5% and SSIM by 57.1%. In *outdoor* scene experiments, the method effectively restores details compared to degraded images captured by the simplified system. The proposed restoration method improves resolution and quality, simplifies system structure, and reduces both cost and operational complexity, facilitating integration and deployment. It is well-suited for applications requiring high portability and low cost, such as portable security monitoring, spaceborne remote sensing, and field exploration.

Article • *Open access*

111 **Noise-tolerance detection of high-frequency dynamic targets with mid-infrared thermal radiation via intra-cavity enhanced upconversion**

Dong S.,Qiao Z.,Hu J., ... Xiao L.,Jia S.

Optics Express, 33(10) 2025 3

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Broadband and highly sensitive mid-infrared (MIR) light detection and imaging play a crucial role in environmental monitoring, gas detection, biomedical applications, and national defense security. However, the detection of high-frequency, long-distance dynamic targets with MIR *thermal* emission is still limited by the long response time of commercial detectors and intense environmental *noise*. Here, we demonstrate a *noise*-tolerance intra-cavity enhanced broadband MIR upconversion system by counting the upconverted lights through a high-sensitivity single photon detector. The intra-cavity enhanced pump power of 1064 nm reaches about 200 W, resulting in a high upconversion efficiency with the MIR spectra ranging from 3 μm to 5 μm. By performing discrete Fourier transformations of the arrival time of each upconverted photon, the frequency-domain imaging manifests a strong *noise*-tolerance feature, even under the ratio of the *noise*-to-signal of up to 100. The excellent performance of our system has also been evidenced by measuring the rotation frequency of a turbojet engine in *outdoor* conditions. Our proposal offers promising applications to recognize long-distance dynamic targets, such as airplanes and unmanned aerial vehicles, in real-world scenarios with the highly desired suppression of intense background *noise* and stray light.

Conference Paper • *Open access*

	Document title	Authors	Source	Year	Citations
112	**Contributions by Richard Raspet on two aspects of the acoustics of rigid framed porous media**	Attenborough K.	Proceedings of Meetings on Acoustics , 56(1), 045001	2025	0

Hide abstract Find It @ UCL View at Publisher Related documents

Two papers co-authored by Richard Raspet are about the surface impedance of grounds with exponential porosity profiles [R. Raspet and J. M. Sabatier, J. *Acoust.* Soc. Am., 99, 147–152 (1996)] and *sound* propagation in porous media containing main pores with porous walls [W. P. Arnott, J. M. Sabatier and R. Raspet, J. *Acoust.* Soc. Am., 90, 3299–3306 (1991)]. The variable porosity ground impedance model enables better fits to *outdoor* short-range measurements over many types of grassland than the widely used semi-empirical single parameter model. Moreover, it has been found to represent the surface impedance of artificial surfaces including felt and perforated rubber mat. A better approximation than that used hitherto for a variable porosity ground is suggested. The theory for pores with porous walls which was developed in the context of *thermoacoustic* stack designs is shown to be related to several designs involving slit like pores with regularly spaced dead end pores, periodically varying widths or microporous walls in the somewhat different context of creating useful *sound* absorbers by additive manufacture.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
113	**A framework for developing adaptive acoustic comfort: Insights from expert interviews**	Dicle S.Y.,Aletta F.,Kang J.	Applied Acoustics, 235, 110642	2025	11

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The ability of individuals to adapt to environmental conditions, known as environmental adaptation, reflects their capacity to modify behaviors and physical surroundings to better align with external demands. In *thermal* comfort assessments, this adaptation process is evaluated using the adaptive *thermal* comfort model. Similarly, the relationship between individuals' adaptability and their perception of *acoustic* environments is also considered to play a significant role in *acoustic* comfort assessments. Based on this foundation, this study investigates the development and assessment of an adaptive *acoustic* comfort model, following on its initial definition: "a model that relates indoor *acoustic* conditions to *outdoor* ones, moderated by other environmental, contextual, and personal factors." Through expert interviews, the research aims to develop a preliminary framework for assessing adaptive *acoustic* comfort, highlighting the importance of environmental adaptation and user tolerance. This framework includes the identification of assessment methods, *acoustic* indicators, and adaptive model variables necessary for testing the model's existence and effectiveness in *acoustic* evaluations. By examining *acoustic* comfort from an adaptive perspective, the study recommends evaluating indoor *acoustic* comfort in relation to *outdoor* conditions. To capture variability in user responses, it suggests having real-world data alongside laboratory studies using specific *acoustic* indicators. Additionally, the model framework emphasizes the necessity of evaluating personal, environmental, and building-related variables through various developed adaptive models.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
114	**3SqAir Project: A Living Lab Towards Sustainable Smart Strategy for Indoor Climate Quality Assurance in Classrooms**	Ogundiran J., Nyembwe J.-P.K.B., Ogundiran J.O., ... Pereira L.D., Gameiro da Silva M.	Atmosphere, 16(5), 584	2025	3

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The indoor climate quality in classrooms at the University of Coimbra, Portugal, was investigated as part of the 3SqAir project, supported by the Interreg SUDOE program. This research focused on two equipped classrooms with different ventilation systems: natural and mechanical ventilation. Both classrooms were continuously monitored for IEQ parameters: *thermal* comfort, indoor air quality, *noise*, and lighting during heating and cooling seasons. Air temperature, relative humidity, CO_2 concentration, particulate matter, nitrogen dioxide, volatile organic compounds, formaldehyde, *sound* pressure level, and illuminance were measured. *Outdoor* weather conditions were also recorded. The primary focus was on air temperature, CO_2 concentrations, and relative humidity, while air change rates (ACH) were estimated using the Tracer Gas Method. The results showed inadequate *thermal* conditions in both classrooms, particularly during the heating season. Most weekly mean CO_2 concentrations were within acceptable limits, while

	Document title	Authors	Source	Year	Citations

ACH were below standard recommendations in four CO_2 decay phases. Simulations of CO_2 decay revealed further air quality gaps in each room. Corrective measures within the 3SqAir project framework were suggested for approval and implementation while monitoring continues. This work represents the first phase in an evolving study towards developing sustainable strategies for improving indoor air quality in classrooms.

Article • *Open access*

115 **Adapting Restorative Urban Design, for Open Spaces Towards, Thermal Heat Island, Reaching Human Mental Health and Well-Being, Case Study: Qaitbay Citadel Plaza, Alexandria, Egypt**

Metwally M.T.,Ali S.S., Mohamed A.F.A.

Future Cities and Environment , 11

2025

5

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Nowadays rapidly evolving urban landscape. These challenges increasingly impact residents' mental health and well-being. Overcrowding, *noise* and air pollution, long commutes, the size of the built environment, landscape and green areas, and the variety of ground surface materials affect *outdoor thermal* comfort in hot climates to reach mental health and well-being. These public spaces are dealing with the issue of an expanding urban heat island, which affects how users behave. This paper aims to support restorative urban design in Alexandria, Egypt, to reach users *thermal* comfort, mental health and well-being at the forefront of public spaces. To attain human mental health and well-being in public spaces. Several studies have been done. Such as: Bowling Questionnaire methodology on about 92 participants to assess user perceptions of Qaitbay citadel plaza according to Gehl's theory. And simulated scenarios to measure material effect on users' *thermal* comfort in the plaza using ENVI-met 4.0 program. As the questionnaire found that: 30% rated accessibility as "poor," and only 75% felt almost unsafe in the area. *Thermal* discomfort was prevalent, with 40% reporting the temperature as "uncomfortable," 63% finding shaded areas inadequate, and 75% attributing discomfort to heat-retaining surface materials. And mitigate urban heat island phenomena. Through the use of different paving material mitigation strategies, green spaces, and the landscape through different scenarios, till reaching the best scenario which decrease the minimum temperature 2.06 °C and the maximum temperature 3.59 °C for users' *thermal* comfort and wellbeing inside the space reaching to a restorative urban public space.

Article • *Open access*

116 **Assessing the Impact of Ambient Noise on Outdoor Thermal Comfort on University Campuses: A Pilot Study in China's Cold Region**

Ning S.,Jing W.,Ge Z.,Qin Z.

Atmosphere, 16(4), 410

2025

3

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This study investigates the impact of different *noise* levels on *thermal* comfort in *outdoor* environments. The research was conducted in two university squares in Xi'an, China, exhibiting distinct *noise* exposures, with twenty volunteers participating in the study. These individuals provided subjective evaluations of *thermal* comfort through questionnaires while situated in environments with disparate *acoustic* conditions in conjunction with the documentation of prevailing meteorological circumstances. The analysis yielded three salient findings. Initially, a marked elevation in perceived warmth was noted in environments experiencing higher *noise* levels, with 35.29% of subjects in the high-*noise* plaza (HP) reporting feeling warm (TSV = 2), which was 11.76 percentage points higher than in the low-*noise* plaza (LP). This included a 5.88 percentage point uptick in the frequency of "hot" (TSV = 3) *thermal* sensations reported in the HP. Furthermore, an intensification of *thermal* discomfort was observed in noisier settings, with the *thermal* comfort vote (TCV) in HP encompassing a spectrum from very uncomfortable to neutral and a predominant 90% of TCVs indicating discomfort, 35.29% of which were deemed very uncomfortable. Lastly, the findings suggest that high-decibel *noise* exposure notably amplifies the perception of heat within a specific high-temperature bandwidth. Beyond this delineated *thermal* threshold, the influence of *noise* on *thermal* sensation substantially diminishes.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
117	**Acoustic Fabry–Perot Resonance Detector for Passive Acoustic Thermometry and Sound Source Localization**	Yue Y.,Dong Z.,Qi Z.-M.	Sensors, 25(8), 2445	2025	1

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Acoustic temperature measurement (ATM) and *sound* source localization (SSL) are two important applications of *acoustic* sensors. The development of novel *acoustic* sensors capable of both ATM and SSL is an innovative research topic with great interest. In this work, an *acoustic* Fabry-Perot resonance detector (AFPRD) and its cross-shaped array were designed and fabricated, and the passive ATM function of the AFPRD and the SSL capability of the AFPRD array were simulated and experimentally verified. The AFPRD consists of an *acoustic* waveguide and a microphone with its head inserted into the waveguide, which can significantly enhance the microphone's sensitivity via the FP resonance effect. As a result, the frequency response curve of AFPRD can be easily measured using weak ambient white *noise*. Based on the measured frequency response curve, the linear relationship between the resonant frequency and the resonant mode order of the AFPRD can be determined, the slope of which can be used to calculate the ambient *sound* velocity and air temperature. The AFPRD array was prepared by using four bent *acoustic* waveguides to expand the array aperture, which combined with the multiple signal classification (MUSIC) algorithm can be used for distant multi-target localization. The SSL accuracy can be improved by substituting the *sound* speed measured in real time into the MUSIC algorithm. The AFPRD's passive ATM function was verified in an anechoic room with white *noise* as low as 17 dB, and the ATM accuracy reached 0.4 °C. The SSL function of the AFPRD array was demonstrated in the *outdoor* environment, and the SSL error of the *acoustic* target with a *sound* pressure of 35 mPa was less than 1.2°. The findings open up a new avenue for the development of multifunctional *acoustic* detection devices and systems.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
118	**Exploring the relationship between elderly women responses, multi-domain factors, and street geometry indicators: A case study in a Mediterranean context**	Mahia L.,Berkouk D., Bouzir T.A.K.,Pigliautile I., Pisello A.L.	Building and Environment , 273, 112647	2025	9

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Urban public spaces must ensure safe conditions for all, including vulnerable groups such as the elderly people. Since environmental characteristics of urbanized areas largely depend on cities' design-related urban geometry, grasping how multi-domain stimuli resulting from this variation impact elderly *outdoor* activities and overall health is fundamental to promoting more sustainable and healthier urban communities. This study combines objective and subjective methods to examine the relationship between multi-domain factors, elderly women's responses, and street geometry during a late afternoon walk in a Mediterranean context, Algeria. Forty-one elderly women engaged in walking activities across different streets, providing subjective feedback on environmental sensation and psychophysical health while being monitored in terms of physiological parameters and environmental conditions. Findings reveal that a high sky view factor, associated with high illuminance mean values and increased sunlight sensation vote (SSV), increased the skin temperature of the forehead (up to 39 °C) and induced an unrelaxed state among elderly women ($P < 0.001$). Moreover, a high height-to-width ratio, associated with high mean values of the physiological equivalent temperature (PET) and increased *thermal* sensation vote (TSV), increased heart rate ($P < 0.05$) and induced a weak state ($P < 0.001$). Additionally, high mean values of the *sound* pressure level (SPL) and increased *acoustic* sensation vote (ASV) caused a weak ($P < 0.001$) and a relaxed state ($P < 0.05$). Furthermore, our study highlighted some health parameters that could be used as predictor indicators of *outdoor* sensations for elderly women who are unable to objectively assess their environment.

Review

	Document title	Authors	Source	Year	Citations
119	**Indoor and outdoor airflow modeling in built and urban environments by water tank and channel experiments: A review**	Wang Y.,Hang J.,Mo Z.	Building Simulation, 18(4)	2025	12

Hide abstract Find It @ UCL View at Publisher Related documents

Water tank and channel experiments have been widely utilized in indoor and *outdoor* airflow studies for their ability to adjust variables and boundary conditions in a controlled environment. This study reviews the experimental setup of water tanks and channels, measurement techniques, and their applications in multi-scale airflow modeling, in

Document title	Authors	Source	Year	Citations

particular indoor airflow, street-canyon flow, city-scale urban heat circulation and meso-scale boundary-layer flows. Water tanks are mainly adopted for *thermally*-dominated airflow studies without considering the background wind, while water channels are applied to dynamic-*thermal* interacting airflow researches. The advantages and disadvantages of commonly adopted measurement techniques in water tank and water channel experiments are summarized in terms of particle image velocimetry (PIV), *acoustic* Doppler velocimeter (ADV), laser Doppler velocimeter (LDV), and magnetic resonance imaging (MRI). Based on the water tank and channel measurements, the exhaled airflow and natural ventilations are major concerns in indoor environments. Wind- and buoyancy-driven flow, ventilation and pollutant dispersion are investigated within and above the street canyons. The urban heat island circulation and the convective boundary layer are intensively studied under different *thermal* and stratification conditions. Current limitations include the oversimplification of building models, the inability to ensure strict flow field similarity, and the neglect of specific dimensionless numbers. Water tank and channel experiments serve as important tools for parametric experiments, allowing for the validation of numerical simulations and comparisons with full scale measurements. In the future, water tank and water channel experiments should more closely align with real atmospheric conditions, such as using more realistic physical models, accounting for actual atmospheric background wind circulations, and considering the non-uniform distribution of heat fluxes in urban areas.

Article

☐ 120 **Mid-Infrared Tunable Diode Laser Absorption Spectroscopy Formaldehyde Analyzer Based on Optical Path Temperature Control | 基于光路温控的中红外可调谐半导体激光吸收光谱甲醛测量仪**

Jiarong L.,Bo F.,Weimin D., ... Weixiong Z.,Weijun Z.

Guangxue Xuebao Acta Optica Sinica , 45(7), 0730002

2025

3

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Objective Formaldehyde (HCHO) is an important air pollutant in both indoor and *outdoor* environments. Its emission sources include the direct releases from industrial emissions, construction materials, and polymeric resin-based furnishings. HCHO can also be generated through the photochemical oxidation of volatile organic compounds (VOCs) in atmospheric environments. Accurate measurement of formaldehyde concentration is crucial for health effect studies, atmospheric chemistry research, and pollution prevention. Tunable diode laser absorption spectroscopy (TDLAS) is one of the important methods for detecting formaldehyde, which offers advantages such as high sensitivity, high time resolution, and in-situ measurement. However, temperature variation in the operating environment can affect the key parameters of a TDLAS system, thereby impacting system stability and measurement sensitivity. To address this issue, two primary temperature control methods are usually employed. One involves directly controlling the temperature of the multi-pass gas cell (MPC) to maintain optical path length stability. The other entails temperature control of the entire optical path of the TDLAS system, including the multi-pass cell, laser, photodetector, mirrors, and lenses. Temperature control has been demonstrated to be an effective means to improve the performance of TDLAS systems. However, precise temperature control specifically tailored for formaldehyde measurement systems has not been reported before. Methods A TDLAS system with optical path temperature control is developed for formaldehyde measurement. The instrument uses a mid-infrared interband cascade laser emitting at 3.5 μm as the probe source. The absorption line of formaldehyde is selected at 2831.64 cm^{-1} with a line strength of $S=5.651\times10^{-20}$ $cm^{-1}/(molecule\cdot cm^{-2})$. A compact dense spot pattern spherical mirror cell is developed with an optical base length of 17.7 cm, which allows the incident laser beam to reflect 312 times in the multi-pass cell, thereby increasing the effective absorption path length to 55.2 m, with a volume of only 350 mL. The Gaussian beam formula is used to fit the beam radius at different distances to obtain the beam waist position. A focusing lens (f=500 mm) is employed to adjust the beam waist position. The laser beam waist is matched to the center of the multi-pass cell, which is located 479 mm away from the laser. To enhance measurement precision, wavelength-modulated spectroscopy (WMS) and rapid background subtraction techniques are used to minimize the *noise* in the spectral signal. A semiconductor cooling temperature control box is developed to maintain the operating temperature of the optical system at 32 ℃. Results and Discussions The temperature inside the temperature control box could be stabilized at 32 ℃ with a high precision of 6 m℃ (Fig. 6) when the temperature control is turned on. Rapid background subtraction is performed using a three-way solenoid valve, which enables swift switching between the background gas and sample gas. The total time consumed to complete a background subtraction measurement is 16 s. The sample gas signal is difficult to distinguish accurately due to interference from background structures. However, by subtracting the background gas spectrum, the 2f signal can be extracted, which significantly improves the detection

Document title	Authors	Source	Year	Citations

sensitivity of the system (Fig. 7). During calibration, the absolute concentrations of diluted HCHO gas in the cell are calculated using direct absorption spectroscopy. The peak-to-peak value of the formaldehyde 2f signal exhibits a linear relationship with volume fraction, which yields a correlation coefficient of 0.9994 (Fig. 8). The performance of the TDLAS instrument is evaluated by measuring a time series of formaldehyde concentrations at a fixed volume fraction. With temperature control enabled, the system achieves a detection precision of 0.25×10^{-9} (1σ, 60 s), which is 4.7 times higher than that when temperature control is disabled (Fig. 10). Conclusions The system utilizes a compact, dense spot pattern spherical mirror optical multi-pass cell with a path length of 55.2 m and a volume of only 350 mL to enhance the effective absorption path. The beam profile of the used interband cascade laser emitting at 3.5 μm is measured, and the laser beam waist is matched to the cell. A semiconductor-based *thermoelectric* temperature control box is designed to precisely regulate the entire optical path, including the interband cascade laser, multi-pass cell, and detector, which achieves a temperature control precision of 6 m℃ at the set point of 32 ℃ . By combining wavelength modulation spectroscopy and rapid background subtraction techniques, the system achieves a measurement precision of 0.25×10^{-9} (1σ, 60 s) under temperature-controlled conditions, which is 4.7 times higher than that under non-temperature-controlled conditions.

Article

121 **Multichannel Polarization Diversity Imaging LiDAR | 多通道偏振分集成像激光雷达** Zhang L.,Sun J.,Cong H., ... Lu Z.,Chen W. Zhongguo Jiguang Chinese Journal of Lasers , 52(7), 0710001 2025 4

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Objective Imaging light detection and ranging (LiDAR) utilizes the range values of target point clouds (or pixels) to estimate the three-dimensional (3D) shape of a target. However, ambiguity arises in determining which point clouds (or pixels) belong to the target when it is in cluttered or partially occluded environments, leading to uncertainties in reconstructing the 3D shape. Polarimetry, which examines the interaction between polarized light and materials, is a promising method for improving the determination of target pixels (or point clouds). The Mueller matrix, a comprehensive description of polarization properties, has been demonstrated as a powerful tool for characterizing and distinguishing targets with varying polarization characteristics. However, these demonstrations of Mueller matrix imaging have primarily focused on biomedical applications and have employed charge-coupled device time-sharing imaging. Additionally, the polarization diversity imaging technique has only been reported in synthetic aperture radar systems, based solely on the polarization imaging of target scattering matrices. We propose a multichannel polarization diversity imaging LiDAR that integrates Risley-prism multibeam scanning, frequency-modulated continuous wave (FMCW) coherent detection, and Mueller matrix polarimetry techniques for target classification. The proposed system eliminates ambiguity in determining which pixels (or point clouds) correspond to the target, a challenge faced by conventional imaging LiDAR systems. Methods To reduce artifacts caused by time-sharing polarization imaging, spatial parallelism and beam synchronization techniques are employed to simultaneously capture multipolarization point clouds using Risley-prism multibeam scanning. To achieve high spatial resolution in point clouds, the system incorporates the FMCW coherent detection, which offers high sensitivity, superior resolution, and a large dynamic range while simultaneously measuring distance and velocity. To address the issue of insufficient dimensionality for target recognition in unstructured environments with conventional imaging LiDAR, the Mueller matrix is introduced into LiDAR systems for target identification and classification. Different objects are distinguished and classified by leveraging variations in their polarization parameters. Results and Discussions A long-range 3D polarimetric imaging experiment was conducted using the proposed LiDAR architecture under *outdoor* conditions. The multidimensional information acquisition capability of the multichannel polarization diversity imaging LiDAR was exploited, and various filtering methods were applied to process the raw point clouds with a large field-of-view angle of ±30° (Fig. 3). To further demonstrate the LiDAR's polarization imaging capability, small-scale buildings (A6 and A7) were selected as targets for recognition and classification (Fig. 4). In the cross-polarized point clouds [Fig. 4(b2)], buildings A6 and A7 were roughly divided into two regions: one with polarized reflectance intensities of 53–100 (colored in green) and the other with polarized reflectance intensities of 210–320 (colored in blue-violet). This indicates that the facades of buildings A6 and A7 are composed of two materials with distinct polarization properties. For more precise recognition and classification of building materials and structural details, Mueller matrix imaging was employed. The facades of buildings A6 and A7 exhibit a clear demarcation line (Fig. 5). Additionally, there is no ambiguity in determining which point clouds (or pixels) corresponded to the target. Different polarization parameters

	Document title	Authors	Source	Year	Citations

were utilized to distinguish and classify various targets, effectively enhancing the accuracy of target identification (Figs. 6 and 7). Conclusions We demonstrated both theoretically and experimentally a multichannel polarization diversity imaging LiDAR based on Risley-prism multibeam scanning, FMCW coherent detection, and Mueller matrix polarimetry techniques for building target recognition and classification. The multichannel polarization diversity and Mueller matrix characterization techniques were applied for the first time in imaging LiDAR. The proposed polarization diversity imaging LiDAR enables simultaneous multichannel distance, velocity, and intensity measurements, significantly reducing artifacts caused by time-shared polarization imaging. After *noise* filtering, building materials were initially distinguished from polarized reflection intensity point clouds for long-range targets. To achieve more accurate recognition and classification of building materials and structural details, the Mueller matrix and its associated parameters were utilized to characterize targets with different polarization properties. The proposed system resolved ambiguity in determining which point clouds (or pixels) belong to the target. Our architecture fully leverages the potential of 3D polarization imaging LiDAR for long-range target detection and classification in unstructured building environments, providing a promising approach for improving the accuracy of 3D perception in urban building target detection and classification.

Article

☐ 122 **Dual-Strategy from a Single Molecule: Low-Cost, Dopant-Free, Stable D-A-D Type Quinoxaline-Based Hole-Transporting Materials for Perovskite Solar Cells in Both Indoor and Outdoor Applications**

Mondal B., Tiwari R., Manna S., ... Singh R., Samanta S.K.

ACS Applied Energy Materials, 8(6)

2025

22

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Indoor photovoltaics represent a sustainable and reliable source of energy in low-power electronic devices, including the rapidly increasing Internet of Things (IoTs). In this context, hybrid perovskites have garnered surging attention as potential photovoltaic materials owing to their exceptional optoelectronic properties, appropriate band gaps, and ease of solution-based fabrication. Unfortunately, the high cost and necessity of hygroscopic dopants for hole-transporting materials (HTMs) such as spiro-OMeTAD is a major bottleneck, hindering the large-scale application of perovskite solar cells (PSCs). Considering the benefits of electron-deficient units in boosting the efficiency and stability of photovoltaic dyes and polymers, herein, we introduce two donor-acceptor-donor (D-A-D) quinoxaline-based HTMs, YN1 and YN2. These materials incorporate a D-A-D structure to strategically modulate HOMO levels, enhance stability, and reduce production costs. Differences in their photovoltaic performances were studied by using photoluminescence quenching, hole reorganization energy, hole mobility, and charge extraction capability. Dopant-free YN2-based PSCs deliver a maximum power conversion efficiency (PCE) of 28.35% at indoor photovoltaics (1000 lx LED illumination, 0.321 mW cm^{-2}) and 15.62% at *outdoor* conditions (AM 1.5G illumination, 100 mW cm^{-2}) which are at par with the gold-standard, doped spiro-OMeTAD. Compared to that, YN1-based devices show moderate efficiencies of 23.23% in indoor conditions and 10.92% in *outdoor* conditions. Interestingly, the YN2-based device outperforms YN1 and spiro-OMeTAD-based devices in long-term operational stability by maintaining 41.3% of initial PCE after 550 h of *thermal* stress at 85 °C with RH ~ 55%. Alongside, due to the facile two-step synthesis process, the price of YN2 is only $35/g, which is cost-effective compared to commercially available high-performance reference HTMs. The economic viability, outstanding photovoltaic efficiency, and long-term stability of YN2 indicate its strong potential for future practical applications.

Article

☐ 123 **Long-lasting far-UVC persistent luminescence for solar-blind optical tagging**

Shan X., Zhang Y., Lv X., Liang Y.

Inorganic Chemistry Frontiers, 12(8)

2025

12

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Far ultraviolet-C (far-UVC; 200-230 nm) luminescent materials have garnered significant interest in recent years, driven by the growing demands for applications such as disinfection and solar-blind imaging due to their distinct wavelength features. However, the research and development of far-UVC persistent phosphors are lacking. Here, we report the realization of far-UVC persistent luminescence in $CaSO_4:Pr^{3+}$ and $CaSO_4:Pb^{2+}$ phosphors, which show emissions peaking at 220 nm and 230 nm with a long persistence time of >24 h after ceasing X-ray excitation. This is

Document title	Authors	Source	Year	Citations

by far the shortest UVC afterglow emission to the best of our knowledge. The far-UVC afterglow from the charged phosphors can be readily detected using a solar-blind UV camera in both indoor-lighting and *outdoor* environments owing to the absence of background *noise* from ambient light. The continuous photostimulation of indoor white LED light and *outdoor* sunlight has different impacts on the far-UVC afterglow performance of $CaSO_4:Pr^{3+}$ and $CaSO_4:Pb^{2+}$ phosphors, which is elucidated by the decay time-dependent *thermoluminescence* (TL) curves under different light conditions. This study expands the field of persistent luminescence to the far-UVC spectral region and will inspire the discovery of more excellent far-UVC persistent phosphors.

Article

☐ 124 **Design and validation of the multi-parameter control strategies for fan-pad cooling systems in rabbit housing during summer | 兔舍夏季湿帘-风机降温系统多参数调控策略设计与验证**

Wang Q.,Lyu W.,Li W., ... Liu W.,Wu Z.

Nongye Gongcheng Xuebao Transactions of the Chinese Society of Agricultural Engineering , 41(5)

2025

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Automatic control systems have been widely used for the fan-pad cooling of livestock houses in summer. However, current regulation can be confined to the single-parameter temperature. In high-humidity and high-temperature regions, the premature or delayed activation and deactivation of water pumps can increase the indoor humidity levels on the heat stress of animals. In low-humidity and high-temperature areas, the activating water pumps can cause significant fluctuations in indoor temperature, thus resulting in *thermal* discomfort for the animals. In this study, a multi-parameter control strategy was designed to tailor for the different climatic conditions. A dual-parameter control strategy was also developed for the high temperature and high humidity climates. The water pump was regulated using two parameters, including the indoor temperature and *outdoor* humidity. Fan operation was then controlled for the optimal indoor temperature. The multi-parameter control strategy was implemented in the low-humidity and high-temperature climates. The water pump was then regulated using three parameters, including the indoor temperature, *outdoor* temperature, and *outdoor* humidity. While the variable frequency of fans was regulated using indoor temperature. Additionally, a “prevent sudden drop measure” was introduced to reduce the fan speed during the initial startup of water pumps. The “temperature fluctuation mitigation measure” was applied to adjust the water circulation time. These strategies were validated in the rabbit farms in Qingdao City, Shandong Province, and Qingyang City, Gansu Province, China. In Qingdao, the meat rabbit house was equipped with a controller to monitor the environmental regulations under different weather conditions. The results showed that the proportion of time experiencing heat stress (temperature-humidity index, THI > 27.8) was reduced to 2.2% during daytime and 0 at night. On high humidity cloudy days with the insufficient cooling potential of the cooling pad, the dual-parameter control strategy effectively cooled the indoor environment, where the water pump was closed to increase ventilation. On sunny days, the cooling pad achieved a theoretical cooling value of (3.3 ± 1.0) °C and an actual cooling value of (0.8 ± 0.5) °C. Importantly, no rise of indoor temperature was observed, when the water pump was deactivated, indicating the effective pump operation and control. In Qingyang, another rabbit farm was used to further validate the control strategies. An automatic controller was installed in a treatment house. While another meat rabbit house with a manually controlled cooling pad and fan operations served as the control house. Environmental parameters, rabbit body temperature, and production performance indicators were collected to compare the performance between them. The findings revealed that the indoor temperature near the cooling pad area decreased by 6.6 °C within 7 min after the initial activation of the water pump. The “prevent sudden drop measure” was then implemented to extend the time. A similar cooling effect was then achieved in 23 min. The “temperature fluctuation mitigation measure” effectively reduced the fluctuations of indoor temperature from 3 °C to within 1 °C, indicating a stable *thermal* environment. Compared with the control, the treatment house reduced the fluctuations of diurnal temperature, the daily average temperature of the rabbit body ($P< 0.05$), a 4.1% decrease in the cumulative mortality rate, and a 4.3% increase in the total sales revenue. These results demonstrate the effectiveness and versatility of the multi-parameter control strategy to enhance the cooling performance of fan-pad cooling systems under varying weather conditions in both high-temperature-humidity and high-temperature-low humidity regions. Stable indoor environments were maintained to reduce the rabbit's body temperature for economic returns.

	Document title	Authors	Source	Year	Citations

This strategy can provide a *sound* reference to precisely control the environmental temperature in livestock houses in summer.

Article • *Open access*

125 **Multi-Level Spectral Attention Network for Hyperspectral BRDF Reconstruction from Multi-Angle Multi-Spectral Images** — Song L.,Li H. — Remote Sensing, 17(5), 863 — 2025 — 7

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With the rapid development of hyperspectral applications using unmanned aerial vehicles (UAVs), the traditional assumption that ground objects exhibit Lambertian reflectance is no longer sufficient to meet the high-precision requirements for quantitative inversion and airborne hyperspectral data applications. Therefore, it is necessary to establish a hyperspectral bidirectional reflectance distribution function (BRDF) model suitable for the area of imaging. However, obtaining multi-angle information from UAV push-broom hyperspectral data is difficult. Achieving uniform push-broom imaging and flexibly acquiring multi-angle data is challenging due to spatial distortions, particularly under heightened roll or pitch angles, and the need for multiple flights; this extends acquisition time and exacerbates uneven illumination, introducing errors in BRDF model construction. To address these issues, we propose leveraging the advantages of multi-spectral cameras, such as their compact size, lightweight design, and high signal-to-*noise* ratio (SNR) to reconstruct hyperspectral multi-angle data. This approach enhances spectral resolution and the number of bands while mitigating spatial distortions and effectively captures the multi-angle characteristics of ground objects. In this study, we collected UAV hyperspectral multi-angle data, corresponding illumination information, and atmospheric parameter data, which can solve the problem of existing BRDF modeling not considering *outdoor* ambient illumination changes, as this limits modeling accuracy. Based on this dataset, we propose an improved Walthall model, considering illumination variation. Then, the radiance consistency of BRDF multi-angle data is effectively optimized, the error caused by illumination variation in BRDF modeling is reduced, and the accuracy of BRDF modeling is improved. In addition, we adopted Transformer for spectral reconstruction, increased the number of bands on the basis of spectral dimension enhancement, and conducted BRDF modeling based on the spectral reconstruction results. For the multi-level Transformer spectral dimension enhancement algorithm, we added spectral response loss constraints to improve BRDF accuracy. In order to evaluate BRDF modeling and quantitative application potential from the reconstruction results, we conducted comparison and ablation experiments. Finally, we solved the problem of difficulty in obtaining multi-angle information due to the limitation of hyperspectral imaging equipment, and we provide a new solution for obtaining multi-angle features of objects with higher spectral resolution using low-cost imaging equipment.

Article • *Open access*

126 **Efficient Light to Heat Conversion in Sb2Se3 Nanorods and the Role of Macro-channel Imprinted Sb2Se3 Loaded Hybrid Membrane for Superior Desalination Performance** — Sen N.,Debnath A., Bhattacharjee S., ... Saha A.K., Chattopadhyay K.K. — Small, 21(7), 2408293 — 2025 — 4

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This report validates Sb_2Se_3 nanorods (NRs) as a potential contender for solar *thermal* heat generation. The water droplet experiment shows Sb_2Se_3's light-to-heat conversion efficiency as ≈57.8% for red (671 nm), 58% for green (532 nm) lasers. Following this PVDF(M)/ Sb_2Se_3 NRs hybrid membranes for solar desalination reached ≈59°C in 15 minutes of illumination. The heat generation is dominated by an electron/hole-*acoustic* phonon scattering mechanism. Despite having superior visNIR absorption and heat localization in Sb_2Se_3 NRs, the hybrid membranes show an evaporation rate of 1.72 kg m^{-2} h^{-1} only, even if mass loading is increased. The hydrophobic Sb_2Se_3 NRs layer restricts water diffusion to hot zones, reducing solar evaporation efficiency. A novel macro channel imprinting strategy in hybrid membranes speeds up water transport to the hot zone. Consequently, optimized macro channel membranes achieve ≈2.37 kg m^{-2} h^{-1} mass loss and 148% solar evaporation efficiency under a 1000 W m^{-2} mercury vapor lamp. Therefore, imprinting macro-channel can be a possible strategy, addressing the hydrophobic materials in desalination applications which can be expanded in other similar materials. Moreover, its *outdoor* sunlight application achieves impressive solar evaporation efficiency (≈108%). The steam generated effectively removes heavy metals, meeting World Health Organization (WHO) potable water standards.

	Document title	Authors	Source	Year	Citations
	Article				
127	**Outdoor space design and its effect on mental work performance in a subtropical climate**	Luo Y.,Ma X.,Li J., ... de Dear R.,Lu S.	Building and Environment , 270, 112470	2025	30

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Due to the stress resulting from high-density urbanisation, a premium is being placed on the restorative benefits of *outdoor* spaces for city dwellers. *Outdoor* spaces are increasingly being used as workplaces for tasks that have usually been performed indoors. However, the interrelationships among physical environmental parameters, perceived workload, and cognitive performance in *outdoor* spaces have not been studied in-depth. This study explores the suitability of *outdoor* urban spaces for short-term mental work during the transitional and hot-humid seasons in China's subtropical region. Four typical *outdoor* spaces in Shenzhen were selected to conduct questionnaire surveys and cognitive tests on a sample of 96 college-age participants. Concurrently, micrometeorological, illumination and *acoustic* environmental measurements were taken. The findings revealed that, firstly, in the transient season, working outdoors achieved similar or even better cognitive performance compared with indoors, with less perceived workload pressure, suggesting the feasibility and potential benefits of working outdoors. However, in the hot season, the subjective comfort, perceived workload and cognitive performance are all rather poor. Secondly, *thermal* conditions significantly affected *outdoor* work. Cognitive performance was superior outdoors compared to indoors when the Universal *Thermal* Climatic Index (UTCI) was within the range of 9 to 32 °C. However, this performance declines sharply when UTCI exceeds 32 °C. Similar trends are also observed for subjective comfort and perceived workload. Thirdly, lighting conditions also influenced comfort and performance, with participants preferring brighter environments (1000–2000 Lux) outdoors compared to indoor settings. Meanwhile, space design played a critical role in mitigating *outdoor* discomfort. Vegetation and shading elements, such as those in Patio Space, reduced *thermal* stress with an average UTCI reduction of 2 °C and improved satisfaction by 20 %, underscoring their importance in creating usable *outdoor* work environments. Moreover, perceived workload was found to be the mediator between physical environment and cognitive performance, with physical burden being the most influential indicator. The results provide actionable insights for designers, urban planners and policymakers to promote sustainable, comfortable, and productive *outdoor* environments in subtropical cities.

	Review • *Open access*				
128	**The health and indoor environmental quality impacts of residential building envelope retrofits: A literature review**	Lozinsky C.H., Casquero-Modrego N., Walker I.S.	Building and Environment , 270, 112568	2025	37

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Retrofitting existing buildings to improve energy efficiency is an important strategy to meet increasingly stringent energy efficiency targets. While the primary objective of energy efficiency retrofits is to reduce energy consumption and greenhouse gas emissions, retrofits can also result in non-energy impacts (NEIs), which contribute to decision-making processes and overall value of the retrofit. NEIs have been studied extensively in retrofitted residential buildings; however, these studies have historically grouped passive (i.e., building envelope) and active (i.e., heating, ventilation, and air conditioning (HVAC) and energy system) upgrades, making it difficult to identify the underlying mechanism(s) of action for each NEI and developing effective retrofit strategies, based on occupant need. The purpose of this study was to better account for NEIs, based on a literature review, summarizing the current state of knowledge on NEIs associated with residential building envelope retrofits. We limited our search to health- and indoor environmental quality-related NEIs. The review identified strong evidence that building envelope retrofits improve *acoustic* comfort, wintertime *thermal* comfort, and respiratory and cardiovascular health outcomes. IAQ outcomes were mixed, with studies reporting both increases and decreases to indoor contaminant concentrations following retrofits. The strength of the effect was generally governed by pre-retrofit contaminant concentrations and whether indoor concentrations were dominated by indoor or *outdoor* sources. Most studies evaluating summertime *thermal* comfort identified increased incidence of summertime overheating; however, none of these studies linked the change in *thermal* conditions to health outcomes. Recommendations for future work include expanding studies to include more market rate housing and the health impacts of summertime overheating in retrofitted buildings.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
129	**The Influence of Multisensory Perception on Student Outdoor Comfort in University Campus Design**	Touhami H.,Berkouk D., Bouzir T.A.K.,Khelil S., Gomaa M.M.	Atmosphere, 16(2), 150	2025	14

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The user's experience is critical in spatial design, particularly in *outdoor* spaces like university campuses, where the physical environment significantly influences students' relaxation and stress relief. This study investigates the combined impact of *thermal*, luminous, and auditory environments on students' perceptions within recreational areas at Bordj Bou Arreridj University Campus. A mixed-method approach combined field surveys and on-site measurements across eleven locations within three distinct spatial configurations. The findings from this study indicate that the auditory environment had the most substantial influence on overall perceptions, surpassing luminous and *thermal* factors. The open courtyard (Area 1) was perceived as less comfortable due to excessive heat and *noise* exposure. The shaded zone (Area 2) was identified as the most vulnerable, experiencing significant *thermal* stress and *noise* disturbances. In contrast, the secluded patio (Area 3) achieved the highest comfort rating and was perceived as the most cheerful and suitable space. Correlation analysis revealed significant interrelationships between physical and perceptual dimensions, highlighting the critical role of factors such as wind velocity, sky view factor, and illuminance in shaping *thermal*, luminous, and *acoustic* perceptions. A fuzzy logic model was developed to predict user perceptions of comfort, suitability, and mood based on measured environmental parameters to address the complexity of multisensory interactions. This study highlights the importance of integrating multisensory evaluations into spatial design to optimize the quality of *outdoor* environments.

Article

130	**Effect of thermal-acoustic-air quality composite environments on overall comfort of urban pocket parks considering different landscape types**	Li W.,Tian L.,Jin X.,... Tao Y., Peng P.	Energy and Buildings , 328, 115167	2025	15

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To bridge the research gap of landscape types' impact on overall comfort, this study studied the relationship between the combined effect of *thermal-acoustic*-air quality and overall comfort in urban pocket parks during winter through surveys and monitoring. Environmental parameters monitored included *outdoor* air temperature, relative humidity, wind velocity, ambient *noise* etc. The (*thermal*, *acoustic*, and air quality) comfort survey data, universal *thermal* climate index (UTCI), weighted equivalent continuous *sound* level (LAeq), and particulate matter with a diameter of less than 2.5 μm (PM2.5) were utilized to objectively evaluate the overall comfort vote (OCV). Furthermore, the comfort index-based and the perception vote-based OCV prediction models were built. Results showed that the water landscape parks had the highest overall and *thermal* comfort, followed by mountain, square, and building landscapes. For water and square landscape parks, the *thermal* comfort vote (TCV) was the most important comfort perception, while the air quality sensation vote (AQSV) was key for building landscape parks. No comfort perception vote index significantly affected OCV for mountain landscape parks. For "acceptable" OCV (=0), minimum UTCI thresholds were −5°C, −1.5 °C, and 5 °C for water, mountain, and building landscape parks, respectively, when LAeq exceeded 56 dBA, OCV remained constantly low (< 0) regardless of UTCI. The comfort index-based model is accurate and reliable for OCV prediction. This study provides the threshold requirements for achieving an "acceptable" overall comfort level in different types of pocket park, and could guide the urban pocket park design.

Article • *Open access*

131	**Performance study and LCA of a ZigZag PV noise barrier: Towards mass-customization of IIPV applications**	Colberts F.,Bouguerra S., Wieclawski A., ... Daenen M.,Vroon Z.	Applied Energy, 378, 124724	2025	3

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Photovoltaic *noise* barriers (PVNB) offer dual functionality in reducing traffic *noise* and generating renewable electricity. In this research, the potential of ZigZag PVNBs has been investigated. The ZigZag Solar product, developed by Wallvision, has proven to offer multiple advantages in energy yield and aesthetics for building façade applications. For *noise* barrier applications, the ZigZag structure could offer interesting features in safety and *noise* cancellation (obtained by filling the ZigZag construction with Rockwool material) on top of the advantages in aesthetics and

Document title	Authors	Source	Year	Citations

energy yield. A ZigZag PVNB has been designed and constructed at the Brightlands Chemelot Campus in Geleen, after which the electrical performance has been automatically monitored under Dutch climate conditions. The measurements have been compared to simulated data, which allowed optimization of the model. As Rockwool material is used in the ZigZag construction, the *thermal* model had to be optimized to reduce significant differences in measured and simulated V_{MPP} data. Temperature measurements by a novel Fiber Bragg technology revealed that temperature differences between measured cell temperature and input temperature for the simulations are between 10 and 20 °C. After optimizing the *thermal* model, the power output of the ZigZag PVNB could be predicted more accurately, resulting in a yearly potential energy yield up to 1066 kWh/kWp. Measured data over the period June 2023 till April 2024 showed an energy yield up to 873 kWh/kWp. A deviation of 18 % between measured yearly energy yield can be related to system losses such as cabling and inverters. Life Cycle Assessment (LCA) of several configurations of a global system, including concrete infrastructure, solar panels, ZigZag cassettes, cabling and converters shows a Global Warming Potential (GWP) score varying from 190 to 290 CO_2 eq/kWh, according to the models developed in this study, indicating its interest compared to the Dutch and German electricity mixes. In addition, the energy required to produce and install the ZigZag PVNB system at various lengths has a predicted payback time of 6–10 years (maximum 30 % of the total expected lifetime). The balance of system, in specific the DC/DC converters followed and battery system) followed by the concrete element on which the ZigZag PVNB was mounted are the largest contributors to the carbon footprint of the ZigZag PVNB demonstrator. The carbon footprint could potentially be reduced by using cleaner battery technologies or energy storage systems.

Article • *Open access*

☐ 132 **Evaluation of PM2.5 bound microplastics and plastic additives in several cities in Taiwan: Spatial distribution and human health risk**

Wu C.-H., Dang T.T.M., Mutuku J.K., ... Huang B.-W., Chang-Chien G.-P.

Science of the Total Environment, 959, 178213

2025

19

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Adverse respiratory health from inhaling microplastics (MPs) in fine particulate matter ($PM_{2.5}$) is a major global problem and a significant cause of human mortality. This study analyzes the concentrations of MPs and plastic additives in $PM_{2.5}$ from five regions of Taiwan using *thermal* desorption/pyrolysis - gas chromatography–mass spectrometry (TD/Py-GC/MS) with a double-shot mechanism. The results showed that the average concentration of MPs varied among regions, ranging from 68.97 to 76.86 ng m^{-3}, while the concentration of plastic additives ranged from 3.78 to 10.97 ng m^{-3}. Notably, polyvinyl chloride (PVC) was the predominant plastic in $PM_{2.5}$, highlighting the need for effective measures to reduce its atmospheric concentration by the Taiwan Environmental Protection Authority (Taiwan EPA). Di-isononyl phthalate (DiNP) and bis-(2-ethylhexyl) terephthalate (DOTP) were the main additives detected, highlighting the urgent need to reduce and replace harmful plastic additives. The study also assessed the toxicological risks posed by $PM_{2.5}$, MPs, and plastic additive concentrations to human health. $PM_{2.5}$ increases the risk of cardiopulmonary disease and lung cancer by 26.4 % and 26.73 %, respectively. The additives present in $PM_{2.5}$ pose a significant danger to *outdoor* workers, with a hazard index (HI) ranging from 1.26 to 2.47 and a cancer risk (CR) ranging from 9.13×10^{-4}–7.61×10^{-3}. The pollution index and the toxicity (hazardous) levels of MPs within $PM_{2.5}$ have reached a critically dangerous level.

Article • *Open access*

☐ 133 **Atmospheric turbulence affects noise annoyance from aircraft flyovers**

Lincke D., Kawai C., Pieren R.

Acta Acustica, 9, 69

2025

0

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Auralization of *outdoor sound* propagation has become an important tool for studying *noise* perception in the contexts of aircraft, wind farm, and drone *noise*. To achieve high realism, these auralizations consider amplitude fluctuations caused by atmospheric turbulence. In a recent publication, a semi-empirical model was introduced which is applicable to relatively long *outdoor sound* propagation such as aircraft and wind farm *noise*, as it considers the saturation effect for amplitude fluctuations in the partially-saturated regime. This article presents a study applying the semi-empirical model for investigating the impact of turbulence-induced amplitude fluctuations on annoyance in auralized aircraft flyovers. For this, two 2-alternative-forced-choice listening experiments were conducted in a controlled laboratory setting: the first tested whether participants could reliably detect audible

	Document title	Authors	Source	Year	Citations

differences between aircraft flyovers under different meteorological conditions; the second assessed which condition was perceived as more annoying. The results show that strong differences in meteorological conditions and the respective atmospheric turbulence can lead to salient audible differences. Relative annoyance ratings tend to increase with stronger atmospheric turbulence. Further, the data suggest that amplitude fluctuations can interact with other characteristics of aircraft *noise* such as fan tones and alter the perceptual impact of these characteristics. Therefore, the study highlights the importance of modeling turbulence-induced amplitude fluctuations in realistic aircraft auralizations, and presumably also wind farm and drone *noise* auralization, as the perceptual impression can be affected in several ways.

Article

☐ 134 **Enhancing autumn greenway walking experience: Exploring the combined effects of noise and thermal environment** — Li Y.,Li L.,Wang W. — International Journal of Biometeorology, 69(1) — 2025 — 2

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The evaluation of *outdoor* green spaces is influenced by diverse sensory perceptions. Traffic *noise* and *thermal* conditions significantly impact greenway-walking satisfaction; their optimization is vital for improving user experience and encouraging *outdoor* engagement. The study examines a typical Beijing greenway during autumn, focusing on strategies to enhance the walking experience under the combined effects of *noise* and *thermal* environments through mobile measurements and surveys. The results show that: 1) The interplay between *noise* and *thermal* factors varies depending on the walking state. Upon arrival, an increase in *noise* significantly worsens *thermal* comfort; higher *sound* levels intensify warm *thermal* sensations, though this effect is not consciously perceived. Upon departure, the effect of *noise* on *thermal* perception is not obvious. In both walking states, *thermal* sensation significantly affects subjective *noise* perception, yet the trends of influence differ. Subjective *noise* loudness increases as *thermal* comfort worsens, showing significant correlation only upon departure. 2) During autumn greenway walks, *acoustic* factors exert a greater impact on Overall Environmental Satisfaction (OES), with subjective *noise* loudness being more influential than *noise* level, followed by air temperature (T_a). Greater *noise* decreases OES, while OES increases initially with T_a and then decreases. The integrated effects of *noise-thermal* factors on OES show significant changes. 3) To enhance the autumn greenway-walking experience, the advised parameters are A-weighted *Sound* Level (ASL) ≤ 59.12 dBA and 15.17 °C ≤ T_a ≤ 18.75 °C. Finally, three design strategies are proposed: reducing subjective *noise* loudness, differentiating design based on walking status and balancing *acoustic-thermal* perceptual preferences.

Article • *Open access*

☐ 135 **Effects of interactions between thermal and acoustic environments on subjective comfort evaluations in outdoor public spaces** — Lin D.,Li T.,Chen Z.,Li L. — Journal of Asian Architecture and Building Engineering, 24(1) — 2025 — 10

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In this study, five *outdoor* open areas were selected as the study sites to investigate the effects of *thermal* and *acoustic* environments, physical activity levels, and pleasantness on evaluations of the *thermal* and *acoustic* comfort in *outdoor* open spaces. Survey questionnaires and on-site measurements were used to quantitatively assess human *thermal* and *acoustic* comfort in *outdoor* spaces. The following findings were obtained in this study. The *acoustic* environment affected the *thermal* sensation, where reducing excessive *outdoor noise* levels could enhance *thermal* comfort. Increasing the physiological equivalent temperature amplified the *acoustic* sensation. Individuals reported the most comfortable *acoustic* perception outdoors at a physiological equivalent temperature of 30.1°C. Reducing both the *thermal* and *acoustic* indices improved the overall comfort to some extent. Decreasing physical activity levels by an appropriate amount enhanced the *thermal* comfort, *acoustic* comfort, and overall comfort. Finally, maintaining a sense of pleasantness was conducive to increasing the *thermal* comfort, *acoustic* comfort, and overall comfort. In states of moderate and pleasurable feelings (pleasantness level = 0, 1, 2, 3), increased *acoustic* comfort and pleasantness combined significantly enhanced the *thermal* comfort.

Article

	Document title	Authors	Source	Year	Citations
136	**Outdoor thermal comfort of urban plaza space under thermoacoustic interactions –Taking Datang Everbright City as an example**	Zhen M.,Yuan X.,Gao L., ... Nan K.,Zhang M.	Building and Environment , 267, 112327	2025	10

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The comfort of urban square environments in terms of *thermal* and *acoustic* conditions directly influences the quality of people's travel experiences and their overall well-being. This study, represented by Xi'an Datang Everbright City, investigates the impact of *thermal-acoustic* environments on human comfort in cold regions of China. Four typical spaces within the city square were selected for meteorological measurements and subjective questionnaire surveys to gauge the *thermal* perception of active crowds. Using the Universal *Thermal* Climate Index (UTCI) as a *thermal* benchmark, the study compares the interaction between *thermal* conditions and *acoustic* environments. Results indicate that the neutral UTCI in Xi'an urban squares is 14.9 °C, with a neutral UTCI range of 10.8∼19.0 °C and a preferred temperature of 17.1 °C. Positive types of *sounds* moderately reduce *thermal* sensation. Moderate environmental *sounds* alleviate *thermal* discomfort to some extent among respondents. In environments such as birdsong and flowing water, people prefer higher temperatures, thereby increasing *thermal* tolerance. Conversely, in *noise* from traffic or construction, individuals prefer lower temperatures, reducing their *thermal* tolerance. In environments with broadcasts or conversations, respondents exhibit slightly higher *thermal* tolerance compared to the average neutral temperature. Finally, based on meteorological characteristics at different survey points and respondent feedback, optimization strategies for design are proposed.

	Conference Paper				
137	**The Assessment of a Dielectric Elastomer Automatic Window Technology to Improve Indoor Air Quality and Thermal Comfort Utilizing Outdoor Air Condition**	Shimul S.A.,Li W., Haghighi M.	Thermal Performance of the Exterior Envelopes of Whole Buildings	2025	0

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The Environmental Protection Agency (EPA) ranks indoor air pollution as one of the top five environmental risks to public health. According to IHME, it is responsible for 1.6 million deaths yearly, while the WHO puts this figure at 4.3 million. According to the EPA, Americans, on average, spend 90% of their time indoors, where the concentrations of some pollutants are often 2 -5 times higher than outdoors. Both the EPA and the NIST suggest that simply opening windows and increasing air exchange can significantly improve indoor air quality (IAQ) without any additional measures. However, for ventilation and *thermal* comfort, we primarily rely on HVAC systems, which account for 35-40% of building energy consumption. Unfortunately, conventional HVAC systems often fail to capitalize on *outdoor thermal* conditions, thereby minimizing heating and cooling loads. Given that windows are the most common and efficient means of indoor-*outdoor* ventilation and *thermal* exchange, this research proposes the development of a Smart Breathing Window (SBW) utilizing Dielectric Elastomer Actuators (DEAs). This innovative ventilation method can automatically detect indoor and *outdoor* air conditions and initiate air exchange accordingly. Compared to conventional motor-based fan ventilation, the Dielectric Actuators used in this window provide several advantages, including low energy consumption, a lightweight design, scalability, various sizes and shapes, *noise*-free movement, and options for different levels of opacity, which could open up a novel method of ventilation. The SBW is programmed to automatically respond to changes in temperature, humidity, oxygen (O_2), carbon monoxide (CO), carbon dioxide (CO_2), toxic gas levels, and particulate matter (PM 2.5) concentrations. This research suggests that SBW can significantly enhance IAQ through automatic indoor-*outdoor* air exchange while also reducing the HVAC load utilizing *outdoor* comfort conditions. Lastly, the research outlines the basic design of SBW, performance experiments and discusses its potential as a new ventilation method.

	Conference Paper				
138	**Enhancing Outdoor-to-Indoor mmWave Coverage Using Static Transmissive Surfaces: Experimental Results**	Bas R.,Unal I.,Sayinti A., ... Kayraklik S.,Hokelek I.	30th European Wireless Conference Ew 2025	2025	0

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Static transmissive surfaces (STSs) offer a complementary passive approach for enabling mmWave signal penetration through energy-efficient building materials without requiring power sources or electronic control circuitry. This paper

	Document title	Authors	Source	Year	Citations

presents design and development of STSs specifically for next-generation millimeter-wave (mmWave) wireless networks, featuring specialized patterning that enhances signal penetration through low-emissivity glass while maintaining *thermal* insulation properties. Performance evaluation is conducted in a two-sided (*outdoor*-to-indoor) environment using commercial 5G network equipment. The received signal strength indicator (RSSI), reference signal received power (RSRP), and signal-to-interference-plus-*noise* ratio (SINR) measurements on a customer premises equipment (CPE) terminal show that the designed STSs provide 13-21 dB enhancement over standard low-emissivity glass, effectively improving indoor coverage for 5G mmWave networks. These results validate STSs as environmentally sustainable solutions for mitigating signal attenuation in modern energy-efficient structures without compromising their *thermal* performance.

Article

☐ 139 **Indoor environmental quality in high school buildings during mild winter season: students' perception and field measurements** — Amoatey P.,Al-Saadi S.,Al-Harthy I.,Al-Jabri K.,Al-Azki S. — Advances in Building Energy Research, 19(2) — 2025 — 5

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The state of indoor environmental quality (IEQ) on school students during the winter season in hot climates has received little attention. This study aims to evaluate the IEQ conditions, namely, *thermal*, visual, *acoustic* comfort, and air quality, for typical private (PSB) and government (GSB) school buildings during winter in a typical hot climate. Subjective and objective assessments were conducted on 304 students distributed in 14 classrooms. The results showed that winter *outdoor* climate might have favoured most of the measured IEQ factors, therefore, meeting the ASHRAE 55 and EN 16798-1 standards. *Thermal* comfort yielded neutral–cool classroom conditions in PSB according to predicted mean vote (PMV) and predicted percentage of dissatisfaction (PPD), whereas in GSB, a neutral *thermal* sensation environment of PMV (0.02–0.2) and a PPD of 5–6% was found across the classrooms. Despite the measured *sound* levels exceeding the critical level of 35 dBA, most of the students rated the classroom's *sound* levels as neutral (41.3–44.8%) and satisfying (27.9–31.3%), which might be due to the long-term acclimatization of the students to the background *noise* levels. The majority of the students (42.7–47.6%) described the indoor air quality as neutral, which raises potential health risks.

Conference Paper

☐ 140 **Quantifying Noise Reduction From Envelope Retrofits Using Publicly-Available Data Sources** — McCord K.H.,Rosenberg S.,Muehleisen R.T.,... Vesty D.,Antonopoulos C.A. — Thermal Performance of the Exterior Envelopes of Whole Buildings — 2025 — 0

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Affordable energy for all in the building sector requires abundant energy supply, efficient equipment, and building envelopes that retain *thermal* energy. Energy efficiency upgrades to building enclosures have an added benefit of *noise* isolation: making the indoor environment quieter by reducing the *noise* coming from outside. This reduction of *outdoor noise* infiltration is not only a matter of comfort but also has significant implications for health and wellness, as various *noise* level thresholds are associated with potential negative health impacts such as cardiac stress. Yet, *noise* isolation is an often-overlooked non-energy benefit (NEB) of energy efficiency, with little research quantifying *noise* impacts of retrofits available in the building retrofit literature. This work explores opportunities to quantitatively tie building energy efficiency measures to non-energy *acoustic* benefits, focusing on residential buildings. This work references publicly-available spectral transmission loss data for a variety of wall assembly components. This information is paired with typical home characteristics and upgrade scenarios in ResStock, a publicly available dataset simulating the U.S. residential building stock and exploring building material and technology upgrade scenarios. This proof-of-concept is demonstrated with the baseline ResStock scenario, representing the current U.S. building stock. This database can be used to estimate *noise* reduction from wall and window upgrades across the building stock Challenges, data limitations, and lessons learned are discussed, including the potential to elevate the importance of *acoustic* considerations in building energy efficiency discussions and practices.

Conference Paper • *Open access*

	Document title	Authors	Source	Year	Citations
141	**DESIGN AND TESTING OF A TRL-3 OPEN ROTOR FOR AEROACOUSTICS AND AEROMECHANICS RESEARCH**	Figueroa-Ibrahim K.M., Gold J.S.,Sutliff D.L., ... Ross M.H.,Morris S.C.	Proceedings of the ASME Turbo Expo , 10	2025	1

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The NASA Advanced *Noise* Control Fan Facility (ANCF) was used with a new open-rotor test article. The goal was to understand the test article design space such that the salient aeroacoustic and aeromechanical characteristics of a modern open-rotor engine design were represented. The facility was operated in an *outdoor* environment where the ambient conditions were uncontrollable. A design advance ratio of J = 0.325 was selected based on a statistical analysis of historical ambient data at the *outdoor* test site. A multi-fidelity design loop allowed a rotor blade design that mimicked the blade loading and tip vortex trajectory of a reference design representing take-off conditions associated with an advance ratio of J = 1.0. Results from *outdoor* experiments involving the rotor blades are presented. The experiments consisted of fixed fan speed, rotor blade pitch and variable advance ratio and angle of attack. The configurations and measurements were limited as this entry was intended to evaluate the proof-of-concept and initial operating capability. Wind, far-field radiated *noise*, and rotor blade strain measurements are described.

Conference Paper

	Document title	Authors	Source	Year	Citations
142	**Precise Laser Metrology for Power Systems: QPD Localization via Gaussian and Noise Compensation**	Mohamed M.S., Mustafa E.M.,Heweage M.	Proceedings of the International Middle East Power System Conference Mepcon , (2025)	2025	0

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Accurate localization of laser spots is critical in many advanced applications, including optical sensor and communication systems within modern power grids, but is hindered by challenges in quadrant photodetectors (QPDs). These challenges stem particularly from the mismatch between ideal uniform spot models and the practical Gaussian intensity profiles. Furthermore, measurement accuracy is degraded by *noise* sources such as shot *noise*, *thermal noise*, background illumination, and dark current, which are prevalent in the *outdoor* and high-EMI environments of power systems. This paper presents an integrated approach that models the Gaussian beam distribution and compensates for major *noise* effects to enhance the reliability of laser alignment for power infrastructure metrology. The mathematical model of light energy on each QPD quadrant is refined, and a *noise* compensation algorithm is developed to mitigate the impact of dark current and ambient light. A lookup table (LUT)-based calibration is also introduced to convert uniform-spot positions into accurate Gaussian-based equivalents. Experimental and simulation results show that the proposed method dramatically reduces positioning errors from hundreds of micrometers to sub-micrometer levels near the spot center. This work offers crucial insights into the relationship between beam waist radius and detector performance, providing a highly reliable framework for high-precision laser alignment and metrology critical for future smart grid monitoring and maintenance.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
143	**Real-Time Long-Wave Infrared Semantic Segmentation With Adaptive Noise Reduction and Feature Fusion**	Bae H.,Kang D.-G., Chang M.,Jeong K.Y., Song B.C.	IEEE Access, 13	2025	1

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Semantic segmentation in the Long-Wave Infrared (LWIR) domain is critical for a wide range of applications including emergency response, industrial safety monitoring, and public building security. However, LWIR images often suffer from inherent pattern *noise* (known as fixed-pattern *noise*) and ambiguous object boundaries that hinder accurate segmentation. To address these challenges, this study presents a novel real-time semantic segmentation framework specifically designed for LWIR images. The framework incorporates a Stripe *noise* Denoising Module (SDM) and a Boundary Enhancement Module (BEM), which leverage frequency-domain filtering and learnable weights to adaptively process noisy input data and improve boundary prediction. In addition, a Multi-Stream Fusion Module (MSFM) integrates multi-scale semantic features with boundary information, thereby enhancing segmentation accuracy across diverse object scales. The proposed method demonstrates state-of-the-art performance in both accuracy and efficiency on multiple datasets, including KERInha and SODA. Extensive qualitative and quantitative

	Document title	Authors	Source	Year	Citations

evaluations further validate its robustness, particularly in scenarios where RGB imagery is unavailable. By eliminating the need for supplementary information such as depth data, this approach facilitates precise indoor and *outdoor* segmentation tasks with lightweight computation, making it highly suitable for real-world applications. Code is available at https://github.com/saha3jet/pytorch_efficientirseg.

Article

144 **Effect of Phase Noise on the Performance of Coherent Visible Light Communication Systems** — Ozbilgin Z.,Uysal M. — IEEE Transactions on Communications, 73(11) — 2025 — 2

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Visible light communication (VLC) builds upon the dual use of illumination infrastructure for wireless access. The recent commercialization of laser diode (LD)–empowered white light luminaires gives the opportunity to build coherent VLC systems. Coherent VLC systems can leverage the high bandwidth of lasers and advanced digital signal processing techniques to significantly improve the data rates. For coherent reception, a local oscillator in the form of a monochromatic laser is used. In practice, lasers suffer from random fluctuations in phase due to spontaneous emission and carrier density fluctuations. In this paper, we analyze the effects of phase *noise* on the performance of a coherent VLC system in the presence of atmospheric turbulence as encountered in *outdoor* and vehicular applications. We assume that a phase locked loop (PLL) is used for phase recovery. Turbulence-induced fading phase is estimated through pilot symbols and provided as input to the PLL. The PLL uses this estimated channel phase to initialize its phase estimate at the beginning of each frame. Based on the statistical distribution of the residual phase, we derive an approximate closed-form expression for the pairwise error probability (PEP) which is then used to obtain a union bound on the error rate. Through comprehensive simulations, we validate the accuracy of the derived expression and quantify the impact of phase *noise* on the error rate performance.

Conference Paper

145 **Experimental and Numerical Acoustic Scaling Investigation of a Low Tip Speed eVTOL Proprotor in Hover** — Fleming J.,Langford M., Walton W.,... Alexander N., Duong T. — Vertical Flight Society 81st Annual Forum and Technology Display — 2025 — 1

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To document *noise* characteristics and provide validation data for *acoustic* modeling of rotor systems appropriate for eVTOL/UAM aircraft, the authors performed an *outdoor* static test of a subscale 5-blade proprotor. The testing was carried out as part of a program to demonstrate feasibility and overall performance of a quiet proprotor system in support of the eVTOL industry. The authors designed a low-tip speed proprotor to approximate performance required by a 4-5 passenger UAM vehicle. A driving design feature was low-tip speed operation (M_{tip} ~0.27) at system disk loadings of 7 to 8 psf (~3.7 N/m^2). The test article was designed as a ground adjustable pitch 5-blade proprotor, with aerodynamic and *acoustic* data collected in *outdoor* static hover testing. The test article diameter of 3 feet (0.91 m) represented a scale factor of approximately 30% to 40% compared to vehicles currently in operation or development. The aerodynamic performance in hover was consistent with other rotor systems tested in the past by the authors (Figure of Merit ~0.72-0.75), and the effects of naturally occurring turbulence on rotor *acoustics* were measured using a 180-deg arc array of ground plane microphones at the Virginia Tech Drone Park test facility. The paper closes with a prop diameter scaling investigation using lattice-Boltzmann method CAA with and without atmospheric turbulence.

Conference Paper

146 **Microclimate Conditions in the SS. Salvatore Church of Bologna** — Saito H.,Manfren M., Fabbri K.,Tommasino M.C., Tronchin L. — Building Simulation Applications — 2025 — 3

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This paper analyses the microclimate conditions in the Church of Santissimo Salvatore in Bologna and their influence on the *acoustics* of the Church and the *sound* of the pipe organ. It is commonly acknowledged that the variation of air temperature, median radiant temperature, relative and absolute humidity could provoke *thermal* expansions of metals which are used for the organ pipes. This paper analyses a monitoring campaign which lasted one year in

	Document title	Authors	Source	Year	Citations

which temperature and relative humidity were stored with a data logger. Moreover, the paper shows the effect in the variation of the microclimate conditions in the church by simulating different numbers of people in the church and different *thermal* conditions *outdoor*. Finally, the paper reports the variations of the *acoustic* parameters simulating the new values of temperature, relative humidity and air velocity.

Conference Paper

147 **Large-Area Surface Acoustic Wave Cleaning for Enhanced Optical Performance of LiDAR Sensors** — Jonas J.C.,Kim H.,Kim S., Kim H.H. — Proceedings of IEEE Sensors — 2025 — 0

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Maintaining optical clarity in LiDAR sensors is critical for autonomous navigation and *outdoor* monitoring, where environmental contamination can significantly degrade sensing performance. This study presents a scalable, noncontact cleaning approach using pulsed surface *acoustic* waves (SAW) on a 5000 mm2 LiDAR lens cover. The system operates at 9 kHz and utilizes two different piezoelectric configurations, i.e., two-piezoelectric (2-piezo) or six-piezoelectric (6-piezo) elements. Its cleaning efficiency was evaluated under 18 waveform conditions, and 2-piezo configuration yielded superior cleaning results, achieving up to 93.8% and 88.1% efficiency by droplet count and mass loss, respectively, due to its concentrated *acoustic* displacement. To enhance cleaning under frozen or muddy conditions, a 20 nm-thick Indium Tin Oxide (ITO) layer was deposited to provide localized heating. Scanning electron microscopy imaging confirmed uniform ITO deposition and efficient *thermal* activation. This work demonstrates an effective, large-area SAW cleaning strategy and lays the groundwork for future ultrasonic frequency (>20 kHz) implementations in *noise*-sensitive environments.

Conference Paper • *Open access*

148 **The Cool, Quiet City machine learning competition: Overview and results** — Miller C.,Ibrahim M., Akbar I.S.,... Quintana M., Fu C. — Journal of Physics Conference Series , 3140(11), 112017 — 2025 — 0

Hide abstract Find It @ UCL View at Publisher Related documents

The prediction of *thermal* and *noise*-based preferences in the urban context is valuable in characterizing interventions to mitigate the challenges of health, productivity, and satisfaction of urban dwellers. The growth of crowd-sourced data and data-driven techniques provides an opportunity to increase the understanding of which machine learning models are most accurate and applicable for this context. This paper outlines the results of a machine learning competition aiming to enhance the accuracy of predicting human comfort in the city context. The competition asks contestants to use contextual data to predict *noise* distraction and *thermal* preference in various indoor and *outdoor* spaces. This competition included the city-scale collection of 9,808 smartwatch-driven micro-survey responses that were collected alongside 2,659,764 physiological and environmental measurements from 98 people using an open-source watch-based platform combined with geolocation-driven urban digital twin metrics. This paper explains the two best solutions to this competition and provides a discussion of the factors that may have contributed to their accuracy of more than 0.7 in multiclass tasks. These solutions notably included the use of LightGBM, XGBoost, CatBoost, and simple Neural Networks while avoiding overly complex solutions such as deep learning or recurrent architectures, which offer limited advantages for structured data classifications.

Article

149 **An Evaluation Method for Ground Resolution of Space-based Optical Payloads | 空基光学载荷地面分辨率评价方法** — Zang L.,Li H.,Chen S., ... Liang D.,Liu W. — Binggong Xuebao Acta Armamentarii , 46 — 2025 — 0

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The image quality of space-based optical payloads is a crucial index for evaluating their imaging performance. Significant differences exist between laboratory test (under ideal conditions) and *outdoor* practical application (in complex environments). It proposes a comprehensive evaluation of the ground resolution of the optical payload using bar targets, conducted through testing both in the laboratory and outdoors, with the field test results serving as the final assessment criterion. Practical testing is conducted based on a space-based optical payload. The research results indicate that the platform vibration, atmospheric turbulence, *thermal noise*, and other factors lead to the

	Document title	Authors	Source	Year	Citations

actual spatial resolution lower than the theoretical value, with their combined effects causing a system performance degradation of 15% - 32%. These findings provide an important basis for the objective and quantitative evaluation of the practical operational capability of space-based optical payloads and point out the main directions for performance optimization.

Article

150 **Phase change composite-enabled flexible thermoelectric cooling for personal thermal-regulation**

Cao L.,Zhang Q.,Cen J., ... Yang C.,He Z.

Journal of Energy Storage , 105, 114757

2025

13

Hide abstract Find It @ UCL View at Publisher Related documents

Thermoelectric coolers (TECs) are expected to be utilized in personal *thermal* management applications due to their environmental friendliness, *noise*-free operation, lightweight design, and fast response time. However, the cooling efficiency of TECs heavily relies on the heat sinks of the hot side, which are typically rigid and bulky, causing them unsuitable for wearable devices. To address this issue, this work developed a phase change flexible heatsink utilizing copper foam as the framework and a ternary composite phase change material as the filler, exhibiting a high *thermal* conductivity of 4.92 W/(m·K) and latent heat of 139.64 J/g. A phase change composite-enabled TEC with excellent flexibility (bending radius of 15 mm) is thus successfully prepared. It can achieve a stable temperature drop of 17 °C over 6 h. An intelligent temperature-controlled headband has also been designed by integrating with a wireless communication module for precise temperature regulation via a mobile phone app interface. This headband ensures optimal comfort by maintaining forehead temperature around 32 °C even when ambient temperatures present large fluctuate. The temperature-controlled headband could be applied to emergency medicine settings, healthcare facilities, and *outdoor* sports activities.

Conference Paper • *Open access*

151 **UASTHN: Uncertainty-Aware Deep Homography Estimation for UAV Satellite-Thermal Geo-Localization**

Xiao J.,Loianno G.

Proceedings IEEE International Conference on Robotics and Automation

2025

5

Hide abstract Find It @ UCL View at Publisher Related documents

Geo-localization is an essential component of Unmanned Aerial Vehicle (UAV) navigation systems to ensure precise absolute self-localization in *outdoor* environments. To address the challenges of GPS signal interruptions or low illumination, *Thermal* Geo-localization (TG) employs aerial *thermal* imagery to align with reference satellite maps to accurately determine the UAV's location. However, existing TG methods lack uncertainty measurement in their outputs, compromising system robustness in the presence of textureless or corrupted *thermal* images, self-similar or outdated satellite maps, geometric *noises*, or *thermal* images exceeding satellite maps. To overcome these limitations, this paper presents UASTHN, a novel approach for Uncertainty Estimation (UE) in Deep Homography Estimation (DHE) tasks for TG applications. Specifically, we introduce a novel Crop-based Test-Time Augmentation (CropTTA) strategy, which leverages the homography consensus of cropped image views to effectively measure data uncertainty. This approach is complemented by Deep Ensembles (DE) employed for model uncertainty, offering comparable performance with improved efficiency and seamless integration with any DHE model. Extensive experiments across multiple DHE models demonstrate the effectiveness and efficiency of CropTTA in TG applications. Analysis of detected failure cases underscores the improved reliability of CropTTA under challenging conditions. Finally, we demonstrate the capability of combining CropTTA and DE for a comprehensive assessment of both data and model uncertainty. Our research provides profound insights into the broader intersection of localization and uncertainty estimation. The code and models are publicly available.

Conference Paper

152 **Behind The Walls: Unlocking Occupant Mindsets and Motivations for Building Envelope Retrofits**

Roiger C.,Coffin S.G., McCord K.,... Fuentes T., Antonopoulos C.

Thermal Performance of the Exterior Envelopes of Whole Buildings

2025

0

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

Upgrading building envelopes to reduce infiltration and increase *thermal* performance is a well-established way to improve energy efficiency in buildings. In addition to the cost savings associated with decreased energy> use, occupants who improve their building envelope can enjoy non-energy impacts of these energy efficiency measures. Increased protection from *outdoor* air contaminants, attenuation of unwanted *noise*, and better ability to maintain *thermal* comfort are potential co-benefits of envelope upgrades. Although these non-energy impacts (NEIs) are frequently cited, research is needed to determine which benefits motivate people to make envelope upgrades. To answer this questions, this paper explores the household decision-making dynamics associated with envelope upgrades using the UPGRADE-E dataset, the largest U.S. household decision-making survey to date (Fuentes et al. 2024). We examine home priorities and considerations that are associated with envelope upgrades. For instance, variables such as home age and occupant prioritization of aesthetics are explored as motivating factors for pursuing exterior envelope upgrades, such as updating siding or roofing. Overall, we adopt a holistic approach to understand how occupants decide to upgrade their building envelope and what additional benefits will co-occur with any upgrades. This research can help envelope-related energy program developers more comprehensively understand how best to reach target audiences regarding envelope upgrades.

Conference Paper

153 **Design and Development for Power Supply Circuit of a Low-Light Detection Imaging System** — Ailing L. — Proceedings of SPIE the International Society for Optical Engineering , 13544, 135440B — 2025 — 0

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The low-light detection remote sensing system is capable of imaging at night and in the morning and evening, which can effectively make up for the shortcomings of conventional satellite dynamic monitoring capability. It can improve China's emergency response speed to major natural disasters and major events, and effectively increase the depth and breadth of the application of earth observation technology in the field of disaster prevention and mitigation. In a pre-project about low-light detection principle prototype development, the Image Intensifier coupled CCD sensor (ICCD) is used as the core imaging device of the low-light detection system, the entire system consists four parts such as a detector ICCD module, the main control and processing unit FPGA module, signal processing circuits and power supply module . Among them, the power supply is used to provide low-*noise* power for the other three modules. The main control pulse which comes from FPGA of master control processing unit, can drive the detector ICCD. And then the ICCD will get low-light radiation through the optical system and finish photoelectric conversion. Ultimately the low-light detection system can output high-speed digital image signal data. On the basis of fully analyzing the input power ripple, electromagnetic interference generated by the layout and wiring of power supply PCB components and *thermal noise* generated by heat dissipation on the imaging performance of the low-light detection system, this paper designs a power supply circuit system which combined several DC/DC modules and a variety of LDOs (low-voltage differential voltage regulators). Starting from the selection of components, PCB layout and wiring, power-up sequence design, etc. The single power supply of the satellite platform is converted into 10 kinds of power supply circuits with different load currents, functions and performance requirements, such as +18V, +13.5V, +12V,+11V, +8V, +7V, +5V, +3.3V, +1.5V and -7V. After single-board debugging and system testing, the output ripple of each power supply is measured to be no more than 20mV, low *noise*, high stability and anti-jamming, which meets the design objectives of the power supply circuit. Ultimately the entire low-light detection system obtain high-quality low-light images in the *outdoor* low-light conditions.

Conference Paper • *Open access*

154 **ALGORITHM FOR OPTIMAL CONTROL OF ENERGY-EFFICIENT SYSTEMS ENSURING ABIOTIC FACTORS FOR ENTOMOLOGICAL PRODUCTION** — Belchenko V., Pishchanska N., Bulgakov V.,... Rucins A., Chernysh O. — Engineering for Rural Development , 24 — 2025 — 0

Hide abstract Find It @ UCL View at Publisher Related documents

Document title	Authors	Source	Year	Citations

The article is dedicated to the development of an optimal control algorithm for energy-efficient systems that provide abiotic factors. The result of the algorithm operation is a final set of conditions, instructions, and actions executed in a specific order, serving as a tool for determining methods for forming energy-efficient systems that ensure abiotic factors in adaptive technologies for cultivating maternal entomocultures. The algorithm consists of six main blocks. The first block defines an automatic air parameter regulation system in microclimate systems based on the type of the air parameter being regulated, the magnitude of the parameter deviation, the method of maintaining the parameter, and the nature of the regulator's influence on the controlled object. The types of systems that meet all the constraints of entomoproduction facilities are considered. The second block determines the conditions for optimal system control using a *thermodynamic* model (TDM). The analysis of these conditions allows the formulation of fundamental principles by which TDM of air conditioning systems (AirCS) ensures optimal air treatment processes, a combination of specific *outdoor* air parameters and indoor air parameters that minimize cooling, heating, water, and air consumption. The third block involves processing the basic information and performing key calculations and operations. The fourth block determines the types of technological schemes for maintaining environmental parameters indoors, considering options such as preheating *outdoor* air, operating without preheating, or using preheating for recirculated air. The fifth block is responsible for presenting the algorithm's output, defining the conditions for controlling energy-efficient systems that provide abiotic factors. The final, sixth block repeatedly performs a specific operation until a predefined condition is met. The information in this block concerns methods of controlling humidification processes – regulating water consumption, bypassing the airflow, or managing intermittent water supply. Capital costs for AirCS in entomological production amount to 20% of the total construction cost, while operating expenses range from 30% to 50% of total operating costs. Therefore, energy saving has been one of the most crucial functions of automatic control systems (ACS) in air conditioning. During the design stage of ACS and the development of technological air treatment schemes, it is necessary to refine the possibilities of process control. As a result of the research, key schemes for maintaining environmental parameters in indoor spaces have been developed to minimize air heat and moisture treatment costs. The developed algorithm enables the generation of a set of conditions, instructions, and actions that must be performed in a specific order. Given detailed initial data and knowledge of ACS equipment management methods, following each step of the developed algorithm provides a technically *sound* ACS solution for air conditioning systems, reducing energy consumption by approximately 25%. The algorithm's outcome serves as a tool for determining methods for forming energy-efficient systems that ensure abiotic factors in adaptive technologies for cultivating maternal entomocultures.

Article

155 **Image Registration and Deep Neurofuzzy Networks for Mitigating Atmospheric Turbulence Effects in Consumer-Based Optical Imaging** — Ali Khowaja S.,Ali U., Dev K.,Lee I.H. — IEEE Transactions on Consumer Electronics, 71(2) — 2025 — 1

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Consumer-based optical imaging systems are characterized as big data processing systems, which are drastically affected by atmospheric turbulences that add geometric distortions and blur effect to the images when used in *outdoor* condition. Physics-grounded simulators have been proposed recently to generate synthetic data but the generalization to the real-world turbulent images is not so good. In this paper, we combine the characteristics of image registration, deep neurofuzzy methods, and channel-attention based discriminative learning strategy to propose image registration, neurofuzzy based denoising, and deblurring network (RND2Net). The RND2Net is designed on a principle that it does not require turbulent image pairs (ground truth images) to train the network, which closely resembles the real-world situation used as consumer devices. The registration module focuses on the region-based fusion techniques while the denoising and deblurring module incorporates deep neurofuzzy network along with dense residual blocks and channel attention mechanism to train the network. The RND2Net is also designed to reduce the *noise* and blur effect from images, while generalizing on the down-stream tasks, such as text recognition. Experimental results show that the RND2Net yields better performance quantitatively as qualitatively on synthetic and real-world datasets in comparison to existing state-of-the-art methods.

Book Chapter

156 **Novel building materials** — Fabiani C.,Pisello A.L., Paksoy H. — Comprehensive Energy Systems — 2025 — 0

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

The selection of building envelope materials plays a crucial role in enhancing building energy efficiency and creating optimal indoor-*outdoor* comfort conditions in the built environment, while also promoting environmental sustainability. Therefore, a comprehensive and interdisciplinary understanding of building envelopes and the latest advancements in building materials is essential for designing environmentally friendly and intelligent structures. This chapter delves into the fundamentals of building physics, including hygrothermal, *acoustics*, and lighting performance within the built environment. It also examines the key properties of various building materials, both traditional and innovative, with a specific emphasis on achieving energy efficiency and environmental sustainability objectives.

Conference Paper

157 **Experimental Prototype of an Indoor Green Façade for Assessing Its Effects on Building Climate** | Convertino F.,Schettini E., Vox G. | Lecture Notes in Civil Engineering , 586 LNCE | 2025 | 1

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Green infrastructures are a sustainable solution for improving overall quality of today's urban and peri-urban living environments. Among these systems, green façades involve the application of vegetation on the buildings' vertical walls. These can provide various benefits, such as the improvement of the building aesthetics, air quality, wall *acoustic* insulation and the energy performance of the building, thus reducing energy costs and promoting microclimatic well-being. Benefits can be achieved in both *outdoor* and indoor applications. These technologies can be applied following different construction solutions and to buildings with various uses. Most of the available research has focused on *outdoor* applications, while this study focuses on an indoor green façade, with the aim of studying its functioning. The first step of this research, the design of the system realized at the University of Bari, is presented. The experimental prototype consists of a sealed chamber to thoroughly investigate *thermal* and gas exchanges, thus analyzing the ability of vegetation to improve the quality of the confined environment. Evergreen plants of Syngonium podophyllum are placed inside the chamber. Several sensors are integrated for detecting and monitoring physical parameters of air, soil, and plants. The system is provided with ventilation. The designed system could be further developed during the research. The analysis of the measured parameters will help to clarify the functioning of the system and to quantify the induced effects. This will be useful not only for designers and experts, but also for all those who wish to implement indoor green façades.

Article

158 **Q-Learning Inspired Geometric Shaping Optimization of APSK Constellations for the Nonlinear LEO-LEO Satellite Channels** | Dong F.,Wei Y.,Zhang H., ... Shi J.,Chi N. | Journal of Lightwave Technology , 43(14) | 2025 | 12

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The low-earth-orbit (LEO) satellite communication has been proposed to support the next generation 6G space-air-ground integrated network owing to its large coverage area and seamless connectivity. To overcome the challenge of long-distance transmission for the free space optical (FSO) links, satellite communication systems often rely on high-power transmission to achieve high signal-to-*noise* ratios (SNRs) and high throughput, albeit at the expense of increased nonlinearity. In this paper, we propose a Q-learning inspired geometric shaping (QLI-GS) optimization scheme for amplitude phase shift keying (APSK) constellation to mitigate nonlinear effects and maximize the transmission distance. A pairwise optimization (PO) algorithm inspired by Q-learning is employed to achieve QLI-GS optimization for the APSK constellation according to an enhanced nonlinear interference *noise* model (ENINM) with an effective memory mechanism, which ensures fast convergence and suppresses nonlinear impairments. Our results show the iteration time of the QLI-GS-APSK is reduced by approximately 99.89% compared to the generalized mutual information (GMI) maximized GS algorithm, and it also offers greater nonlinear tolerance than conventional multi-ring APSK and traditional PO-APSK. By leveraging a fiber-optic-sampled bidirectional dual-terminal FSO equivalent link, the QLI-GS-32APSK can experimentally achieve a data rate of 100 Gbit/s over a 230 km distance based on a single-wavelength system. A maximum distance of 580 km at a data rate of 60 Gbit/s using QLI-GS-8APSK has been successfully demonstrated under the 7% hard decision-forward error correction (HD-FEC) threshold of $3.8e^{-3}$, with a corresponding link budget of 62 dB. Additionally, the robustness of QLI-GS optimization under different atmospheric

	Document title	Authors	Source	Year	Citations

turbulence conditions has been further validated. We demonstrate a single-carrier 100Gbit/s QLI-GS-32APSK transmission over a 4.3 km *outdoor* FSO link with the Fried's coherence length of 11 cm, achieving 0.9 dB shaping gain over conventional 32APSK. The results highlight the promising potential of the proposed QLI-GS-APSK scheme in addressing nonlinear challenges in satellite communications.

Article

159 **Simulation study of dynamic air exchange rates in multizone environments in naturally ventilated residences | 住宅多区域动态自然通风换气量的仿真研究** Jian Y.,Gai X.,Fan C.,Liu S. Qinghua Daxue Xuebao Journal of Tsinghua University, 65(5) 2025 1

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[Objective] Natural Ventilation in real residential environments is characterized by dynamic multizone airflow.Ignoring either of the two features can lead to inaccurate assessments of natural Ventilation in residences. The multiple tracer gas method has traditionally been used to investigate this multizone airflow within residential settings. However, it is impractical for large-scale applications owing to the high costs associated with the measurement process, potential disruption to occupants' daily lives, and the need for stähle Ventilation conditions. As a result, accurately measuring air exchange rates remains a significant challenge. A more in-depth study of the measurement method for dynamic multizone airflows is urgently required. This study proposes a Simulation method for identifying dynamic multizone airflows in naturally ventilated residences.[Methods] This method utilizes COz emitted by occupants as a tracer gas to study multizone airflow in residential buildings. It considers all feasible multizone airflow patterns using a traversal approach and the air volume conservation principle. Furthermore, to strike an optimal balance between effectively tracking the dynamic characteristics of natural Ventilation and minimizing *noise* sensitivity, a transient indoor C02 mass balance equation associated with the Kaiman filter is applied to each zone. The resulting time series of air exchange rates can be presented for each airflow pattern. These rates are then evaluated to identify the pattern that most closely aligns with the actual airflow pattern and the corresponding *outdoor*-indoor air exchange rates and interzonal airflow rates. Furthermore, two Validation experiments were conducted in an unoccupied two-bedroom apartment with controllable Ventilation patterns to validate the method. Subsequently, the method was employed in an occupied apartment» utilizing measured indoor C02 concentrations and occupancy data for each zone to produce the time series of air exchange rates.[Results] The comparison among the calculated air exchange rates using the proposedmethod and experimental data indicates that 85% of the calculated values have absolute errors within + 0. 2 h, and 95% fall within + 0. 4 h. Furthermore, 75% of the calculated values have relative errors within + 10%, and 95% are within + 20%. The calculated air exchange rates and airflow directions closely match the experimental conditions, indicating that the method proposed in this study effectively represents the multizone aspects of natural Ventilation in residential environments. Moreover, the applicability of the method to real residences is demonstrated through its application in an occupied apartment. The calculated air exchange rates for each zone during the measurement period, after filtering out anomalous results, fall within a reasonable ränge. These results present the airflow patterns that characterize the multizone nature of dynamic natural Ventilation.[Conclusions] Natural Ventilation is complex owing to its multizone nature and time dependence, leading to data scarcity. This method effectively addresses this gap by quantifying the multizone representation of dynamic airflows in residences on a large scale. Understanding indoor - *outdoor* air exchange rates and interzonal airflow rates is pivotal, as these Parameters significantly influence indoor *thermal* conditions and air quality. In this regard, this study offers a valuable and practical approach to comprehensively understanding natural Ventilation and its effects on occupants' health conditions in real residential environments.

Conference Paper

160 **Impact of Increased Ventilation Rates in Classrooms on Energy Usage and System Functionality in a US Midwest Climate** Passe U.,Farrar J.,Mahmoodi D. Ieq Conference, C60 2025 0

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The COVID-19 pandemic brought building ventilation into the focus of public health. Low ventilation rates increased the rate of respiratory infection. Ventilation in ASHRAE climate zone 5A directly impacts energy use due to the extremely cold winters and warm-humid summers. This paper assesses the spatial, energy, and HVAC needs of

Document title	Authors	Source	Year	Citations

ventilation upgrades for five major school types in an Iowa school district. The schools were selected based on time and type of construction, location and orientation of classrooms, and type of HVAC system. Recent ASHRAE Standard 62.1 - 2016 for Ventilation for Acceptable Indoor Air Quality required a combined *outdoor* air rate of 6.7 L/s/p or 14.2 CFM/p based on the default occupant density in classrooms (age 9 plus). Based on review of post-COVID-19 literature, a revised 'Minimum Fresh Air Rate per Person' (MFApP) of at least 20 L/s/p or 42.4 CFM/p is recommended by ASHRAE 241 - 2023. To determine the necessary building operations changes for these increased MFApP rates, *thermal* models were developed in the energy simulation tool ClimateStudio for one classroom per school. Load calculations and assembly systems reflected construction documentation data. ClimateStudio uses a preset MFPApP rate of 4.72 L/s/p or 10 CFM/p based on ASHRAE 62.1 - 2022, which was compared to 6.7 L/s/p or 14.2 CFM/p and 20 L/s/p or 42.4 CFM/p with respect to annual and seasonal building energy consumption. Furthermore, the pressure drops across the ductwork were assessed in the current HVAC systems to understand if ducts and fan motors could handle these increased ventilation rates. The results reveal that as MFApP rates increase, site EUI increases by up to 30% for the highest CFM/p depending on the system. In most cases ductwork and fan motors would need to be replaced to continue efficiency and keep *noise* levels low. This paper thus assesses energy use and HVAC systems impact of improved ventilation rates to allow the school district to dedicate funds effectively and keep energy cost increases as low as possible while improving Indoor Environmental Quality (IEQ).

Conference Paper

161 **The Impact of Personal and Contextual Factors on Multi-domain Subjective Comfort Responses in Classrooms** — Morandi F.,Gasparella A.,Tzempelikos A.,Pittana I.,Cappelletti F. — Ieq Conference, C56 — 2025 — 0

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Subjective assessment of the indoor environmental quality (IEQ) and multi-domain comfort depends on environmental parameters as well as on contextual and individual characteristics. The latter is even more important in educational buildings, where students share the same space, affecting their comfort and response in multiple domains. This work presents the impact of personal (i.e., gender, clothing level, body mass index, age, use of glasses) and contextual (i.e., sitting position inside the classroom and orientation/*outdoor* view) factors on the students' sensation votes (*thermal*/air quality/visual/*acoustic*) collected in 50 classrooms (825 questionnaires) during regular lectures. First, we identified which environmental variables were significantly correlated to sensation votes in each of the four domains and studied the role of personal/contextual factors on cross-domain interactions through multi-linear regression. Then, the impact of categorized levels of personal/contextual factors on sensation votes was studied through covariance analysis and post-hoc tests. This analysis determined which of these factors would enable different responses when exposed to the same environmental conditions. The results showed that gender and age affect multi-domain sensation votes, while body mass index and clothing level affect *thermal* sensation. Males have a larger sensitivity to environmental conditions in the *thermal* and IAQ domains. Clothing adaptation varies depending on gender, sitting position and body mass index. Younger students express higher *thermal* sensation votes and are more sensitive to noisy environments. In addition, sitting distance from windows affects *thermal*, IAQ, and visual sensation.

Conference Paper

162 **AI Driven Air Quality Analysis for Health: An Experimental Review** — Samyuktaa S.,Chandra J.,Immanuel A. — Proceedings of the 2025 International Conference on Emerging Techniques in Computational Intelligence Icetci 2025 — 2025 — 0

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Air pollution, both indoor and *outdoor*, was linked to 6.7 million premature deaths in 2020, including over 237,000 children under the age of 5, according to WHO. Indoor Air Pollution (IAP) is a crisis of public health that affects billions of people by exposing them to IAP pollutants like particulate matter (PM2.5), volatile organic compounds(VOCs), polycyclic aromatic hydrocarbons (PAHs), and carbon monoxide (CO). The most common cause of IAP varies from

	Document title	Authors	Source	Year	Citations

incense burning and biomass fuel to ventilation, leading to a horrific human health effect by causing respiratory disease, cardiovascular disease, sick building syndrome, and mental impairment. This review brings together evidence from various studies on the effects of indoor air quality on the environment, health, and productivity. Apart from pollutant exposure, determinants of well-being, i.e., *thermal*, *acoustic*, and visual comfort, are the subject of this article. Developments in artificial intelligence (AI), the Internet of Things (IoT), and computational modeling have revolutionized Indoor Air Quality monitoring to detect pollutants and exposures in real-time. All these technologies have the potential to intervene effectively but are intimidating through the prism of high cost, sensor calibration, and the need for large-scale epidemiological studies. To restrict indoor air pollution risks, inter-disciplinary studies need to be adopted to combine effective ventilation technologies and advanced pollutant control systems. Large-scale applications of clean fuel like solar, biogas, electricity, liquefied petroleum gas (LPG), and efficient biomass stoves need to be employed to restrict home air pollution. The present review calls for an emergent public campaign and policy intervention to enhance indoor air quality, health, and well-being.

Article

☐ 163 **Research on Ultra-Long-Distance Quasi-Line-of-Sight High-Speed Ultraviolet Communication Systems | 超长距离准视距高速紫外光通信系统研究**

Cao F.,Yang T.,Meng N., ... Leng Y.,Zhang Z.

Guangxue Xuebao Acta Optica Sinica , 45(19), 1906003

2025

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Objective Ultraviolet (UV) wireless optical communication (OWC) has recently emerged as a complementary technology to traditional radio-frequency (RF) communications, particularly due to its capabilities in providing non-line-of-sight (NLOS) coverage, high security, and reliable performance in challenging environments such as dense forests, smoke-filled areas, and rugged terrains. With increasing demands in 5G/6G networks for higher bandwidth, lower latency, and enhanced security, UV communication, especially within the deep UV range (200 — 280 nm), has attracted considerable research interest. Although significant advancements have been made in deep UV light-emitting diode (LED)-based systems, existing practical implementations are generally limited to communication links under 1 km and data rates below 1 Mbit/s. Furthermore, traditional strategies to enhance communication reliability—such as spatial diversity architectures or high-redundancy coding—introduce significant system complexity, cost, and power consumption. Therefore, this study aims to overcome these limitations by establishing an integrated UV LED-based communication system capable of achieving real-time, ultra-long-range (>1.5 km), and high-speed (up to 2 Mbaud) transmission without reliance on spatial diversity or complex redundancy coding. Methods In this study, an integrated UV LED wireless optical communication system employing a single-source, singledetector configuration combined with time-domain diversity techniques is proposed and experimentally validated. To mitigate severe channel fading caused by atmospheric turbulence and scattering, a novel adaptive dual-threshold demodulation algorithm and a time-diversity combining technique are developed. The system consists of a high-powerdensity UV LED transmitter module operating at 280 nm wavelength, driven by an NFPC6J chip and modulated through a DD311 driver. A total internal reflection (TIR) collimation lens with a beam divergence of approximately 10° is integrated to reduce divergence losses. At the receiving end, a narrow-band optical filter (Edmund-67813) centered at 280 nm with an optical density (OD) ≥4 is implemented along with additional structural shielding to effectively suppress background interference. A photomultiplier tube (PMT, Hamamatsu R7154) with single-photon detection capabilities serves as the detector, followed by a two-stage amplification scheme consisting of a transimpedance amplifier (LTC6268-10) and an automatic gain control (AGC) amplifier (AD8336). Signal digitization is achieved using a 14-bit, 125 MSa/s analog-todigital converter (ADC, AD9246), with data subsequently processed in real-time using a Zynq-7020 FPGA-based embedded system. The field programmable gate array (FPGA) implements essential signal-processing tasks, including frame synchronization, time-diversity combining through equal-gain combining (EGC), adaptive dual-threshold decision making, and dynamic AGC management. Communication modules are integrated within compact enclosures measuring 8 cm×7 cm×13 cm, designed to facilitate rapid deployment and integration with smartphones for real-time voice and text communications. Theoretical justification for the system relies on modeling atmospheric channel dynamics based on Poisson statistics of photon arrivals and Gaussian-shaped PMT output pulses. The temporal independence of atmospheric fading events is leveraged through time-diversity, effectively enhancing the received signal-to-*noise* ratio (SNR) and reducing bit-error rates (BER). Results and Discussions *Outdoor* field experiments are conducted over a 1.51 km free-space communication link under realistic conditions (temperature: 23.2 — 38.3 ℃ ; humidity: 90%; cloudy conditions), with experimental equipment positioned at opposite ends of a

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roadway (Fig. 3 and Fig. 4). Measurements show significant attenuation and random signal fluctuations due to atmospheric effects, including photon-count variations, pulse-arrival timing uncertainties, PMT gain fluctuations, and multiphoton pulse overlap (Fig. 5). The performance of the proposed time-diversity scheme is analyzed through waveform comparisons at various repetition counts (Fig. 6). Results indicate that increasing the repetition count substantially smooths signal fluctuations and effectively mitigates deep-fading events. Additionally, statistical analyses of the amplitude distributions show a transition from multimodal distributions toward Gaussian-like single-mode distributions as repetition count increases (Fig. 7), consistent with the central limit theorem and indicative of reduced signal variance and improved robustness. Bit-error-rate performance is evaluated across symbol rates from 1.25 to 2 Mbaud. Results reveal that error-free transmission is reliably achieved at symbol rates up to 1.5 Mbaud when employing five or more time-diversity repetitions. At the highest tested symbol rate (2 Mbaud), a repetition count of eight maintains the BER at approximately 10-4 (Fig. 8). These experimental results validate the effectiveness of the proposed time-diversity and adaptive threshold techniques in achieving stable, high-speed, ultra-long-distance UV LED communication without spatial diversity or redundant coding schemes. Conclusions In conclusion, this study successfully demonstrates — for the first time — real-time, ultra-long-range (1.51 km), high-speed (up to 2 Mbaud) UV wireless communication using a single LED source and single PMT detector combined with an innovative time-domain diversity and adaptive dual-threshold demodulation approach. The proposed methodology significantly reduces hardware complexity and cost, overcoming previous constraints associated with traditional spatial diversity and redundancy coding approaches. Experimental validation under realistic *outdoor* conditions confirms that the system achieves robust and reliable communications, satisfying the bit-error-rate requirements for realtime voice transmission. These results establish a strong foundation for future practical implementations, particularly in emergency response and tactical communication applications. Future work will explore online adaptive threshold determination, low-complexity forward error correction (FEC) coding, and further optimization of LED devices and optical systems to enhance system performance, energy efficiency, and practical deployability.

Conference Paper

☐ 164 **Turbulent Transition Model Validation with Multiple Rotors for Industry Application** | Min B.-Y.,Wake B.E. | AIAA Aviation Forum and Ascend 2025 | 2025 | 1

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The Langtry-Mentor transition model with k-ω Shear Stress Transport (SST) model has been assessed for multiple rotors. In this assessment, an adverse impact of the sustaining term has been identified and a modification has been introduced. It was found that the original sustaining term behaved poorly for the separated flows investigated herein, making separation worse and leading to a premature stall in hover. The modified sustaining term uses F1-$\mu_t$blending to deactivate the sustaining term inside boundary layer. The Langtry-Mentor transition model with the modified sustaining term has been validated for multiple rotors with various scales and characteristics, including the XV-15 rotor, the Hover Validation and *Acoustic* Baseline (HVAB) rotor, and Sikorsky's model-and full-scale rotors. Helios, with OVERFLOW as near-body solver, was used for these calculations. The validation showed very good correlation with measured hover figure of merit data for all rotors from low thrust to high thrust conditions, using a consistent transition model and sustaining-term settings. It was found that the modified sustaining term removed the adverse impact of the original sustaining term.

Article

☐ 165 **An efficient multi-modal perception based metal fragment detection algorithm in outdoor environments** | Li X.,Liu C.,Liang F.,Li G. | Signal Image and Video Processing, 18(12) | 2024 | 2

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Current multi-modal based target detection methods for metal debris in *outdoor* environments easily suffer from specific modal *noise* interferences and insufficient multiple feature fusion, which result in low recognition effectiveness. To address these challenges, we propose a novel algorithm for metal fragment detection based on visible light and *thermal* infrared images. Specifically, we firstly design a Bidirectional Frequency-Domain Adaptive Attention Gating (BAFA-Gate) module, to capture the different channel feature correlation and corresponding feature weight vectors among varying modalities, which can be beneficial to reducing modal *noise* interferences and further improving detection accuracy. In addition, we construct a feasible Global Feature Fusion module (GFF), to perform

	Document title	Authors	Source	Year	Citations

both intra-modal and inter-modal fusion of normalized features, which can help to obtain global contextual information and enhance the feature fusion capability. Finally, a series of experiments are conducted using the constructed dataset, and the corresponding experimental results validate the effectiveness and feasibility of the proposed method.

Article • *Open access*

166 **The Adjusting Effects of Trees on Cfa-Climate Campus Acoustic Environments and Thermal Comforts in the Summer**

Lu W.,Chen Y.,Zhou T., ... Yin H.,Liu T.

Acoustics, 6(4)

2024

3

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This study explores the effects of trees on the *acoustic* and *thermal* environment in addition to people's responses to trees in different contexts. Through field measurements conducted during the summer of 2023 at the campus of the Southwest University of Science and Technology in Mianyang, residents' neutral points were locally found to be 52.2 dBA (*acoustic*) and 23.8 °C (*thermal*). Further, at their maximum, the trees were able to reduce heat stress by 4 °C (indicated by the physiologically equivalent temperature—PET) and the *noise* level by 10 dBA (indicated by the A-weighted *sound* pressure—LAeq); this was achieved by trees with a crown diameter of 20 m. Subjective *acoustic* and *thermal* responses varied depending on the context. *Acoustically*, their neutral LAeq values toward the *sounds* of traffic, teaching, sports, and daily life were 46.9, 52.5, 51.0, and 52.7 dBA, respectively. *Thermally*, pedestrians' neutral PET values were 24.2, 26.1, 22.3, and 25.1 °C, respectively, under the same conditions. These phenomena might be a consequence of the effects of *sound* frequencies. Future urban forestry research should focus on planting for environmental quality improvement.

Article • *Open access*

167 **The effects of acoustic-light-thermal environment quality parameters on pedestrians' overall comforts in residential districts**

Zeng F.,Liang S.,Zhang J., ... Liu T.,Wang S.

Scientific Reports, 14(1), 19625

2024

13

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This study explored subjective responses towards various environmental quality parameters. It would be helpful to approach an overall comfort improvement relating to physics. In order to investigate the combined effects of *sound*, light and heat on the overall comfort of pedestrians, which could be evaluated by overall comfort vote, two residential areas in Yubei District, Chongqing (a Cfa city in China) were selected for field measurements and questionnaires. Three were three key findings were concluded based on subjective responses to environmental parameters via multiple linear regression. First, the average *outdoor* neutral temperature, mean neutral *sound* level and neutral illumination intensity were determined to be 26.6 °C (determined by physiologically equivalent temperature), 56.5 dBA (determined by A-weighted equivalent continuous *sound* level) and 21.4 klx (determined by LUX), respectively. Second, considering the effects of both physiology and psychology, *thermal* perceptions varied for *acoustic* and light environments. Residents partially had lower neutral temperatures at the noisy condition. Third, environmental quality factors had a significant effect on overall comfort. Positive correlation between physiologically equivalent temperature and overall comfort vote was found (sig < 0.000); while the effects of *sound* pressure and light intensity were not always significant. Hence, *thermal* stress played a significant role in people's overall comfort. This study has explored the effects of three environment quality parameters on human perceptions. It provided better understanding against dwellers' feelings under complex circumstances. Future urban design and planning works should consider the cooling factors regarding contextual *acoustical* and visual environments.

Article • *Open access*

168 **Advanced Monocular Outdoor Pose Estimation in Autonomous Systems: Leveraging Optical Flow, Depth Estimation, and Semantic Segmentation with Dynamic Object Removal**

Ghasemieh A.,Kashef R.

Sensors, 24(24), 8040

2024

4

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	Document title	Authors	Source	Year	Citations

Autonomous technologies have revolutionized transportation, military operations, and space exploration, necessitating precise localization in environments where traditional GPS-based systems are unreliable or unavailable. While widespread for *outdoor* localization, GPS systems face limitations in obstructed environments such as dense urban areas, forests, and indoor spaces. Moreover, GPS reliance introduces vulnerabilities to signal disruptions, which can lead to significant operational failures. Hence, developing alternative localization techniques that do not depend on external signals is essential, showing a critical need for robust, GPS-independent localization solutions adaptable to different applications, ranging from Earth-based autonomous vehicles to robotic missions on Mars. This paper addresses these challenges using Visual odometry (VO) to estimate a camera's pose by analyzing captured image sequences in GPS-denied areas tailored for autonomous vehicles (AVs), where safety and real-time decision-making are paramount. Extensive research has been dedicated to pose estimation using LiDAR or stereo cameras, which, despite their accuracy, are constrained by weight, cost, and complexity. In contrast, monocular vision is practical and cost-effective, making it a popular choice for drones, cars, and autonomous vehicles. However, robust and reliable monocular pose estimation models remain underexplored. This research aims to fill this gap by developing a novel adaptive framework for *outdoor* pose estimation and safe navigation using enhanced visual odometry systems with monocular cameras, especially for applications where deploying additional sensors is not feasible due to cost or physical constraints. This framework is designed to be adaptable across different vehicles and platforms, ensuring accurate and reliable pose estimation. We integrate advanced control theory to provide safety guarantees for motion control, ensuring that the AV can react safely to the imminent hazards and unknown trajectories of nearby traffic agents. The focus is on creating an AI-driven model(s) that meets the performance standards of multi-sensor systems while leveraging the inherent advantages of monocular vision. This research uses state-of-the-art machine learning techniques to advance visual odometry's technical capabilities and ensure its adaptability across different platforms, cameras, and environments. By merging cutting-edge visual odometry techniques with robust control theory, our approach enhances both the safety and performance of AVs in complex traffic situations, directly addressing the challenge of safe and adaptive navigation. Experimental results on the KITTI odometry dataset demonstrate a significant improvement in pose estimation accuracy, offering a cost-effective and robust solution for real-world applications.

Article

☐ 169 **Mini-infrared thermal imaging system image denoising with multi-head feature fusion and detail enhancement network** — Wu H.,Chen B.,Guo Z., He C.,Luo S. — Optics and Laser Technology , 179, 111311 — 2024 — 16

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The mini-infrared *thermal* imaging system (MITIS) has been widely applied in *outdoor* photography, military observation, medical auxiliary diagnosis, and so on. However, due to the limitations of infrared sensor performance, imaging system size, and environment interference, the infrared images captured by the current MITIS generally encounter *noise* interference and low contrast issues. To solve these issues, we propose a multi-head feature fusion and detail enhancement network, namely MdNet, for complex features (texture and edges) retaining and denoising of infrared images obtained from the MITIS. Specifically, MdNet consists of a multi-head feature fusion block (MFFB), detail enhancement block (DEB), and reconstruction convolution. The MFFB is designed to extract and fuse multi-scale feature information, including multi-scale and feature fusion blocks. The DEB is developed to enhance the *noise* feature and reduce the loss of detailed information by three multi-level local feature attention blocks and residual learning. Quantitative and qualitative experimental results demonstrate that the MdNet achieves competitive performance in removing synthetic and natural *noises* compared with other state-of-the-art infrared image denoising methods and provides an effective solution for MITIS image denoising.

Review • *Open access*

☐ 170 **Indoor Air Quality in Naturally Ventilated Primary Schools: A Systematic Review of the Assessment & Impacts of CO2 Levels** — Honan D.,Gallagher J., Garvey J.,Littlewood J. — Buildings, 14(12), 4003 — 2024 — 44

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Indoor air quality (IAQ) in schools significantly impacts occupant health and academic performance, especially in naturally ventilated (NV) classrooms, where CO_2 levels are often elevated. This systematic review synthesises findings

Document title	Authors	Source	Year	Citations

from 125 studies, examining CO_2 as an indicator of ventilation rates (VRs) and its impact on IAQ, health, and academic performance in NV primary school classrooms. This analysis highlights seasonal and temporal variations in CO_2 concentrations, revealing a median CO_2 concentration of 1487 ppm across 2444 classrooms, with 81% exceeding the recommended 1000 ppm threshold. Influencing factors include VR, occupant density, generation rates, and occupant behaviours. Increased VRs consistently lowered CO_2 levels and enhanced IAQ. CO_2 concentrations correlated with particulate matter, volatile organic compounds, bioeffluents, microbial concentrations, and bacteria and fungi levels, but not with traffic-related pollutants like NO_2, which is associated with asthma prevalence. Elevated CO_2 levels consistently correlated with fatigue, headaches, respiratory symptoms, reduced academic performance and absenteeism, suggesting potential socio-economic benefits of increased VRs. However, effective IAQ management requires balancing ventilation with considerations of *thermal* comfort, *noise*, and *outdoor* pollutants. The findings highlight the need for standardised IAQ indices and CO_2 monitoring protocols, offering insights for future research, intervention design, and investment aimed at enhancing classroom environments.

Article • *Open access*

171 **Crowdsourced data as a strategic approach to include the human dimension in outdoor environmental quality assessments** — Cureau R.J.,Bou-Zeid E.,Pigliautile I.,Pisello A.L. — Science of the Total Environment, 953, 175925 — 2024 — 7

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Outdoor environments extend living spaces as venues for various activities. Comfortable open public spaces can positively impact citizens' health and well-being, thereby improving the livability and resilience of cities. Considering the visitors' perception of these environments in comfort studies is crucial for ensuring their well-being and promoting the use of these spaces. However, traditional survey methods may be time- and resource-consuming to gather significant sample sizes, usually focusing on selected homogeneous samples. Crowdsourced data, then, has emerged as an alternative for assessing human perception, as it eases the collection of subjective feedback and potentially amplifies impact and inclusivity. This study presents a strategic approach for analyzing publicly available and willingly reported crowdsourced data from a digital mapping platform in *outdoor* comfort evaluations, aiming to verify whether these data are informative regarding environmental quality perception and to identify the environmental factors that people are most sensitive to. Urban parks located in New York City served as a case study. A multi-source, interdisciplinary information framework combined crowdsourced reviews with environmental data used to determine prevailing *thermal* conditions. Overall perception of parks was well-rated, revealing that their attractions and activities are probably the most appealing characteristics for park attendance. Regarding environmental perception, *acoustic* and *thermal* factors are clearly the most influential. *Acoustics* were well-rated, while the main aspect regarding the *thermal* domain is the recognition of shading as a mitigator for hot conditions. Environmental data provided complementary insights, particularly concerning the range of *thermal* sensations experienced in urban parks. The findings confirm that willingly reported crowdsourced data can provide valuable insights into urban crowd environmental perception, presenting a potentially suitable and effective method to include the human perspective in environmental quality assessments, as well as to evaluate and predict environmental-related risks.

Article • *Open access*

172 **Influence of Subjective Factors on Window Use in Maternity Hospitals in Spring** — Xiao M.,Deng W.,Ma H., ... Wang Y.,Pan S. — Sustainability Switzerland, 16(22), 9808 — 2024 — 1

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Poor indoor air quality in maternity hospitals can spread respiratory diseases; however, limited research exists on modifiable factors like occupant behavior. This study explores subjective drivers of window-opening in maternity wards, using surveys and on-site measurements. Results show 71.4% of respondents stay less than 14 days, leading to dynamic and irregular window behavior. Comfort, particularly *thermal* comfort, air quality, and circulation, is the main driver for window operations. Especially at low temperatures, pregnant women's comfort plays a critical role, while other factors increase in importance as temperatures rise. The results show that environmental factors drive window-opening, while indoor comfort drives closing. Occupants are more tolerant to humidity than *thermal* discomfort, but window adjustments are random. Moreover, respondents prioritize others' needs in closing windows,

	Document title	Authors	Source	Year	Citations

where *outdoor noise* often serves as a key factor. The time also had a greater impact on both window opening and closing decisions, while field measurements confirm that time influences window-opening more than closing. A Multivariate Analysis of Variance (MANOVA) analysis of the questionnaire data for 'comfortable sensation' showed that only 'air circulation' and 'air humidity' were correlated with comfort, whilst 'heat sensation' was the least correlated. The possible reason is that the questionnaire was conducted in the spring, when temperatures were within the comfort range, and therefore, occupants were more concerned with other factors. Window operations in maternity wards are a collective strategy for natural ventilation. This study shows that the factors affecting window use in maternity hospitals are different from other buildings, providing useful ideas for improving maternity ward design.

Article

173 **Investigation of surface thermal characteristics induced by the acceleration of a submerged vehicle** — Wang B.,Du Y.,Qian Z.,Luo F.,Gao C. — Physics of Fluids, 36(11), 115171 — 2024 — 3

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Surface *thermal* characteristics are significant targets for non-*acoustic* detection technologies of underwater vehicles. This study aims to investigate the *thermal* characteristics induced by underwater vehicles during acceleration. The acceleration process of the self-propelled vehicle was obtained by simulation. Experimental and simulation studies were conducted using the temperature stratification measured in an *outdoor* pool as the background, while also considering the presence of a cool surface skin layer of the water. A comparative analysis between experimental and simulation results shows that the position and morphology of the surface *thermal* characteristics are consistent, and the area error of the *thermal* characteristics is within 15%. The findings reveal that, during the acceleration process of the underwater vehicle, the wake can still be divided into two stages despite the increasing speed. In the initial stage, the wake is relatively weak, making it difficult to form noticeable *thermal* characteristics on the water surface. Additionally, the study shows that at shallower depths, the acceleration state of the vehicle can be directly inferred from the *thermal* characteristics on the water surface, allowing for rough localization of the acceleration position. In contrast, at greater depths, direct determination becomes more challenging.

Article • *Open access*

174 **Effects of perceived environmental quality and psychological status on outdoor thermal comfort: a panel study in Southern China** — Lam C.K.C.,Pan H.,Nie W.,... Yin Z.,Han J. — Sustainable Cities and Society, 112, 105578 — 2024 — 39

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Urban residents are exposed to multiple environmental stimuli. Past studies have examined the influence of heat stress on *outdoor thermal* comfort, but the cross-modal effects of various sensory stimuli on *outdoor thermal* comfort remain unclear. This study used chi-square tests and logistic regression to examine the cross-modal effects of perceived environmental quality (*acoustic*, visual and air quality) and psychological status (irritability, perceived tiredness) on *outdoor thermal* comfort. We conducted *thermal* walk experiments in Guangzhou, China, in September and October 2022. This panel study consisted of mobile environmental sensors and *thermal* comfort surveys by fourteen university students on three walking routes, representing urban renewal sites, new urban areas, and urban blue-green spaces. The 'very bright' (OR=15.76, p<0.001) and 'glaring' groups (OR=4.09, p<0.001) were likelier to feel hotter than the 'dark' and 'not glaring' groups. People who found the *noise* level annoying were likelier to feel uncomfortable (OR=1.61, p=0.004). Moreover, people perceiving air quality as 'very good' were less likely to find the *thermal* environment unacceptable (OR=0.27, p=0.035). The irritated (OR=4.18, p<0.001) and tired groups (OR=1.81, p=0.002) were likelier to find the *thermal* environment intolerable. Our findings highlight the need to consider multisensory comfort and psychological status to inform sustainable urban design.

Article

175 **Analysis and Suppression of Hybrid Errors in Snapshot Diffractive Computational Spectral Imaging | 快照式衍射计算光谱成像混合误差分析与抑制** — Shen X.,Shao R.,Ma S.,... Ben Y.,Xue C. — Guangxue Xuebao Acta Optica Sinica, 44(19), 1911003 — 2024 — 2

Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

Objective Spectral imaging is a multidimensional information acquisition technology that combines traditional imaging with spectral analysis. Traditional spectral imaging technologies are often complex and costly, making them difficult to popularize in dynamic or transient scenes. In contrast, snapshot spectral imaging technology can capture spatial and spectral data within a single integration cycle of the imaging system. With the development of micro-nano optics, diffractive optical elements (DOEs) have been applied to snapshot spectral imaging due to their small size and high design flexibility, further reducing device volume and hardware costs. However, existing snapshot spectral imaging technologies based on DOEs are susceptible to the effects of diffractive lens fabrication accuracy and various errors during imaging. Moreover, they require sophisticated spectral image reconstruction algorithms, limiting their widespread application under practical conditions. To fully utilize the advantages of diffraction lens, we conduct in-depth research on their imaging and designing principles, error analysis, image acquisition, reconstruction, and deep learning algorithms. A new type of snapshot differentiable coded spectral imaging system is proposed, which can optimize the design of optical systems and achieve high-quality reconstruction of spectral images. The system demonstrates promising results in simulation and practical image restoration, showcasing its practical value. Methods We introduce a novel approach using a hybrid diffractive-refractive lens scheme, which effectively reduces the microstructure density of DOEs (Fig. 1). This not only shortens the system focal length and decreases DOE fabrication complexity but also enhances the imaging signal-to-*noise* ratio. Furthermore, it employs a deep unfolding framework alongside an improved Transformer model (DUF-DST, Fig. 2) to facilitate the reconstruction of diffraction spectral images. Building upon this framework, we conduct a comprehensive analysis of error sources in snapshot diffraction spectral imaging systems. This includes fabrication errors during DOE preparation (Figs. 3 to 4), component assembly discrepancies during imaging (Figs. 5 to 6), as well as sensor and environmental *noise* factors (Fig. 7). Through rigorous quantitative validation experiments, we quantify the magnitude of each error and assess their impact on imaging and final reconstruction outcomes via meticulous modeling and simulation. Finally, throughout the DOE design and reconstruction model training phases, we employ a joint optimization method to effectively mitigate these error sources. Results and Discussions To validate the effectiveness of the aforementioned optical model and spectral image reconstruction method, we conduct simulation tests by establishing a comprehensive image degradation model and reconstruction network framework based on actual experimental parameters. The DOE utilized in this paper is devised using an end-to-end joint optimization method (Fig. 8), which takes partial machining errors into consideration during the design optimization process. Through degradation-reconstruction testing on 30 scenes, the reconstructed results achieve an average peak signal-to-*noise* ratio (PSNR) of 37.16, a structural similarity index (SSIM) of 0.9881, and a spectral angle mapper (SAM) of 0.0591 (Fig. 9). Comparison with results from four other mainstream image reconstruction models demonstrates that the DUF-DST model employed here exhibits superior reconstruction performance (Fig. 10). Furthermore, to verify the effectiveness of the error suppression method proposed in this paper, a series of indoor and *outdoor* experiments are conducted (Figs. 12 to 18). These experimental scenarios closely resemble real-world application environments and encompass various analyzed errors and *noise*. Reconstruction of the original images captured is performed using a reconstruction network optimized based on error considerations. Experimental results indicate that the reconstruction model employed in this paper achieves high-quality restoration of spectral images, and the proposed error suppression method significantly enhances the robustness of the reconstruction algorithm to errors and *noise* in actual imaging processes. Conclusions Addressing the inadequate consideration of errors in the imaging process by current diffractive spectral imaging technology, which leads to limited imaging effects, we introduce a snapshot diffractive spectral imaging system along with a hybrid error suppression method. It systematically examines errors (height map error and graphic structure location error) arising from diffraction lens fabrication and component assembly, as well as system and environmental *noise*. Based on these error terms, a diffraction degradation model is constructed, and a deep unfolding network is used to reconstruct the diffraction-blurred images. By jointly training the degradation model and reconstruction network, the reconstruction algorithm's generalization ability to errors and *noise* is significantly enhanced. Relevant simulations and indoor/*outdoor* experiments demonstrate that the imaging model, with error suppression and the proposed reconstruction algorithm, effectively enhances the imaging quality of the system in practical application scenarios, achieving high-quality reconstruction of spectral images within a single integration cycle.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
176	**Development of a Building Evaluation Framework for Biophilic Design in Architecture**	Tabassum R.R.,Park J.	Buildings, 14(10), 3254	2024	34

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Biophilic design has gained prominence in architecture as a strategy to integrate natural elements into built environments, aiming to improve human comfort, well-being, and environmental sustainability. This paper critically reviews existing biophilic design frameworks, identifying gaps in their integration of health outcomes and the relationship between indoor and *outdoor* environments, with a primary focus on indoor environmental quality (IEQ) factors such as air quality, natural light, *thermal* comfort, and *acoustics*. The review was conducted systematically, comparing frameworks using criteria such as health benefits, human–nature connections, and user satisfaction. Key findings highlight a lack of comprehensive frameworks that explicitly link biophilic design to measurable improvements in human health, comfort, and satisfaction. To address this gap, the study proposes an enhanced biophilic design framework that bridges IEQ-focused indoor environments with urban *outdoor* elements, prioritizing human comfort alongside environmental sustainability. The framework also emphasizes the role of post-occupancy evaluation (POE) in assessing the real-world performance of biophilic interventions, particularly in achieving user satisfaction and ensuring long-term effectiveness. This research contributes to the advancement of biophilic design by offering actionable strategies for architects, urban planners, and policymakers to adopt biophilic principles that create resilient, healthy, and sustainable spaces. Additionally, the study emphasizes the need for empirical testing and validation of the proposed framework, including POE and user surveys, to assess its real-world impact on human well-being and satisfaction, thus setting the stage for future research.

Article • *Open access*

177	**Optomechanical design analysis and development of space-based visible coded spectrometer based on curved prism dispersion**	Jia X.,Wang F.,Liu J., ... He X.,Liu Y.	Journal of Optics India , 53(4)	2024	1

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The coded spectrometer, a relatively new instrument, has the advantage of high signal-to-*noise* ratio, giving it great application potential. We report the optical design of a space-based visible coded spectrometer based on curved prism dispersion (SVCS-CPD) and further explore the design and analysis of its support structure. A static simulation and modal analysis, sinusoidal vibration mechanical analysis, and random vibration mechanical analysis were performed to demonstrate the excellent mechanical and *thermal* stability of the support structure. To realize rapid assembly and integration of the off-axis optical system, a point source microscope and three-coordinate measuring machines were employed. In addition, tests of optical design parameters and *outdoor* push-scan imaging were conducted to validate the high imaging performance of the SVCS-CPD. The results showed that the modulation transfer functions of the SVCS-CPD in the central and marginal fields of view were 0.61 at 31.25 line pairs/mm and 0.53 at 31.25 line pairs/mm, respectively. In addition, the spectral smile and spectral keystone were < 0.3 pixels. The development of the SVCS-CPD can provide a useful reference for similar spaceborne dispersion spectrometers.

Article • *Open access*

178	**Integrated Approach for Human Wellbeing and Environmental Assessment Based on a Wearable IoT System: A Pilot Case Study in Singapore**	Salamone F.,Sibilio S., Masullo M.	Sensors, 24(18), 6126	2024	10

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This study presents the results of the practical application of the first prototype of WEMoS, the Wearable Environmental Monitoring System, in a real case study in Singapore, along with two other wearables, a smart wristband to monitor physiological data and a smartwatch with an application (Cozie) used to acquire users' feedback. The main objective of this study is to present a new procedure to assess users' perceptions of the environmental quality by taking into account a multi-domain approach, considering all four environmental domains (*thermal*, visual, *acoustic*, and air quality) through a complete wearable system when users are immersed in their familiar environment. This enables an alternative to laboratory tests where the participants are in unfamiliar spaces. We analysed seven-day data in Singapore using a descriptive and predictive approach. We have found that it is possible to use a complete wearable system and apply it in real-world contexts. The WEMoS data, combined with

	Document title	Authors	Source	Year	Citations

physiology and user feedback, identify the key comfort features. The transition from short-term laboratory analysis to long-term real-world context using wearables enables the prediction of overall comfort perception in a new way that considers all potentially influential factors of the environment in which the user is immersed. This system could help us understand the effects of exposure to different environmental stimuli thus allowing us to consider the complex interaction of multi-domains on the user's perception and find out how various spaces, both indoor and *outdoor*, can affect our perception of IEQ.

Article • *Open access*

179 **Effects on the Indoor Environment in a Stable for Horses in Winter: A Case Study** — Kic P.,Wohlmuthova M.,Starostova L. — Agriculture Switzerland, 14(8), 1287 — 2024 — 3

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The aim of this article is to show the most significant factors influencing the indoor environment in winter considering the operating conditions of an older stable modified for housing 12 horses and an indoor riding arena for teaching and sports purposes. This research focused on assessing the influences affecting the internal environment from the point of view of the construction of the building and ventilation control in the operating conditions of working and non-working days. The analysis of the results showed that the massive masonry structure has sufficient *thermal* insulation and accumulation, which was manifested by good temperature stability inside the stable of 7.2 ± 1.7 °C when the outside air temperature was −4.80 ± 1.5 °C. At low *outdoor* air temperatures of −6.44 ± 0.4 °C, the following conditions were found: a higher relative air humidity (76.0 ± 5.3%), a high CO_2 concentration 2317.1 ± 931.7 ppm, and a high airborne dust concentration PM_{10} = 231.94 ± 19.13 μg·m^{-3} and $PM_{2.5}$ = 160.13 ± 6.28 μg·m^{-3}. Therefore, it is necessary to improve the solution and function of the stable ventilation. The small size of the windows and their uneven distribution (average daylight factor e_i from 0.313 ± 0.154 to 0.835 ± 0.309) caused insufficient daylight in some individual boxes.

Conference Paper • *Open access*

180 **Optimization of indoor quality and thermal comfort for university classrooms using data-based machine learning** — Jiang Q.,Liu J.,Yang X. — E3s Web of Conferences, 546, 02011 — 2024 — 6

Hide abstract Find It @ UCL View at Publisher Related documents

Improving indoor environment quality on university classrooms is a hot research topic. The on-site experiment was conducted on indoor environmental quality regarding temperature, humidity, air pollutants, light and *acoustics* during different seasonal conditions. The result shows that nearly 25% of indoor particulate matters exceeded the GB18883 standard when the *outdoor* environmental pollution was severe under natural ventilation conditions in autumn. More than 20% of students experienced symptoms of drowsiness, dizziness, chest tightness, poor breathing, as well as depression and irritability. From the analysis of occupant demand, indoor air pollution and *thermal* comfort are the most anticipated areas for students to improve their learning environment. This paper proposes an optimal IEQ prediction model integrated with students' satisfaction and indoor environmental features using machine-learning classification algorithms. The back-propagation neural network shows the high prediction accuracy among different algorithms. The traditional PMV-PPD model shows an accuracy rate of only 28% for *thermal* sensation prediction, while the highest prediction accuracy obtained through machine learning algorithms is about 75%. Moreover, the influence of individual's *thermal* adaptation ability, including gender, long-term *thermal* experience, and psychological factors, and environmental factors was analyzed in this study.

Article

181 **Estimation of the effect of thin-plated surface on a three-staged looped thermoacoustic engine for low-grade heat recovery** — Zhang L.,Kang H.,Ding X.,... Wen J.,Zhang P. — Applied Thermal Engineering, 248, 123366 — 2024 — 6

Hide abstract Find It @ UCL View at Publisher Related documents

This paper introduces two novel structures of looped *thermoacoustic* engines designed for *outdoor* low-frequency *sound* sources. The system consists of three *thermoacoustic* engines in a loop and a stub for *sound* radiating, which is characterized by reliability, high efficiency, and compactness. Two thin plates are used to prevent the Gedeon flow,

	Document title	Authors	Source	Year	Citations

facilitate *sound* propagation, and adjust the *acoustic* field. The system with a straight stub achieved a maximum *Sound* Pressure Level of 126.09 dB at 1 m away, surpassing highest reported records by 25.96 %. The system also achieves a high output efficiency of 3.86 %. The system with tapered stub and a 0.1 mm silicone plate, achieved an SPL of 120.76 dB at 1 m. To maintain the predetermined *acoustic* field, it is important to consider a thin plate with small Young's modulus and thickness. The effect on the *acoustic* field must be considered when the thickness exceeds 0.8 mm. The addition of the 0.28 mm thin plate in system with tapered stub resulted in a 7.8 W increase in *acoustic* power compared to that of system without the thin plate.

Article • *Open access*

182 **The Working Environment in Primary Healthcare Outpatient Facilities: Assessment of Physical Factors and Health Professionals' Perceptions of Working Environment Conditions**

Cezar-Vaz M.R., Bonow C.A., Vaz J.C., ... de Souza J.L., Vaz J.

International Journal of Environmental Research and Public Health, 21(7), 847

2024

1

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The objectives of this study were to assess the adequacy of physical parameters/factors (temperature, relative humidity, *noise*, and illuminance levels) of the work environment in PHC facilities, to evaluate the association between the adequacy of these measured physical parameters and the physical characteristics of the PHC facilities and their surroundings and to assess the association between health professionals' perceptions about exposure to physical risks in the PHC work environment and the adequacy of physical parameters measured in the same facilities. The study monitored 23 PHC facilities in southern Brazil and interviewed 210 health professionals. Data analysis involved Pearson's chi-square, Fisher's exact test, Spearman's correlation, and multivariate linear regression analysis was used to control for confounding factors. The significance level was set at 5% ($p \leq 0.05$). The combination of temperature and relative humidity presented *thermal* comfort levels outside the adopted criteria for adequacy in consultation (*outdoor* relative humidity, $p = 0.013$) and procedure rooms (front door open, $p = 0.034$). Inadequate *sound* comfort (*noise*) levels in the morning shift were found in the vaccination (front door open, $p = 0.021$) and consultation rooms (movement of people, $p = 0.016$). In PHC facilities where reception rooms had insufficient lighting, internal curtains were opened less frequently ($p = 0.047$). The analysis of health professionals' perceptions of physical factors demonstrated that physicians more frequently perceive the physical risk of temperature and humidity ($p = 0.044$). The higher the number of nurses ($p = 0.004$) and oral health technicians in the PHC facilities ($p = 0.031$), the greater the general percentage of adequacy of monitored physical parameters. It was also confirmed that the higher the perception of moderate or severe physical risk among health professionals, the lower the general percentage of the adequacy of the physical parameters of the work environment of the PHC facilities evaluated ($r_s = -0.450$, $p = 0.031$). This study's evidence contributes to a better understanding of physical conditions and future occupational interventions to ensure the comfort, safety, and well-being of PHC workers.

Article

183 **Sandwich composites from recycled plastic reinforced with krajood (Lepironia articulata) fiber for building applications**

Homkhiew C., Srivabut C., Boonchouytan W., Rawangwong S.

Iranian Polymer Journal English Edition, 33(6)

2024

8

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Abstract: This research aimed to showcase an economical and environmentally friendly approach to producing sandwich-structured composites from recycled plastics and krajood waste. The effects of the contents and particle sizes on mechanical and physical properties, including *thermal* conductivity and *sound* absorption coefficient, were investigated. The experimental results were analyzed using Analysis of Variance (ANOVA). The results confirmed that the isocyanate glue at 25% (by weight) and krajood fiber at 75% (by weight) at 40 mesh enhanced the flexural, compressive, and screw withdrawal properties the most. The increase in krajood fiber contents ranged from 75 to 85% (by weight) and increased the water absorption percentages. Core layer samples with high conductivity at 0.085 W/mK are considered suitable for building insulation applications, whereas sandwich composites at 0.189 W/mK are used for *outdoor* materials. In addition, the *sound* absorption coefficient and *noise* reduction coefficient were used to evaluate the application efficiency of sandwich-structured composite materials. The highest *sound* absorption

	Document title	Authors	Source	Year	Citations

coefficient and *noise* reduction coefficient were found using a *sound* frequency of 1000 Hz. The highest *noise* reduction coefficient value of 0.78 was found in the core layer sample, meaning that the sample absorbs 78% of *sound* and reflects the remaining 22% of *sound* waves back into a room. Thus, the outcomes of this study offer recommendations for the manufacture of sandwich composite materials from recycled polypropylene reinforced with krajood fiber in applications for construction and building products. Graphical Abstract: (Figure presented.)

Article

184 **Structural, acoustical, and thermal evaluation of an experimental house built with reinforced/hollow interlocking compressed stabilized earth brick-masonry**

Abdel Gelil Mohamed N., Moustafa A., Darwish E.A.

Journal of Building Engineering, 86, 108790

2024

19

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With the increasing demand on affordable construction that addresses the global challenges of the modern world, earthen construction is being reintroduced and investigated as a potential sustainable and affordable building method. Compressed stabilized earth blocks (CSEBs) are a common form of enhanced earthen building materials. Previous studies focused on the structural performance of CSEB walls and prisms consisting of one type of bricks with the same mixture, analyzed their *thermal* and *acoustic* properties as units only, or evaluated the *thermal* performance of solid unreinforced CSEB masonry. The present research is the first complete structural, *acoustical*, and *thermal* investigation of an experimental house built with a novel reinforced/hollow interlocking compressed stabilized earth brick-masonry (ICSEB) using varying stabilizer contents, compliant with the Egyptian code for building with stabilized soil—Part One: Building with compressed stabilized soil units (EG-SE2016). Methods depend on experimental testing, codified theoretical calculations, finite element analysis, and live measurements. According to the structural experimental testing program, reinforcement was found to triple the load bearing capacity of ICSEB columns and, although minimal rebars were used in the ICSEB walls, it has improved their load bearing capacity by 23% for single walls (125 mm) and 58% for double walls (250 mm). In addition, doubling the thickness of the unreinforced ICSEB walls increased the load bearing capability by 164%, while raising the thickness of the reinforced wall from 125 mm to 250 mm increased the load bearing capacity by more than threefold (3.38). Based on the codified theoretical calculations and findings of the finite element analysis, it is safe to recommend reinforced double ICSEB walls as a reliable construction for a two-story residential building. *Acoustically*, double ICSEB walls attained the *noise* reduction index value of 40 dB, which is required by the Egyptian code for *acoustic* insulation and *noise* control in residential buildings, to protect against the average *outdoor noise* levels of 65–70dBA. *Thermally*, reinforced double ICSEB walls kept the indoor air temperature and relative humidity values within the *thermal* comfort zone, 23.9–25.7 °C and 28–36%, respectively, despite the fluctuating *outdoor* temperatures and relative humidities (16–37 °C and 7–88%, respectively). Indoor air temperatures were 1.5–2 °C lower than those of concrete masonry walls. Additional research is required to evaluate resistance to environmental aspects, such as exposure to water, wind, and corrosion, and to further investigate their *thermal* and *acoustical* properties.

Article • *Open access*

185 **A Perceptual Assessment of the Physical Environment in Teaching Buildings and Its Influence on Students' Mental Well-Being**

Wen Q., Zhou Q., Ye H., ... Shan J., Huang Z.

Buildings, 14(6), 1790

2024

8

Hide abstract Find It @ UCL View at Publisher Related documents

Numerous studies have examined the impact of the built environment on mental health, yet there remains an underexplored area concerning how microenvironments within educational buildings affect students' mental well-being from a physical environment standpoint. This paper fills this gap by utilizing data from 440 valid questionnaires to develop regression models that assess students' perceptions of physical environment factors in college teaching buildings and their impact on anxiety likelihood. This study examined the physical environment of the teaching building's interior, courtyard, and semi-*outdoor* areas. Findings indicate that students' perceptions of specific physical environment factors—such as classroom ventilation ($p < 0.01$, OR = 0.330), lighting ($p < 0.01$, OR = 0.444), *noise* conditions ($p < 0.01$, OR = 0.415), courtyard *thermal* comfort ($p < 0.01$, OR = 0.504), and the views from semi-*outdoor* areas ($p < 0.01$, OR = 2.779)—significantly influence the likelihood of experiencing anxiety. Optimal physical conditions are linked to reduced student anxiety. The suitability of the physical environment of teaching buildings is interrelated,

	Document title	Authors	Source	Year	Citations

and it is urgently necessary to address issues related to unsuitable lighting in window areas of classrooms, as well as problems with ventilation, lighting, and *noise* caused by the corridor layout within teaching buildings. These insights are crucial for the design and renovation of academic buildings to enhance students' mental well-being.

Article • *Open access*

186 **A field study of investigation of indoor environmental quality status in Indian offices: Concerns and influencing building factors**

Shukla P.,Mishra S., Goswami S.

Journal of Building Engineering , 86, 108648

2024

7

Hide abstract Find It @ UCL View at Publisher Related documents

Maintaining a high-quality indoor environment in office buildings is important as it impacts the office workers' health, well-being, and productivity. The objective of this paper is to present the state of indoor environmental quality (IEQ) in Indian offices. It discusses the findings of a comprehensive IEQ evaluation of 30 air-conditioned Indian offices; highlights the areas of concern and discusses the building aspects that were influencing the IEQ conditions. Building details collected from the project teams and the IEQ data on 17 parameters under 4 categories (indoor air quality (IAQ), lighting and views, *thermal* comfort, and *acoustics*) that were collected via spot measurements and on-site inspections, were analyzed using a data-driven approach to determine the IEQ status. The measured data was examined with reference to relevant Indian and international standards. Study findings reveal a prevalence of sub-standard indoor environment in Indian offices and the main concerns were IAQ (high levels of CO2, TVOC, PM & NO2), lighting conditions (low illuminance and low EML levels), lack of access to good quality external views and high levels of interior generated background *noise*. The research highlights several IEQ issues in Indian offices that need to be addressed to maintain conditions as per the recommendations of the relevant national and international IEQ standards. The paper also describes the IEQ evaluation protocol. This protocol and the findings of this study will provide guidance to building owners and facility managers in investigating and improving the indoor environmental conditions in office spaces.

Article

187 **Creating comfortable outdoor environments: Understanding the intricate relationship between sound, humidity, and thermal comfort**

Qin Z.,Lu B.,Jing W., ... Qiu J.,Dong Y.

Urban Climate, 55, 101967

2024

17

Hide abstract Find It @ UCL View at Publisher Related documents

This study provides insight into the impact of interactive *sound* and humidity environments on *outdoor thermal* comfort in open spaces in cold regions. Executed in Xi'an, China, the research involved a synthesis of questionnaire-based subjective assessments and meteorological data collection. Four pivotal findings emerged from the study. Initially, environments characterized by concurrent *acoustic* and humid conditions demonstrated a proportional increase in *Thermal* Sensation Vote (TSV) corresponding to rising humidity levels. Secondly, in these environments, an increase in *sound* levels was linked to an escalated *thermal* sensitivity among individuals, with the most significant sensitivity, quantified at 0.1769, observed at *sound* intensities between 60 dB (A) and 80 dB (A). Thirdly, the influence of ambient *sound* on TSV and *Thermal* Comfort Vote (TCV) was observed to vary with ambient humidity. Variations in *sound* levels had a pronounced effect on both TSV and TCV within a humidity range of 25% RH to 70% RH, but this correlation dissipated in higher humidity conditions (70% RH to 85% RH). Finally, the study indicates that environments with lower *sound* levels contribute to the mitigation of extreme *thermal* stress, particularly when ambient humidity is maintained between 25% RH and 70% RH.

Article

188 **Effects of thermal and acoustic environments on human comfort in urban and suburban campuses in the cold region of China**

Chen P.,Dai Y.,Zhen M.

Environmental Science and Pollution Research , 31(21)

2024

14

Hide abstract Find It @ UCL View at Publisher Related documents

In this study, we investigated the effects of *thermal–acoustic* environments on human comfort in a cold region by focusing on Xi'an in China as a representative city. Four typical open spaces were identified within two universities, with one located in an urban area and the other in a suburban area. Surveys were conducted using questionnaires

Document title	Authors	Source	Year	Citations

and environmental parameters were measured to assess *thermal–acoustic* perception. The physiological equivalent temperature (PET) and *noise sound* pressure levels were employed to measure the *thermal–acoustic* environments in the four open spaces. The results showed that the neutral PET was 19.1 °C in Xi'an from March to May, and the neutral temperature range was 14.9–23.4 °C. The preferred temperature was 20.0 °C. *Acoustic* sensation votes were lower in suburban areas than urban areas. Respondents in urban areas exhibited significantly higher sensitivity to the *thermal* and *acoustic* environments compared with those in suburban areas. Sensitivity to the *thermal* environment decreased as the environmental *noise* levels decreased. The temperature perceived as most suitable increased as the sensitivity to the *thermal* environment decreased. *Acoustic* comfort evaluations were generally higher in colder and warmer environments. Warmer environments heightened the sensitivity to the *acoustic* environment and this effect gradually decreased as the *sound* pressure levels increased. Thus, based on empirical analyses, we compared the effects of *outdoor thermal–acoustic* parameters on comfort to provide experimental data support for further *outdoor thermal* comfort research.

Article

189 **Microseismic Observation Enabled by High-Sensitivity Micromechanical Interferometers** — Li C.,Sun Z.,Zhou L., ... Huang X.,Yang B. — IEEE Transactions on Industrial Electronics, 71(5) — 2024 — 8

Hide abstract Find It @ UCL View at Publisher Related documents

High-sensitivity accelerometry has historically been employed to explore Earth sciences and advance resource-related industries, such as petroleum exploitation. The devices capable of such measurements, however, traditionally involve precise mechanical structures and consequent large dimensions. In this article, combined by micromechanical systems and grating-assisted interferometry, a closed-loop optical accelerometer with nano-g (ng) sensitivity is developed for microseismic observations. In order to lower the intrinsic *thermal noise*, a modified 8.84-Hz micromechanical spring-mass oscillator is designed and fabricated. Additionally, an active control system is equipped to stabilize the device temperature, and thus, enhance the low-frequency stability. Field tests demonstrate that the prototype has achieved a scale factor of 1195 V/g, along with a dynamic range of 1.12 mg and a bandwidth of about 7 Hz. Further, under the controlled 45 ± 0.01°C temperature condition, complete terrestrial microsesimic peaks from 0.1 to 3 Hz are tracked. After calibration by a reference seismometer, the device self-*noise* is determined as 2 ng/Hz from 0.2 to 4 Hz with a bias stability of 7.9 ng. The transducer and control circuits are both packaged within a 13.6 cm x 13.6 cm x 12.8 cm insulated enclosure, making it advantageous for *outdoor* observations.

Article • *Open access*

190 **Field Investigation on Adaptive Thermal Comfort in Rural Dwellings: A Case Study in Linyi (China) during Summer** — Tang H.,Gao Y.,Tan S., Guo Y.,Gao W. — Buildings, 14(5), 1429 — 2024 — 16

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A large number of people in China still live in rural villages. The indoor environment of these rural dwellings directly affects the quality of life of the occupants. Nevertheless, constrained by the quality of dwelling construction, rural buildings have poorer indoor environments and, at the same time, have a higher operating energy consumption. However, inadequate attention has been given to the summer *thermal* environment in cold regions. This work has been carried out around the *thermal* environment of rural residences in cold regions during summer. Field measurements, questionnaires, and data analysis were used in this study. We recorded the indoor and *outdoor thermal* environment parameters on a typical summer day in the Linyi rural area. Moreover, the subjective sensations and *thermal* adaptive behaviors of the participants were recorded in detail with a questionnaire. Linear regression showed that the neutral temperature for residents in summer was 27.52 °C, with acceptable temperatures ranging from 25.14 °C to 29.9 °C. Age and gender differences were found to affect the occupants' sensation of *thermal* comfort and humidity, as well as their *thermal* adaptive behavior. In addition, a *thermal* adaptive model has been constructed in the study, which will further enrich the *thermal* adaptive investigation and provide a scientifically *sound* reference for the renovation and development of the local rural areas.

Article

	Document title	Authors	Source	Year	Citations
191	**Heat-sink-free solar-driven thermoelectric device for micro energy scavenging**	Yan B.,Li Y.,Guo Y., ... Yang Z.,Zhang C.	Nano Energy, 123, 109382	2024	20

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Harvesting energy from nature, such as mechanical energy, *thermal* energy, and biomass energy, has become an effective way to supply power for self-driven sensing and Internet of Things (IoT) applications. Among that, solar-driven *thermoelectric* (STE) device takes advantage of all-solid-state energy conversion, such as no *noise*, easy access, and free of wear, exhibiting great potential. Here, we propose a solar-driven *thermoelectric* device for harvesting solar energy, based on Micro Electromechanical System (MEMS) technology. Specifically, the optimization of solar absorptances of structural layers significantly improved the photo-*thermo*-electric conversion performance. Furthermore, the electrical output strongly depends on temperature difference dominated by the absorptance difference. Therefore, the reported device can be free of traditional heat sink, which greatly simplifies its structure and expands the working conditions, especially in microdevices. Particularly, a continuous output voltage of 0.34 V/cm^2 is achieved with concentrated solar radiation, demonstrating its potential under no heat sink conditions. Besides, a fast response (4.2 ms under 660 nm laser) is achieved benefitting from the micron-size solar absorber, showing its significance in sensing applications. Finally, the proposed STE device is utilized to harvest *outdoor* solar energy, displaying promising potential in solar energy harvesting devices and providing an insightful perspective for *thermoelectric* applications.

Review • *Open access*

	Document title	Authors	Source	Year	Citations
192	**Personal Strategies to Reduce the Cardiovascular Impacts of Environmental Exposures**	Bonanni L.J.,Newman J.D.	Circulation Research, 134(9)	2024	9

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Ubiquitous environmental exposures increase cardiovascular disease risk via diverse mechanisms. This review examines personal strategies to minimize this risk. With regard to fine particulate air pollution exposure, evidence exists to recommend the use of portable air cleaners and avoidance of *outdoor* activity during periods of poor air quality. Other evidence may support physical activity, dietary modification, omega-3 fatty acid supplementation, and indoor and in-vehicle air conditioning as viable strategies to minimize adverse health effects. There is currently insufficient data to recommend specific personal approaches to reduce the adverse cardiovascular effects of *noise* pollution. Public health advisories for periods of extreme heat or cold should be observed, with limited evidence supporting a warm ambient home temperature and physical activity as strategies to limit the cardiovascular harms of temperature extremes. Perfluoroalkyl and polyfluoroalkyl substance exposure can be reduced by avoiding contact with perfluoroalkyl and polyfluoroalkyl substance-containing materials; blood or plasma donation and cholestyramine may reduce total body stores of perfluoroalkyl and polyfluoroalkyl substances. However, the cardiovascular impact of these interventions has not been examined. Limited utilization of pesticides and safe handling during use should be encouraged. Finally, vasculotoxic metal exposure can be decreased by using portable air cleaners, home water filtration, and awareness of potential contaminants in ground spices. Chelation therapy reduces physiological stores of vasculotoxic metals and may be effective for the secondary prevention of cardiovascular disease.

Conference Paper

	Document title	Authors	Source	Year	Citations
193	**A lightweight FL-based UAV detection model using thermal images**	Nair A.K.,Sahoo J.,Raj E.D.	ACM International Conference Proceeding Series, 15	2024	0

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With the growing popularity of lightweight flying systems, unethical usage of UAVs has become a concern from privacy and security perspectives. Existing detection approaches are ill-suited for such small devices, making it a tedious task to track their activities. In this work, we tackle this issue by proposing an FL-based UAV classification model utilizing *thermal* images in *outdoor* setting. Compared to other approaches, *thermal* images are more reliable for UAV detection as UAVs release *thermal* signatures, maintaining their visibility irrespective of environmental factors in *thermal* images. The proposed model is agnostic to major issues such as occlusion, luminance, and movements. With a primary focus on maintaining user privacy and system security, an initial DL model was formulated, further

	Document title	Authors	Source	Year	Citations

shifting it to a federated setting. Additionally, Lapclain *noise*-based differential privacy was integrated into the FL model to enhance the system privacy. Extensive experiments performed validated the model's capabilities in accurately predicting the presence of UAVs in the given scenario with an accuracy of 96% and 92% for the DL and FL models, respectively.

Article

194 **Preparation and properties of high blending phosphogypsum-desulfurization ash-waste soil based functional prefabricated autoclaved aerated concrete slabs**

Wang C.-Q.,Cheng L.-X., Wang Z.-Y.,... Huang D.-M., Wei S.

Construction and Building Materials , 423, 135879

2024

25

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There is a huge stock of industrial solid waste piles such as phosphogypsum (PG), desulfurization ash (DA), and waste soil (WS) in China, and their utilization rate is insufficient. These have potential risk for environmental safety due to pollutants contained in them. In this study, after proportioning design, raw material pretreatment, bubble regulation and multi-temperature section maintenance process, successfully prepared a high blending solid waste based functional prefabricated autoclaved aerated concrete slabs (FPAS) with excellent functional performance, and the optimal ratio of each raw material is PG: WS: DA: cement: lime = 8:50:20:10:12, and the dry mass ratio is up to 75%, the compressive strength of the slabs is up to 5.2 MPa, the standard dry density is 606 kg/m^3, the dry shrinkage value is 0.36 mm/m, and the post-freezing strength is 4.0 MPa, which is in line with the basic requirements in the standard “Autoclaved aerated concrete slabs” (GB/T15762), and at the same time, its fire-resistant integrity is qualified for 240 min, and the coefficient of *thermal* conductivity is 0.15w(m.k), airborne *sound*-weighted *sound* insulation is 46 dB, also can show different functional characteristics. In order to investigate the environmental safety of heavy metals (HMs) in the slabs, the simulation experiment of HMs leaching from *outdoor* stacking of slabs was carried out. The results show that the HMs in the slabs do not pose obvious risks to the natural environment and human body, but three of the characteristic HMs, Cr, Cd and Hg, have significant leaching patterns, which can be expressed by a second-order kinetic model. Finally, the life cycle assessment and cost analysis of the high blending PG-DA-WS based FPAS and the conventional aerated slabs were carried out. The main environmental damage category of the high blending PG-DA-WS based FPAS is land ecological risk, and it has obvious environmental and economic benefits compared with the conventional aerated slabs.

Article

195 **A study of acoustic-light-thermal effects on pedestrians’ overall comfort in a Cfa-climate campus during the summer**

Xiao A.,Cheng B.,Zhang J., ... Liu T.,Zhu F.

Journal of Thermal Biology , 121, 103839

2024

21

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The environmental quality, in terms of *acoustic*, visual, and *thermal* environments, significantly affects people's comfort levels. Along these lines, in this work, their comprehensive impact on people's overall comfort was systematically explored. Pedestrians' *outdoor* neutral points on various environmental parameters were found by performing linear regressions. Similarly, people's *thermal* perceptions (indicated by neutral temperatures, NT) were found to vary for both *acoustic* and light environments. They would be increasingly heat sensitive (R^2 increases) in a noisier environment while the NTs varied for either *sound* or light intensity levels. From our analysis, it was demonstrated that people's overall comforts were negatively correlated with these parameters in different degrees. This work provides valuable insights for future urban design and planning studies to create better *outdoor* environments.

Article • *Open access*

196 **Strategic Integration of a Vegetative Component on a Metal Roof Base: An Evaluation of Its Impacts on Thermal and Acoustic Performance in the Tropics**

Aw S.B.,Leng P.C., Ling G.H.T.,... Law N.H.K., Goh S.Y.Z.

Buildings, 14(4), 915

2024

2

Hide abstract Find It @ UCL View at Publisher Related documents

	Document title	Authors	Source	Year	Citations

This paper attempts to ascertain the *thermal* and *acoustic* impacts of introducing a vegetative roof layer on insulated and uninsulated metal roofs for tropical climates, through field measurements in Skudai, Johor, Malaysia, that were conducted for both dry and wet days. Four small-scale roof modules were tested, namely an uninsulated metal roof (uiMDR), an insulated metal roof (iMDR), and two identical corresponding modules with an additional vegetative component (uiGR and iGR, respectively). *Outdoor* ambient temperature (T_{amb}) was the most influential correlated variable affecting the roof outer surface temperature (R_{TOS}) in 50% of the assessed scenarios. On the selected dry day, the inter-quartile ranges (IQR) of iGR, iMDR, uiGR, and uiMDR were 6.21 °C, 8.32 °C, 6.69 °C, and 1.66 °C, respectively; the IQRs were 1.6 °C, 4.11 °C, 2.59 °C, and 1.78 °C, respectively, on the selected wet day. Based on design U-value calculations, iGR was better than iMDR and uiMDR for both dry and wet days. The U-value of uiGR was also better than iMDR under dry-day conditions. The Wilcoxon signed-rank test also indicated a statistically significant difference in the roof inner surface temperature (R_{TIS}) measurements (p-value = 0.00) during Malaysian daylight hours, between 8.00 a.m. and 6.00 p.m., regardless of the weather. In terms of *sound* level reduction under dry-day conditions, the Kruskal–Wallis and Wilcoxon signed-rank tests showed statistically significant differences in *sound* level reductions, with iGR and uiGR performing better than iMDR and uiMDR (p-values = 0.00). The *sound* level reductions for iGR, iMDR, and uiGR were 51%, 32%, and 31%, respectively, while uiMDR experienced *sound* level amplifications by 6%, possibly due to the *acoustic* resonance effect. This proof of concept may encourage a broader application of extensive GRs in Malaysia using metal roofs, beyond the conventional RC base construction method.

Article • *Open access*

197 **Pulse Accumulation Approach Based on Signal Phase Estimation for Doppler Wind Lidar** — Liang N.,Yu X.,Lin P., ... Luo F.,Tong S. — Sensors, 24(7), 2062 — 2024 — 8

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Coherent Doppler wind lidar (CDWL) uses transmitted laser pulses to measure wind velocity distribution. However, the echo signal of CDWL is easily affected by atmospheric turbulence, which can decrease the signal-to-*noise* ratio (SNR) of lidar. To improve the SNR, this paper proposes a pulse accumulation method based on the cross-correlation function to estimate the phase of the signal. Compared with incoherent pulse accumulation, the proposed method significantly enhances the correlation between signals from different periods to obtain high SNR gains that arise from pulse accumulation. Using simulation, the study evaluates the effectiveness of this phase estimation method and its robustness against *noise* in algorithms which analyze Doppler frequency shifts. Furthermore, a CDWL is developed for measuring the speed of an indoor motor turntable and the *outdoor* atmospheric wind field. The phase estimation method yielded SNR gains of 28.18 dB and 32.03 dB for accumulation numbers of 500 and 1500, respectively. The implementation of this method in motor turntable speed measurements demonstrated a significant reduction in speed error—averaging 9.18% lower than that of incoherent accumulation lidar systems. In experiments that measure atmospheric wind fields, the linear fit curve slope between the measured wind speed and the wind speed measured via a commercial wind-measuring lidar can be reduced from 1.146 to 1.093.

Article • *Open access*

198 **Optical System Design and Athermalization of Telephoto Lens | 长焦镜头光学系统设计及无热化研究** — Zhang L.,Lu Q.,Duan F., Dai X.,Qiao D. — Guangxue Xuebao Acta Optica Sinica, 44(8), 0822004 — 2024 — 7

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Objective The actual working environment of vehicle-mounted LiDAR is complex, including seasonal long-time limit temperature and rapid changes in indoor and *outdoor* high and low temperatures. These temperature variations possibly change the internal optics and structure of the lens, which results in image plane drift and reduces imaging quality. For telephoto lenses, the image plane drift can be more obvious with the ambient temperature changes. The current passive athermalization design method has the problems of complex structure and large volume caused by multi-layer lens barrels, or the introduction of diffractive elements and aspheric lenses to increase the production cost, and the narrow temperature range of *thermal*-free, which is difficult to adapt to the practical applications in the complex environment of vehiclemounted LiDAR. Therefore, it is necessary to reduce the image plane temperature drift and improve the environmental adaptability of telephoto lenses with an advanced athermal design method. Methods The telephoto lens which can accurately capture distant targets and magnify the details is integrated with a line array detector to improve the resolution. Meanwhile, based on the fact that the total length of the optical system of

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the telephoto structure is smaller than the focal length, the volume of the telephoto receiving optical system is compressed to a certain extent to realize the requirements of lightweight miniaturization and high resolution of the vehicle-mounted LiDAR. Aiming at the problem that the telephoto lens is susceptible to temperature, we improve the two-group compensation design method of the passive optical and mechanical athermalization to maximally offset the optical focal length change of the optical parts from the *thermal* difference brought by *thermal* expansion and contraction of the structural parts, and to reduce the image plane drift, thus realizing the athermalization of the telephoto lens. Finally, the image plane drift of the as-designed lens is less than the depth of focus over a wide temperature variation range from -40 to 100 ℃. This is conducive to ensuring the imaging quality of the lens, and the designed structure has a simple preparation process and is easy to engineer and produce. Results and Discussions Different combinations of optical materials and optical focal length distributions are determined, structural components of different *thermal* expansion coefficients (TCEs) are matched, and the *thermal* difference of the optical system compensates for each other, with the system athermalization design achieved. Without the *thermal* expansion and contraction of the barrel holder taken into account, the focal shift of the lens with temperature change is always minimized and the image plane drift is 0.075 mm when the temperature increases to 100 ℃ (Fig. 7). The *thermal* expansion and contraction of the barrel holder is considered as a material to compensate for *thermal* aberration to make the sensor detecting surface always in the image plane. In the wide temperature range from – 40 to 100 ℃, with the temperature change, the receiving optical system obtained from the selected optics and structural component materials has almost no significant focal shift, even when the temperature is as high as 100 ℃, and meanwhile the amount of focal shift is only 0.021 mm, smaller than its depth of focus at room temperature (0.074 mm), and the field curvature and distortion of this optical system have small changes (Fig. 8). The MTF at 30 lp/mm is all larger than 0.5 for each field of view (FOV), and the focal plane shifts are all small, which indicates that the designed lenses can maintain *sound* image quality over a wide range of temperatures from –40 to 100 ℃ (Figs. 9 and 10). The diffuse spot radius in the full FOV is smaller than 7 μm, which reveals that the focal shift of the lens is little affected by temperature (Fig. 11). The results of photographing vehicles traveling on the road show clear imaging of the vehicles and obvious feature areas such as the outer contours of the vehicles (Fig. 14). The above results prove that the imaging quality and temperature adaptability of the lens can be guaranteed by the above athermalization design to compensate the system *thermal* difference. Conclusions We employ the telephoto lens with a long focal length and small FOV to subdivide the scanning area and integrate a line array detector to achieve an image-level imaging effect. Based on the characteristic that the total length of the optical system of telephoto structure is smaller than the focal length, a receiving optical system with a telephoto ratio of 0.38 is designed, which has a smaller lens length and lower cost and meets the requirements of vehicle-mounted LiDAR in terms of high resolution, light weight, and small size. Given the large temperature difference in the working environment of vehicle-mounted LiDAR and the image plane drift of the telephoto lens, a passive optical and mechanical athermalization is implemented to confirm the reasonable combination of multi-plane spherical glass lens and structural components. Finally, a four-piece telephoto lens optical system with a simple structure and a focal shift of 0.021 mm less than the depth of focus of 0.074 mm over a wide temperature range from –40 to 100 ℃ is designed. The MTF of each FOV at 30 lp/mm is larger than 0.5 and the diffuse spot radius in the full FOV is smaller than 7 μm. The vehicle imaging is clear, and the outer contour of the vehicle and other characteristics of the area are obvious, which achieves athermalization and shows favorable environmental adaptability.

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☐	199	**Understanding how indoor environmental classroom conditions influence academic performance in higher education**	Brink H.W.,Lechner S.C.M., Loomans M.G.L.C., Mobach M.P.,Kort H.S.M.	Facilities, 42(3-4)	2024	17

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Purpose: This study aims to qualitatively examine the relationship between the indoor environmental quality (IEQ), lecturers' and students' perceived internal responses and academic performance. Design/methodology/approach: To capture user experiences with the IEQ in classrooms, semi-structured interviews with 11 lecturers and three focus group discussions with 24 students were conducted, transcribed, coded and analyzed using direct content analysis. Findings: The findings show that lecturers and students experience poor *thermal*, lighting, *acoustic* and indoor air quality (IAQ) conditions that may influence their ability to teach and learn. Maintaining acceptable *thermal* and IAQ conditions was difficult for lecturers, as opening windows or doors caused *noise* disturbances. In uncomfortable conditions, lecturers may decide to give a break earlier or shorten a lecture. When students experienced discomfort, it

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may affect their ability to concentrate, their emotional status and their quality of learning. Research limitations/implications: The findings originate from a relatively small sample, which might have limited the number and variety of identified associations between environment and users. Practical implications: Maintaining acceptable air and *thermal* conditions will mitigate the need to open windows and doors. Keeping doors and windows closed will prevent *noise* disturbances and related distractions. This will support the quality of learning in classrooms. This study reveals the end users' perspectives and preferences, which can inspire designers of new school buildings in higher education. Originality/value: This study emphasizes the importance of creating and maintaining optimal IEQ conditions to support the quality of teaching and learning. These conditions are particularly relevant when classroom occupancy rates are high or *outdoor* conditions are unfavourable.

Conference Paper

200 **A review: Outdoor visible light communication on modulation and receiver denoising scheme** — Othman N., Beson M.R.C., Aljunid S.A., Endut R. — Aip Conference Proceedings, 2898(1), 030038 — 2024 — 4

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A rapid growth of wireless communication system has boosted the exponential development on smart technology while driving a substantial increase in traffic volume. With a highly congested and demand on RF bandwidth spectrum, Visible Light Communication (VLC) seem as a complementary technology towards RF based system which provide wider bandwidth, free and unregulated channel, poses no health hazard, high security and low power consumption leads VLC to be one of the promising technology in data transmission. By maintaining the main purpose of illumination, the exploitation on the existing LED-based lighting system is used as a transmission mechanism to photodetector or image sensor which is used to detect the variations of the light intensity via free space optical channels. Different VLC modulation schemes is review in this paper to see the pros and cons of each scheme implemented in *outdoor* application specifically for vehicle-to-vehicle (V2V) VLC system. Various type of *noises* is presented in this paper with the denoising scheme to eliminate or reduce the shot and *thermal noise* caused by natural sunlight and artificial roadside infrastructure light which degrades the performance of VLC system.

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